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NG9-1-1 PSAP and ECC Specifications for the NENA i3 Solution

ABSTRACT: This standard specifies functions and interfaces for NG9-1-1 Public Safety Answering Points and Emergency Communications Centers conforming to the NENA i3 Standard for Next Generation 9-1-1.

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The CRM team will provide the remainder of the information on this page.

ANS Candidate Document (include on draft document, if applicable)

NENA-STA-023.1-YYYY

NENA Board of Directors Approval:	Month DD, YYYY
ANSI Approval:	Month DD, YYYY (include if an ANS document)
Next Scheduled Review Date:	Month DD, YYYY

Prepared by:
National Emergency Number Association (NENA) Agency Systems Committee, NG9-1-1 PSAP Working Group

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EXECUTIVE OVERVIEW

This document serves as the comprehensive technical specification for the interfaces and system functionality for Functional Elements (FEs) and Services comprising the NENA i3 Solution for the Next Generation 9-1-1 Public Safety Answering Point (PSAP) and Emergency Communications Center (ECC). At a high level, NENA-STA-023 defines the system and technical capabilities the NENA i3 solution uses to receive, process, and log 9-1-1 emergency service requests. This standard utilizes many of the Internet Engineering Task Force (IETF) standards to maximize interoperability across Internet Protocol (IP)-connected systems. The initial publication of this standard comprises four main functional capabilities:

- Call Handling
- Incident Handling
- Dispatch
- Internal and External Data Exchange interfaces

Future revisions of this document will include additional functionalities (see the Recommendation for Additional Development Work section **Error! Reference source not found.**4).

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This standard defines the system and functional capabilities that exist in public safety organizations that implement the NENA i3 standard. A public safety organization that follows this standard is referred to in the NENA i3 Standard for Next Generation 9-1-1, NENA-STA-010 [7] **Error! Reference source not found.** as an "i3 PSAP". This document uses the term Public Safety Answering Point (PSAP) when the function being described is the delivery and handling of emergency calls. Elements of this standard apply to Emergency Communication Centers (ECCs) that may or may not have call answering responsibilities as a primary function yet do provide significant coordination between public safety organizations utilizing NG9-1-1 systems, thus necessitating their adoption of these standards to maintain the highest level of interoperability. An Agency can be operated by a 9-1-1 Authority or can be a service provided by a public or private entity like an ambulance or tow truck service, so Agency is used in this broader sense when private providers are included in the context. This standard addresses the challenges of maintaining effective emergency services end-to-end in an all-IP environment that is undergoing significant technological advancements from the legacy E9-1-1 systems. This document should be used to assist in planning, managing critical interoperability issues, and mitigating service impacts due to performance and task overload, all of which are crucial for the quality and delivery of resilient NG9-1-1 services.

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9-1-1 stakeholders are encouraged to mandate the use of these specifications when developing, acquiring, configuring, implementing, and operating their NG9-1-1 system solutions. By adopting this standard, Agencies, PSAPs and ECCs can improve service quality, enhance resilience, and ensure interoperability, leading to better outcomes in emergency response. NG9-1-1 Authorities should refer to NENA Request for Proposal Considerations Information Document, NENA-INF-021, [26] for information and advice on creating NG9-1-1 PSAP/ECC Requests For Proposals and Requests For Information.

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2. MUST NOT: This phrase, or the phrase "SHALL NOT," means that the definition is an absolute prohibition of the specification.
3. SHOULD: This word, or the adjective "RECOMMENDED," means that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.
4. SHOULD NOT: This phrase, or the phrase "NOT RECOMMENDED," means that there may exist valid reasons in particular circumstances when the particular behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before implementing any behavior described with this label.
5. MAY: This word, or the adjective "OPTIONAL," means that an item is truly optional. One vendor may choose to include the item because a particular marketplace requires it or because the vendor feels that it enhances the product while another vendor may omit the same item. An implementation which does not include a particular option "must" be prepared to interoperate with another implementation which does include the option, though perhaps with reduced functionality. In the same vein an implementation which does include a particular option "must" be prepared to interoperate with another implementation which does not include the option (except, of course, for the feature the option provides.)

These definitions are based on IETF RFC 2119 [2] as updated by RFC 8174 [3].

UNIT OF MEASUREMENT

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In the description of the interface or object in NENA documents, a table may appear that describes the parameters to the interface and the data objects. These tables are informative; the Artifact file is the normative documentation of the interface and data objects unless otherwise specified in the document. If there is a discrepancy between the table and the Artifact file, the Artifact file controls unless otherwise specified in the document.

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Business rules are contained in this document for associated Artifacts. These business rules MUST be followed in implementing the standard.

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REASON FOR ISSUE/REISSUE

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Document Number	Approval Date	Reason For Issue/Reissue
NENA-STA-023.1-202Y	[Month DD, YYYY]	Initial Document

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1 INTRODUCTION

This standard describes Functional Elements (FEs) and Services, but not specific system implementations so it is possible providers of system solutions for the NENA i3 solution could bundle different FEs into different system implementations to meet the required standard (e.g., Call Handling and Incident Handling and Dispatch could be bundled into a single system solution or could be deployed as separate system solutions.)

The elements defined in this standard are depicted in the high-level diagram below showing three broad components of the NENA i3 solution. Two of the three components in the diagram are not defined in this standard but are included here for context. Calls[1] into the NG9 1 1 ecosystem begin in an Originating Network of commercial service providers (shown on the left below). Emergency Calls are accepted into the NENA i3 component labeled NGCS-ESInet below and are routed to the appropriate PSAP according to the NENA i3 Standard for Next Generation 9-1-1, NENA-STA-010 [7]. These Emergency Calls and Incidents are then processed by the NG9-1-1 PSAP or ECC using the services and functions defined by this standard.

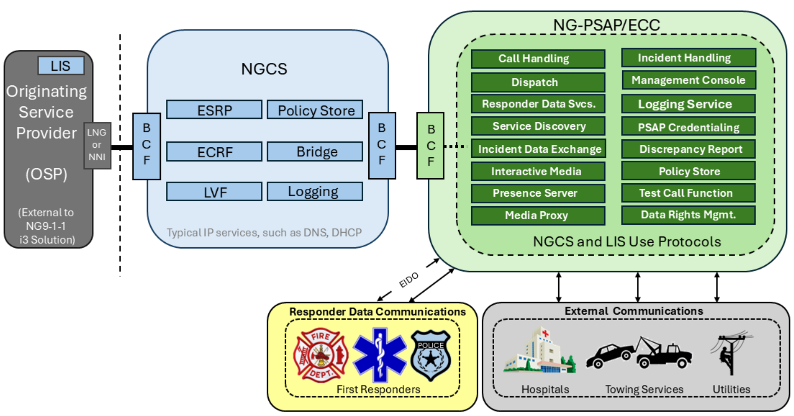

Figure 1 Simplified Overview Diagram of NG9-1-1 Solution

1.1 Key Sections of the Standard

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- **Call Handling** – This FE is responsible for the functionality and interfaces related to the processing of calls and includes the interface to the NGCS over which emergency calls are received and as well as any local interfaces to the other NG9 1 1 PSAP or ECC communication systems and networks (an administrative telephone switch (PBX), Public

[1] As defined in Terms and Abbreviations, the term "Call" includes text messages and non-interactive calls.

Switched Telephone Network (PSTN), etc.). Call Handling optionally includes an Interactive Media Response component.

- **Incident Handling** – This FE is involved when an incident is created, typically when a call is received, or an incident is reported via other means. An NG9 1 1 call is accompanied by a unique Incident Tracking Identifier used to track the Incident throughout its lifecycle. This FE utilizes Emergency Incident Data Objects (EIDOs) to exchange Incident information with other FEs.
- **Dispatch** – This FE recommends appropriate resources (emergency responders, equipment, etc.) available for an Incident, assigns emergency responders and other resources to an Incident, manage the inventory and status of resources, provides and receives relevant resource information for the ECC and responders, and monitors and logs all transactions associated with resources.
- **Management Console** – This FE supports important management functions for the PSAP or ECC and utilizes important event mechanisms specified in NENA STA 010 [7] such as PSAP Service State, Queue State, Discrepancy Reporting, the Policy Store, and the LogEvent Interface.
- **Web Services** – Includes Application Programming Interface (API) descriptions of the structure and function of web services used in PSAP and ECC operations, including setting queue states and managing agent activities, including discrepancy reporting and queue state management.
- **Other Functional Elements, Services, and Interfaces** – Includes detailed descriptions of functional elements like Presence Server, Media Proxy, Incident Data Exchange, Responder Data Services, and a Test Call Function.
- **NENA Registry System (NRS) Considerations** – Includes information on managing registry entries, policies for managing the registry and its content, and guidelines for setting initial values in the registry.

2 GENERAL SPECIFICATIONS

This document specifies Functional Elements (FEs), that are typically components of a larger system.

The specifications in this section apply to multiple Functional Elements or Services, or to the NG9 1 1 Public Safety Answering Point (PSAP) or Emergency Communications Center (ECC) as a whole.

2.1 Format of Timestamps and Date/Time Values

Timestamps contained in JavaScript Object Notation (JSON) objects and XML documents specified in this document governed by this specification SHALL be represented as specified in the Timestamp section of NENA-STA-010 [7] unless otherwise specified by the controlling standard (e.g., RFC 4119 [57], as updated by RFC 5139 [58]).

2.2 Service Names

A Service is an implementation consisting of one or more instances of one or more FEs that act as one to implement functions defined in this document. Services are identified by a name in the IANA "serviceNames" registry [xxx]. This document requests IANA to add new Service names in the "serviceNames" Registry section. Systems that implement more than one service

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will have more than one Service Name from the Service Names registry. Service names are used in Data Rights Management (DRM) policies as specified in Authorization and Data Rights Management (DRM).

2.3 Interface Names

FEs and Services defined in this document have one or more defined interfaces. Interfaces have names in the IANA "Interface Names" registry [xxx]. This document requests IANA to add new Interface Names in the "Interface Names" registry section of this document. Interface names are used in DRM policies as specified in Authorization and Data Rights Management (DRM).

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2.4 Service Availability

While Next Generation 9 1 1 systems have the same need for availability as traditional 9 1 1 systems, the NG9 1 1 architecture provides more flexibility to achieve the necessary availability. Traditional metrics such as "five-nines" and other carrier-grade measures of availability continue to be relevant and may still be utilized, but the resiliency that is inherent in an IP-based platform lends itself to looking at the availability of the entire call process, rather than component-by-component. Common statistical process control methods are an appropriate way to measure both service quality and availability of entire processes, such as NG9 1 1 call handling and dispatch.

Traditional Telco-based 9 1 1 systems have used carrier grade availability standards, calling for individual components built to "five-nines" availability standards (5.26 minutes of downtime per year), often deployed in redundant pairs. Equipment is housed in special facilities that have robust power and sophisticated environmental controls. While a valid approach to ensuring availability comes at a cost, and there are other means to achieve it. This document does not impose any specific availability requirement. Implementers MUST decide what availability Service Level Agreement (SLA) their systems will be able to offer and 9 1 1 authorities and ECCs need to consider the cost to achieve the availability they require. See the Emergency Services IP Network Design Information Document, NENA INF 016 [8] for information on network availability. The NENA APCO ECC Service Capability Criteria Rating Scale, [7] provides a self-evaluation assessment tool for reviewing PSAP capabilities against models representing the best level of preparedness, survivability and sustainability.

Five-nines availability in IP-based systems, such as NG9 1 1, is typically achieved by having more than two redundant components, with each component having attained less availability than the "five-nines" standard while, at the same time, utilizing less robust power and less sophisticated environmental controls. In this context a PSAP can may be considered a component. In this document, we define the availability metric of interest to be that an Emergency Call can be answered by someone who can manage appropriate assistance.

With NG9 1 1 call diversion, multiple PSAPs may be used to achieve an acceptable level of availability. In NG9 1 1, call handling diversion policy SHOULD be defined by an agreement between cooperating Agencies. Within the NG9 1 1 environment, a PSAP MUST be prepared to (under local policy control) receive and process, to the best of their ability, diverted calls that do not fall under any formal agreement. Unmanned backup PSAPs increase availability, but the time to bring the backup PSAP online is a significant factor in that availability. Backup PSAPs are viable options and can be utilized by implementers where appropriate to achieve the target Availability.

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2.5 Physical Considerations

The physical considerations for a PSAP are like those for any facility of a similar size hosting computing and networking infrastructure. NENA's *E9-1-1 PSAP Equipment Standards*, NENA-STA-027 [10] contains physical and electrical environmental requirements that are also applicable to NG9 1 1 PSAPs. The Installation, Maintenance and Administration section of that document is also applicable to a PSAP.

NG9-1-1 PSAPs depend on the reliable operation of various equipment, including computers, communications equipment, and power sources. All the products and equipment required for NG9-1-1 need a suitable physical environment that does not threaten or impair their operation.

NENA's *PSAP Site Characteristics Technical Information Document*, NENA-INF-024 [9] provides advice on site characteristics appropriate to support the mission of a PSAP in NG9-1-1, transitional, and E9-1-1 implementations. That document covers a broad range of topics that include equipment area and structural requirements, environmental, electrical, lighting, acoustic, and fire and electrical protection requirements. This information SHOULD be reviewed by those preparing an existing or new facility for installation and support of NG9-1-1 equipment and personnel.

NENA-INF-024 [11] is a guide for PSAP managers and for the designers, manufacturers and installers of PSAP Customer Premises Equipment (CPE). It includes recommendations for:

- Equipment Area
- Structural Guidelines
- Equipment Room Environmental Guidelines
- Electromagnetic Interference
- Acoustics
- Fire Protection
- Electrical Protection (Grounding, etc.)
- Alternating Current Power Guidelines
- Physical Access

There are also recommendations for checklists.

NENA-INF-024 [11] provides only recommendations and does not address PSAP service availability. See the PSAP Service Availability section of this document for details on PSAP availability options. The availability of a PSAP is completely dependent upon the availability of the IP networks coming into it. Dual fiber entrances with diverse approaches to the building will protect against a digging operation or external event that could take out one of the fiber connections. The entrances should have known diverse paths to multiple IP network service providers. PSAPs should also employ other diverse paths to network connections, including cable, wireless, microwave, etc. Additional information is available in NENA-INF-016 [8]. Implementing an "active-active" distributed PSAP configuration, where two sites take calls simultaneously and one can take over for the other in case of a failure is highly desirable. This redundancy can be accomplished by staffing two sites simultaneously or having mutual aid agreements with neighboring PSAPs and implementing the appropriate rules in the Policy Routing Function that serves the PSAPs. Alternately, a less desirable option would be a backup site that can take calls when the primary site becomes unavailable.

If PSAP equipment is housed in an external facility, that facility may have more stringent requirements than those described in NENA-INF-024 [11]. PSAP equipment SHOULD NOT be

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housed in a facility that does not, at a minimum, follow the recommendations in NENA-INF-024 [11].

Securing physical access to buildings, rooms, and restricted areas is critical, and implementing multiple levels of physical security is REQUIRED. The *Security for Next Generation 9-1-1 Standard (NG-SEC)*, NENA-STA-040 [12] provides details on the basic requirements for NG9-1-1 security, and the 9-1-1 authority MUST incorporate these requirements into the physical security plan. Securing facilities with physical locks is not sufficient. Written policies and procedures MUST be employed to ensure that personnel maintain a secure physical environment and MUST be conveyed to employees often enough to ensure that all employees understand the policies and procedures and the importance of adhering to them.

Properly managing and monitoring the physical facility is as important as managing and monitoring the technology inside. Written policies and procedures MUST be employed to ensure that the physical facility provides the appropriate physical infrastructure and must include regular inspection and testing of all backup systems and services. NENA-INF-040 *Managing and Monitoring NG9-1-1 Information Document* [83] provides advice on managing critical infrastructure and services, including:

- Primary power and battery backup power facilities
- Generator power facilities and fuel
- Separate fusing to service equipment racks, cabinets, etc.
- Appropriate building, antenna, equipment and cabinet grounding
- System and data redundancy status
- Environmental conditions for hardware infrastructure, such as temperature and humidity
- Fire alarm maintenance and testing
- Water detection alarms
- Flood protection infrastructure
- Physical security infrastructure, such as biometric access controls, CCTV surveillance, guards, man traps.
- Services that support facility staff, like water supply and air conditioning/heating facilities. Human requirements will differ from those required for system hardware.

Important alerts and notifications SHOULD be proactively monitored per policy including National Weather Service alerts and other alerts of external events that could impact the physical facility in a way that could degrade the PSAPs ability to provide emergency services.

2.6 Agency Network

This document incorporates the use of the ESInet as defined in NENA-STA-010 [7] for the Agency ESInet (i.e. the network that hosts implementations of FEs and Services defined in this document). The ESInet needs to follow the recommendations documented in Emergency Services IP Network Design Information Document NENA-INF-016 [8].

This document recommends that Agency ESInets avoid single purpose networks unless no other alternatives exist. While the architecture for legacy PSAPs has typically consisted of multiple physical networks for disparate functions, Agency ESInets favor a single, redundant, physical network separated into logical Virtual LANs (VLANs) to address any desired separation or

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security. The Agency ESInet is expected to support multiple vendors with separate SLAs, management, and security policies.

Due to the distributed nature of i3 networks, the architecture of the Agency ESInet may allow interconnections to networks with different security policies. For example, if connectivity to the FBI's Law Enforcement Online databases and systems or to the National Crime Information Center (NCIC) is required, then compliance with the FBI's Criminal Justice Information System (CJIS) Security Policy [13] or a similar policy in other countries will have to be supported. Agency ESInets that communicate with local, state and national networks will likely have to accommodate various and, possibly, conflicting, security policies.

The Agency ESInet design SHALL adhere to the following specifications:

- NENA i3 Standard for Next Generation 9-1-1, NENA-STA-010 [7]
- NENA Security for Next Generation 9-1-1 (NG-SEC), NENA-STA-040 [12]

Additional information on ESInet design can be found in NENA-INF-016 [8], some of which is applicable to the design of Agency ESInets.

Some requirements for an Agency ESInet are different from those for ESInets beyond the PSAP border. Not every IP address inside a PSAP needs to be a publicly routable address. An Agency ESInet could be switched rather than routed.

2.7 Quality of Service (QoS)

ESInets use An Architecture for Differentiated Services, RFC 2475 [14] to classify network traffic. In order to maintain acceptable Quality of Service, this document extends the scope of the i3 Architecture ESInet and requires that the Agency ESInet MUST implement DiffServ as specified in the Emergency Services IP Network section of NENA-STA-010 [7].

SIP elements in a PSAP MUST support RTP: A Transport Protocol for Real-Time Applications (RTCP), RFC 3550 [15] and Secure Real-time Transport Control Protocol (SRTCP) [16] as defined in NENA-STA-010 [7]. User Agents (UAs) in a PSAP MUST be able to differentiate RTP failure from real silence by the caller and attempt to reestablish the stream as specified in the Transport section of NENA-STA-010 [7].

2.8 Border Control Function (BCF)

A BCF SHALL be deployed between the Agency ESInet and serving ESInet. A BCF SHOULD be considered with other non-ESInet networks. In order to prioritize traffic, BCFs serving PSAPs MUST police Differentiated Services code points for traffic entering a PSAP that is not already marked. BCF functionality and interfaces are described in detail in the BCF section of NENA-STA-010 [7] and PSAPs should take advantage of this functionality to protect their service. A PSAP MAY use a BCF provided by the NGCS or MAY have its own BCF. A Virtual PSAP (i.e., a PSAP where personnel and the Functional Elements (FEs) of the PSAP architecture are not physically collocated) MAY require a BCF between any part of that virtual PSAP and the network that connects them together.

2.9 Security Requirements

As specified in the PSAP Network section above, the PSAP network is an ESInet. i3 PSAPs SHALL adhere to security standards defined in the Security section and the Service Interfaces section of NENA-STA-010 [7]. PSAPs SHALL conform to the applicable specifications in NENA-STA-040 [12].

2.10 PSAP Credentialing Agency (PCA) Requirements

The ECC Services (as defined in this document), Agents, and Elements that provide or participate in NG9-1-1 Services MUST each have an appropriate valid certificate traceable to the PCA as described in STA-010 [7]. The *NG9-1-1 Interoperability Oversight Commission* (NIOC) 122 is the independent oversight governance body for the NG9-1-1 Public Key Infrastructure (PKI), implemented by the PCA.

A PSAP or ECC applies to the appropriate certificate authority to request certificates and SHOULD contact its state or regional 9-1-1 Authority for information on the process. Certificates traceable to the PCA will have to be periodically renewed per the certificate authority's policy. Renewal will incur some expenses and some PSAP resources will be required to manage the renewal process.

In addition, secure storage for the certificates and a means for securely accessing the certificates must be maintained. Methods and procedures for recovering from a "key compromise" security incident must be developed, maintained and communicated within the organization.

2.11 Authentication and Trust

Authentication of Agents accessing elements described in this document MUST be implemented with a universal Single Sign On (SSO) paradigm as described in the Authentication section of NENA-STA-010 [7].

Implementations MUST conform to the provisions in the Trusting Asserting and Relying Parties section of NENA-STA-010 [7].

2.12 Policy

Policies are stored into and retrieved from a Policy Store using the web service specified in the Policy Store Web Service section of this document. A policy (or policy document or policy object) consists of a set of rules and is sometimes termed a "ruleset". A default rule MUST be included in every policy ruleset. Policies defined in this document are in eXtensible Access Control Markup Language (XACML) format and are used to specify Data Rights Management (DRM) rules. See NENA-STA-010 for full detail on NG9-1-1 XACML policies, including digital signature requirements.

Policies are named for the function that the policy affects, (e.g., the InstantRecallPolicy for an Instant Recall Service). FEs and Services defined in this document MUST use the policies in the Policy Store to control access to their interfaces and data as specified in the Authorization and Data Rights Management section of this document.

2.13 Authorization and Data Rights Management (DRM)

Authorization and Data Rights Management in the NG9-1-1 PSAP is based on eXtensible Access Control Markup Language (XACML), [17] as described in the Authorization and Data Rights Management section of NENA-STA-010 [7]**Error! Reference source not found.**. Examples provided in this document use XACML 3.0. New implementations conforming to this standard

Deleted: 106

Commented [MLS6]: Must be updated in STA-010, it's out of date, doesn't reflect current standards for SSO.

Commented [MS6R2]: Did Guy make this comment on a call? The i3 v 3.1 doc says "The mechanism used to implement the SSO requirement may change in version 4 of this document. OpenID is under consideration as a mechanism." Maybe that was the response to this comment?

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SHOULD use XACML 3.0 [85]. Existing implementations MAY use XACML 2.0. A future version of this document will require XACML 3.0.

XACML policies are stored in a Policy Store and are accessed through the Policy Store web service defined in the Policy Store Web Service section of this document. Each XACML policy defines a "Target", which describes who or what the policy applies to (by referring to attributes of users, roles, operations, objects, dates, etc.), and one or more "rules" to permit or deny access. Access is defined to mean some combination of Read, Update, Create, Delete, and Execute permissions (for interfaces). Rules can "permit" or "deny" access.

In policy rules:

- Subject attributes can include the id (as an agentId), agency (as an agencyId), role (from the IANA Agent Roles or Agency Roles registries), and agencyType (from the agency locator record). The attributes agencyId, agency, role, and agencyType are string types.
- Resources can be data structures or interfaces. Interfaces are named by the ServiceId and an Interface Name from the IANA Interface Names registry [xxx], separated by a tilde (~). Data items are named by the json object name from the OpenAPI interface description that defines them and the element tag (if access is controlled by element) separated by periods. For example:

  PresenceServer~SIMPLEpresence.presence.tuple.status.activityState.CallHandling

  If no element name is specified, the access to the entire data structure is controlled by the rule.
- Actions are "Read", "Create", "Update", "Delete", and "Execute".

Every Service MUST have a configurable default behavior for the case where the Policy is not available, perhaps due to the Policy Store being unreachable or when no default rule is specified in the policy. In that case, the Agency's local policy will determine whether "permit" or "deny" will be returned.

All Services specified in this document MUST have a XACML policy that governs access to its interfaces. As an example, to specify a rule for access to a web service interface defined in this document, the Resource in the rule is the Service Name or ServiceId and the Interface Name, separated by a tilde ("~"), for example:

- "LoggingService~Logging" or "logger.state.pa.us~Logging"

If the Resource in a rule is a Service Name from the IANA serviceNames registry or a ServiceId, then the rule applies to all interfaces on that Service, for example:

- "LoggingService" or "logger.state.pa.us"

Rules for access to all interfaces specified in this document are defined in this way.

If an interface returns a data object, the object resource in the rule is specified by the object name and, optionally, a specific element name within the object, for example, MyObject.myElement.

Access permissions are dependent upon the identity and role of the requester. Therefore, all attempts to access Service interfaces resources SHOULD be made over TLS using the

Commented [MS7]: XACML 2.0/3.0 definitions:
"**Subject** - An actor whose attributes may be referenced by a predicate".
"**Target** - The set of decision requests, identified by definitions for resource, subject and action, that a rule, policy or policy set is intended to evaluate"
"**Resource** - Data, service or system component"

Commented [MS8]: To be created by Omnibus RFC, can't reference it until it exists.

Commented [MLS9]: TODO: i3 decided we'll use interface names, and added text that says if the Service Name is specified without an interface name, then the rule applies to all interfaces on the Service. **Make sure all our text agrees with i3's new text!**

Commented [MS10]: TODO: I think this should be "logger.state.pa.us~Logging.MyObject.myElement"? Or do we not have to identify the service and interface because they are implied somehow? Policy Type arguably implies the Service Name.

Commented [MS10R2]: TODO: raise this issue with i3. It has the example "<eido.AgencyDataComponent.agencyName>"

DRAFT

requester's PCA certificate or other accepted certificate2. It is possible that the Service could receive a request to access a service interface in the clear, i.e. no TLS is used. In that case the DRM Policy's default rule (see the Policy section) MUST be used to determine access permissions.

2.14 Element State Requirements

All FEs in a PSAP or ECC MUST implement the notifier side of the Element State event package as specified in the Element State section of the NENA-STA-010 [7]**Error! Reference source not found.**. When the FE receives a SUBSCRIBE request specifying the emergency-ElementState event package, it consults the elementState policy specifying its own Element Identifier as the target to determine if the requester is permitted to subscribe. It returns 603 (Decline) if not acceptable. If the request is acceptable, it returns 200 OK. The FE MUST implement event rate filters as described in Session Initiation Protocol (SIP) Event Notification Extension for Notification Rate Control, RFC 6446 [18]. Upon acceptance, the FE sends an initial NOTIFY message containing the following information:

Commented [MS11]: New edit. The requirements are in STA-010, which we referenced. This is informational only.

- elementId: Element Identifier
- reason: reason for change
- state: current state of the Element, MUST be a value found in the elementState registry.

Upon any change to the Element's state and respecting any rate filter the subscriber may have specified, the FE MUST generate a NOTIFY message containing the updated state. Only entities that need to know the state of individual FEs in a Service would typically use Element State. Examples would be the Management Console or a load balancer that manages requests sent to a Service. Entities that only need to know the overall state of the Service would typically subscribe to Service State instead (see below).

All FEs MAY support the subscriber side of elementState.

2.15 Service State Requirements

Commented [MS12]: TODO: this text can be reviewed when the Service Discovery text has been finalized and approved.

A Service, as defined in this document, represents a set of one or more FE instances, addressable with a single Service Identifier, that provides the functionality and interfaces specified for the FEs in this document.

A Service Identifier refers to the collection of elements that define a Service. Service Identifiers are Fully Qualified Domain Names (FQDNs). An example of a Service Identifier is "foo.allegheny.pa.us". The Service Name is not required to be part of the FQDN. Each service has a unique Service Identifier. Implementations that include more than one service in an element have more than one Service Identifier.

Service Names come from the IANA "Service Names" registry. A Service's resources include the interfaces of the FEs the Service incorporates and data elements available on that interface. Services have Data Rights Management (DRM) policies identified by a Service Name from the IANA "Service Names" registry or a ServiceId, and the owner of the policy. See the Policy section of this document for details on policies and individual FE sections for details on the Interface Names used in DRM rules.

Deleted: service

[2] External entities like private tow truck or ambulance services could have certificates issued by some trusted certificate authority other than the PCA that is deemed acceptable by the Agency.

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All Services in the PSAP or ECC MUST implement the notifier side of the ServiceState/SecurityPosture event package defined in the Service State section of NENA STA 010 [7].

When a Service receives a SUBSCRIBE request specifying the emergency-ServiceState event package, the Service consults the serviceState policy that specifies its Service Identifier as the target to determine if the requester is permitted to subscribe. It returns 603 (Decline) if not acceptable. If the request is acceptable, it returns 200 OK. The Service MUST implement event rate filters as described in RFC 6446 [18]. Upon acceptance, the Service MUST send an initial NOTIFY message containing the following information:

- serviceId: Service Identifier
- reason: reason for change
- state: current state of the Service; MUST be a value found in the IANA Service State registry.

Upon any change to the Service's state and respecting any rate filter the subscriber may have specified, the Service MUST generate NOTIFY messages containing the updated state and send them to authorized subscribers.

2.16 Agent States

Agent states allow the Agency to monitor and track the availability of its emergency personnel. If implemented consistently within the Agency, agent state information can be used to ensure incoming communications are routed quickly and efficiently to available personnel and can provide meaningful reporting information for workforce management. The state of an agent specifies the availability of that agent to accept communications from internal and external sources. The Agent State can be determined by comparing the Agent's current Activity State with the policies for the Agency. The term Activity State is used in this document to refer to state data that includes the types of communications supported by an FE being used by an Agent, along with the communication types the Agent is currently engaged in on that FE. See the Presence Server FE Interfaces section for details on Agent State determination.

The current Activity State for an agent is published by the various FEs to a Presence Server FE. Using an extension to the SIP Presence model, the Presence Server FE publishes the state of the Presentity (i.e., the agent being monitored) as a representation of the agent's availability to accept communications. The Presentity (a combination of the words "presence" and "entity") refers to an entity that has presence information associated with it such as status, reachability, and willingness to communicate.

Activity States are supported in FEs used by the agents, such as the CHFE, the Collaboration FE, and the PTT System Services FE. The Activity State describes the state of the agent's activity for that FE. Therefore, the overall state of an agent is made up of activity states from all FEs relevant for the agent.

Commented [MLS13]: Dan objects to including ActivityState for something that is not even defined in this version. Noted by the group, but consensus is to keep Collaboration FE in ActivityState to support existing apps and in the Future Work Items table for the next revision.

An agent may be logged in or logged out. If the agent is Logged In, then Activity State information can be published for the agent. If the agent is Logged Out, then no Activity State information can be published for the agent.

The actual Activity State data structure is defined in the section Presentity Activity States for the Presence Server.

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The Activity State data structure can carry information about the following FEs:

Table 2-1 FEs Supported in Activity State Data Structure

FE	Description
CallHandling	Call Handling (CHFE)
Collaboration	A "chat" message function[3]
PttSystemServices	Push-To-Talk System Services FE

The Activity State data structure can carry information about the following media types that can be supported by one or more of the above FEs. Some FEs support multiple media types. Some media types are supported by multiple FEs, possibly using different protocols. Protocol differences between FEs aren't significant because they would not affect policy rules. Media type and the FE that is using it are significant to Activity State because they can affect the policy rules that govern the Agent's availability for other communication sessions. The following types of media are supported by Activity State:

- **Voice**: Streaming audio. Supported by the Call Handling and PTT System Services FEs, and by some Collaboration applications.
- **Video**: Streaming video. Supported by Call Handling and by some Collaboration applications.
- **Instant Message**: A message sent as a block of text. Supported by Call Handling and Collaboration.
- **Real-Time Text**: Streaming text sent one character at a time. Supported by Call Handling and some Collaboration applications.

The combination of the Activity State from each of the relevant FEs and the policy of the Agency determines an agent's availability for new assignments such as calls, chat, or Incidents. Availability may be different for internal purposes as opposed to external purposes. The Presence Server FE advertises an agent's availability according to Agency policy.

Therefore, the concept of an agent's availability is dictated by the operational procedures of the Agency. Some Agencies may support the notion that an agent is capable of being involved in multiple communications simultaneously while other Agencies may restrict agents to only handle one form of communication at a time. For example, an agent that is active with a voice call may be considered unavailable for all other types of communications at one Agency but may be permitted to accept text communications in another Agency. Note that agent availability may be reported differently depending upon the entity to which the report is being sent. For example, an Agent may be shown as available for internal communications, but not for communication with another Agency.

Ultimately, it is the responsibility of each Agency to determine the criteria for the availability of an agent for new assignments such as calls, chat, or Incidents. The Activity States defined by this specification simply provide a mechanism to identify the states of an agent for the various available communication media. It is beyond the scope of this document to specify the operational procedures that an Agency may use to determine agent availability.

DRAFT

[3] A future revision of this document will define a standardized Collaboration FE that will support interoperable individual and group "chat" communications. This section defines Agent Activity States that will be used by this to-be-defined Collaboration FE and MAY be used by any other chat application that wishes to report Agent Activity States.

2.17 Agent Roles

This document defines an <agentRoles> extension element to the standard PIDF <status> element that contains the entire role string from the Subject Alternate Name element of the Agent's certificate. This Role string specifies all the Roles and Role Modifiers applicable to the Agent. Roles and Role Modifiers are defined in the IANA "Agent Roles" Registry. See the PIDF Extension for Agent Role section of this document for details of the <agentRoles> extension element.

3 INTERFACES

In this document, we make use of three kinds of interfaces:

- Web services, typically using a RESTful interface
- Simple HTTPS [20] GET (and in some cases, PUT or POST) with retrieval of data structures based on a Uniform Resource Identifier (URI)
- SIP interfaces, including SIP SUBSCRIBE/NOTIFY

Services and Elements that implement interfaces defined in this document MUST enforce Data Rights Management policy for the interfaces, and any data elements the interfaces can return, as specified in the Policy section of this document.

3.1 Web Services

Unless specified otherwise, web services defined in this document provide RESTful interfaces that use a simple *HyperText Transfer Protocol Over TLS (HTTPS)*, (RFC 2818) [20] GET (and in some cases, PUT or POST) with retrieval of JSON data structures based on a Uniform Resource Identifier (URI). The normative definition of each interface is in a YAML Ain't Markup Language (YAML) file on NENA's Repositories for interfaces and schemas [4].

Web services in this document MUST conform to specifications in the NENA-STA-010 [7] Service Interfaces section that apply to such services, for example, HTTPS Transport, Status Codes, and Versions.

3.2 Discrepancy Report Proxy Web Service

The Discrepancy Report Proxy web service is implemented by the Management Console FE (MCFE) and allows FEs and Services within the PSAP to submit Discrepancy Reports (DRs) to be filed on behalf of the PSAP's FE or Service by the MCFE. The MCFE uses the information in the request to submit the DR to the appropriate Discrepancy Reporting Web Service as defined in the Discrepancy Reporting section NENA-STA-010 [7].

HTTP method: PUT

Resource name: .../DiscrepancyReportProxy

The body of the request MUST contain two elements:

Table 3-1 Discrepancy Report Proxy Web Service Parameters

Name	Required	Description
serviceAgencyId	Yes	ServiceId or AgencyId of the Agency to send the DR to
discrepancyReport	Yes	The DR to be filed against the identified Service or Agency.

The MCFE uses serviceAgencyId to obtain an Agency's dscRptSvc URI that accepts DRs from the Service/Agency Locator Service in the NGCS. See the Management Console section for details.

Status Codes

200 OK
400 Bad Request
500 Internal Server Error

3.3 Queue State Web Service

The server side of the Queue State web service is implemented by the CHFE. The client side of the Queue State web service is implemented by the Management Console and is used to manually request a change in the QueueState of a given queue. Examples include changing a diversion queue from Standby to Active, or temporarily disabling a queue for some reason.

Note that the CHFE that manages the queue MAY reject requests to set the state of the queue to certain values if the requested state conflicts with a state dictated by another rule or action such as a change in PSAP ServiceState.

Commented [MS14]: Pick up here on 6/25/26 or after NENA conf. Changed 4.8.2.6 Queue State Subscription table to only specify queue states for certain Service States. There are related edits in 4.8.2.1 made at the end of the 6/18/26 call that have not been reviewed. Review this, then 4.8.2.1, then continue reviewing changes below here.

HTTP method: PUT

Resource name .../QueueState/<QueueName>

Where <QueueName> is the Queue Identifier of the queue the request applies to.

The body of the request MUST contain this element:

Table 3-2 Queue State Web Service Parameters

Name	Required	Description
queueState	Yes	The requested QueueState.

The requested queueState MUST be limited to one of these values from the IANA Queue State registry [xxx]: Active, Standby, or Disabled. Other queueState values can't be manually set and MUST NOT be used.

Commented [MS15]: This pointed to [24], which is wrong. It will be one of the registries the Omnibus creates, so we have to wait for it.

Commented [MS15R2]: We can point to draft if we want to: https://www.ietf.org/archive/id/draft-rosen-ecrit-emergency-registries-02.html

Status Codes

200 OK
400 Bad Request
422 Unknown Queue or queueState Not Allowed
500 Internal Server Error

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3.4 Agent Activity State Web Service

The Agent Activity State Web Service is implemented by the Presence Server FE, and is used by FEs that do not support the Session Initiation Protocol (SIP) to publish the Activity State of their agents to the Presence Server FE. Note that FEs that do not support SIP are not able to subscribe to Activity State changes.

HTTP method: PUT

Resource name .../AgentActivityState

The body MUST contain a PIDF of the form specified for Presentity Activity State PUBLISH bodies in the Presence Information Document (PIDF) subsection of the Presence Server FE section of this document.

See the section of this document entitled Presentity Activity States for the Presence Server for details on Activity State.

Status Codes

200 OK
400 Bad Request
500 Internal Server Error

3.5 Policy Store Web Service

Each PSAP MUST have a Policy Store Web Service that stores and retrieves policies as specified in NENA-STA-010 [7]. Policies defined in this document are in XACML format and are used to specify Data Rights Management (DRM) rules. The format of the Policy Store and the methods used to manipulate policies are specified in the Policy Store Web Service section of NENA-STA-010. Policies are identified by a Service Name from the IANA "Service Names" registry or a ServiceId, and the identity of the agency that owns the policy. This document requests IANA to add new values to the Policy Types registry.

The client side of an authentication with the Policy Store Web Service identifies the agency storing or retrieving Policy rule sets.

A PSAP MAY implement its own Policy Store, or MAY use a Policy Store provided by the NGCS or another Agency. The Policy Store used by an FE or Service MUST be discovered as specified in the Service Discovery section of this document.

PSAP FEs and Services MUST use the policies in the Policy Store to control access to their interfaces and data as described in the Authorization and Data Rights Management section of this document. Policies required by the document are detailed in the Policy section.

3.6 Media Proxy URI Request Web Service

The Media Proxy URI Request web service is implemented by the Media Proxy FE. It accepts a URI of the media to be forwarded and returns the Media Proxy URI for that media, which MUST then be used for all access to, and operations on, the media.

HTTP method: POST

Resource name .../MediaProxyUri

Commented [MS16]: Hyperlink

Commented [MS16R2]: TODO: remove unneeded Policy Types and use Service Names instead. May still need Ihfe and Eido Policy Types because they don't have a "Service".

Commented [MLS17]: Dan and Brian and Suresh think responder devices MUST be able to log LogEvents (not siprec). incidentId can change between retrievals.

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The body of the request MUST contain a mediaProxyURIRequest JSON object that includes the original URI of the media the client wishes to access and if applicable, a related Incident ID and/or related emergency call ID:

Table 3-3 mediaProxyURIRequest JSON Object

Name	Required	Description
originalUri	Yes	The URI for which the client is requesting a Media Proxy URI.
incidentId	Conditional	An emergency Incident Identifier as defined in NENA-STA-010. Required if the request is related to an Incident. Will be included by the Media Proxy when it logs media and LogEvents.
callId	Conditional	An emergency Call Identifier as defined in NENA-STA-010. Required if the request is related to a call. Will be included by the Media Proxy when it logs media and LogEvents.

Commented [MS18]: The original URI is always used when a streaming or file media URI is provided in an EIDO. The receiving Agency's Media Proxy would provide a local URI when requested.

The Media Proxy only supports an originalUri that is an HTTP URI (GET method only), or an RTSP URI, and MUST return "415 Unsupported Media" if it receives a MediaProxyUriRequest for other protocols or methods.

The incidentId member MUST be included if the media is related to an Incident and MUST be omitted if the media is not related to an Incident. An empty incidentId member will be interpreted as omitted.

The callId member MUST be included if the media is related to a call and MUST be omitted if the media is not related to a call. An empty callId member will be interpreted as omitted.

On success, the return includes a mediaProxyUri JSON object that consists of:

Table 3-4 mediaProxyURI JSON Object

Name	Required	Description
returnedUri	Yes	A URI created by the Media Proxy for the media that will be forwarded from the original URI. The client will retrieve the media using this returnedUri.
expirationTimestamp	Yes	The date and time the Media Proxy URI expires.

Commented [MS19]: The MP would associate that identifier with the original URI and expiration.

The returnedUri is created by the Media Proxy. One way to create a URI is to use an FQDN with a locally unique string appended that represents the original media source:

https://MySvc.myDomain.gov/xm1pxTqar3 or rtsps://MySvc.myDomain.gov/rt3X5zunR2

The returnedUri MUST be a secure URI (HTTPS or RTSPS) even if the original URI was a non-secure URI.

The expirationTimestamp is the time until which the Media Proxy guarantees the validity of the returnedUri and MUST be at least one hour in the future.

Status Codes

200 OK

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403 Forbidden
415 Unsupported Media Type
500 Internal Server Error

3.7 Shared Communications Resource Web Services

A system that implements shared communication resources, such as a Push-To-Talk (PTT) system, implements two web services to allow the Logging Service to query for available communication resources (e.g., talkgroups and radio channels), and to retrieve resource details necessary to request recording of media on resources of interest. These two web services are used by the Logging Service for all supported shared communication resources:

- ListSharedCommunicationResources – returns a list of unique resource identifiers for resources available to be recorded.
- GetSharedCommunicationResourceObject – returns the SharedCommunicationResource object for a given unique resource identifier that contains detailed information about the resource.

The Logging Service then calls a web service to register interest in the resource associated with a given resource identifier. This revision of the document defines only one such web service:

- PTT-P25ResourceRegisterInterest – tells a Project 25 (P25) radio system that the Logging Service requests to record media from a shared resource such as a talkgroup or conventional radio channel.

These web services are defined in the subsections below.

3.7.1 ListSharedCommunicationResources

ListSharedCommunicationResources returns a list of resource identifiers that identify shared communication resources available on the system. The resource identifiers must be unique in the shared communication system. The list of resource identifiers returned may be restricted by the server based on the requester's identity or role.

HTTP method: GET

Resource name .../ListSharedCommunicationResources

There are no parameters to ListSharedCommunicationResources.

Status Codes

200 OK
401 Unauthorized
404 Not Found
500 Internal Server Error

The web service MUST return 401 Unauthorized if the requester is not authorized to use the service, 404 Not Found if no communication resources exist, and 500 Internal Server Error if the server encountered an unexpected condition that prevented it from fulfilling the request.

On a successful GET, the web service MUST return 200 OK and MUST include in the body of the response an "AvailableCommunicationResources" JSON object containing an array of "resourceIdentifiers", each of which is a string that identifies a recordable resource on the system:

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Table 3-5 AvailableCommunicationResources JSON Object

Name	Required	Description
resourceIdentifiers	Yes	An array of strings, each of which uniquely identifies a resource that can be recorded on the system.

3.7.2 GetSharedCommunicationResourceObject

GetSharedCommunicationResourceObject returns the SharedCommunicationResource object that matches an individual resourceIdentifier provided as a parameter.

HTTP method: GET

Resource name .../GetSharedCommunicationResourceObject/{resourceIdentifier}

Table 3-6 GetSharedCommunicationResourceObject resourceIdentifier Parameter

Name	Required	Description
resourceIdentifier	Yes	A resourceIdentifer returned by ListSharedCommunicationResources.

Status Codes

200 OK
401 Unauthorized
404 Not Found
500 Internal Server Error

On a successful GET, the web service MUST return 200 OK, and the body of the response MUST contain the matching SharedCommunicationResource object. For a P25 system, the returned object will be a PTT-P25 SharedCommunicationResource JSON object, which is defined in the Shared Communication Resource Data Components section of NENA-STA-021 [22].

Commented [MS20]: Fixed - the JSON object is "PTT-P25" in EIDO-JSON. Added a reference to the section heading.

Table 3-7 SharedCommunicationResource Object

Name	Required	Description
SharedCommunicationResource	Yes	The SharedCommunicationResource JSON object associated with the resourceIdentifier in the request

If the resource does not exist, the web service MUST return 404 Not Found.

If the requester is not authorized, the web service MUST return 401 Unauthorized

If an unexpected condition prevents the server from fulfilling the request, the web service MUST return 500 Internal Server Error.

Commented [MS21]: Terminology is from the RFC that defines the status code.

Deleted: internal

3.7.3 PTT-P25ResourceRegisterInterest

PTT-P25ResourceRegisterInterest is implemented by a P25 PTT system. It allows a Logging Service to request that a PTT System initiate recording of media and metadata when a transmission occurs on a given resource, or to request a PTT system to cancel registration of a previously registered resource.

DRAFT

This revision of the document supports only the PTT-P25 SharedCommunicationResource, which is defined in the Standard for Emergency Incident Data Objects (EIDO), NENA STA 021 [22]. For P25 systems, a shared resource can be either a talkgroup ID or channel ID.

The Logging Service first obtains details on shared communication resources that can be recorded using ListSharedCommunicationResources and GetSharedCommunicationResourceObject, then calls PTT-P25ResourceRegisterInterest to request the P25 system to initiate recording on each resource of interest as specified in the PTT System Services FE section of this document.

HTTP method: PUT

Resource name .../PTT-P25ResourceRegisterInterest

The body of the request contains a “PTT-P25RegistrationRequest” JSON object that identifies the resource for which recording is requested along with the Logging Service’s SIPREC URI and LogEvent URL:

Table 3-8 PTT-P25RegistrationRequest JSON object Request Parameter

Name	Required	Description
resourceIdentifier	Yes	The unique identifier of the resource to be registered or unregistered for recording.
registerFlag	Yes	Indicates type of request: “Register” requests that the resource be registered for recording, “Unregister” requests a previously registered resource to be unregistered for recording.
siprecUri	Yes	The Logging Service SIPREC URI to use.
logEventUrl	Yes	The Logging Service LogEvent URL to use.

Status Codes

200 OK
400 Bad Request
401 Unauthorized
404 Not Found
500 Internal Server Error

If the resource does not exist, the PTT system MUST return 404 Not Found.

If the requester is not authorized, the PTT system MUST return 401 Unauthorized.

If the request is malformed, the PTT system MUST return 400 Bad Request.

If the PTT system cannot complete the request for some internal reason, it MUST return 500 Internal Server Error.

Otherwise, the request is handled as follows:

- If registerFlag is “Register” and the resource is not currently registered for recording, the PTT system MUST return 200 OK, register the resource for recording if the requester

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is authorized, and thereafter initiate recording for any PTT activity on the given resource if possible to do so.[4]

- If registerFlag is "Register" and the resource is already registered for recording, the PTT system MUST return 200 OK and reregister the resource with the siprecUri and logEventUrl provided in this request.
- If registerFlag is "Unregister" and the resource is currently registered for recording, the PTT system MUST return 200 OK, unregister the resource, and no longer initiate recording on it.
- If registerFlag is "Unregister" and the resource is not currently registered for recording, the PTT system MUST return 400 Bad Request and ignore the request.

4 FUNCTIONAL ELEMENTS AND SERVICES

This document defines the set of Functional Elements (FEs) that provide standard functionality and interfaces in an NG9 1 1 PSAP. An FE (Functional Element) is an abstract building block that consists of a set of interfaces and operations on those interfaces to accomplish a task. Mapping between FEs and physical implementations may be one-to-one, one-to-many, or many-to-one. FEs can be bundled into solutions and MAY use non-standard interfaces internally, but the required standard functionality and interfaces specified in this document MUST be supported to achieve interoperability with external FEs. Some FEs, functions, and interfaces herein are denoted as optional; all others are required in order to conform to this standard.

A Service consists of one or more instances of an FE that act as one to provide the functionality specified for the FE. Services have names in the IANA serviceNames registry. The Service names are used in Data Rights Management (DRM) rules that control access to the Service and its resources. All FEs defined in this document are Services and have DRM rules.

Field Code Changed

4.1 Discrepancy Reporting

NENA STA 010 [7] defines a Discrepancy Reporting web service that accepts Discrepancy Reports (DRs) that allow an FE, Service, or Agency to report a problem to another entity.

All FEs and Services in the Agency that are capable of filing DRs MUST implement the client side of the Discrepancy Report Proxy web service defined in the Web Services section of this document. The FE or Service MUST submit the DR to the Discrepancy Report Proxy web service of the Management Console FE (MCFE) when filing DRs against entities inside or outside the PSAP.

4.2 ECC Service Discovery

Commented [MLS22]: U-NAPTR text contributed by Dan M.

Commented [MS22R2]: We agreed to stick with U-NAPTR for NG PSAP, but i3 is still using the S/AL.

The ECC Service Discovery mechanism defined in this section is used only internally within the ECC.

[4] A transmission on a recorded resource might not cause the PTT system to initiate recording if the PTT system or the network is temporarily overwhelmed.

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To discover services outside the ECC, the service discovery mechanism defined in STA-010 (i.e. the Service/Agency Locator) MUST be used as specified in the Service/Agency Locator (S/AL) section of this document.

The ECC Interface Discovery mechanism works as follows:

- Send a Domain Name System (DNS) Naming Authority Pointer (NAPTR) query to DNS with an Application Unique String (AUS) formatted as follows:
    - <interface>.<service>.<Agency Identifier> to get the base URI to query for supported major versions (for web services only). The Agency Identifier MUST be the PSAP's or ECC's own identifier. Then use the desired <version number> as follows:
        - <version number>.<interface>.<service>.<Agency Identifier>
            - Where <version number> is the desired version (for web services only)
            - Where <interface> is a token from the IANA Interface Names Registry [xxx]
            - Where <service> is a token from the serviceNames Registry
    - DNS will return a URI-Enabled NAPTR (U-NAPTR) record (per RFC 4848) which specifies the Uniform Resource Identifier (URI) for that interface on that Service

For example, to target version 1: NAPTR query for "1.logevent.Logging.example.com" would return:

```
IN NAPTR 100  10   "u"    "emergency-service:https"
           "!.*!https://logger.example.com/secure:8443!"  ""

IN NAPTR 200  10   "u"    "emergency-service:http"
           "!.*!http://logger.example.com:8080!"  ""
```

For example: NAPTR query for "siprec.Logging.example.com" would return:

```
IN NAPTR 100  10   "u"    "emergency-service:sips"
           "!.*!sip://logger.example.com/secure!"  ""

IN NAPTR 200  10   "u"    "emergency-service:sip"
           "!.*!sip://logger.example.com!"  ""
```

The specification for every interface will define the UAS to be used for the U-NAPTR query and how the returned URI is to be used. The typical REST/JSON web service using HTTP(S) would follow the OpenAPI standard as specified in the associated YAML schema for the interface. When SIP is specified ("emergency-service:sip" or "emergency-service:sips") then the interface would typically follow the mechanism described in RFC 3263 to properly resolve the SIP(S) URI.

4.2.1 IANA Considerations

U-NAPTR Registrations

This document registers the following U-NAPTR application service tag:

- Application Service Tag: emergency-service
    - Defining Publication: The specification contained within this document.

This document registers the following U-NAPTR application protocol tags:

- Application Protocol Tag: sip

Commented [MS23]: Get ref after it is created

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When it comes to Service Discovery, an Agency may want to restrict who can resolve UAS associated with interfaces restricted for internal uses. This can be accomplished using Access Control Lists (ACL) and Split-horizon DNS. ACL could be used to restrict access to internal clients of a service (such as "siprec.Logging…") which is not accessible from the outside. Split-horizon can be used to return different IP addresses to internal vs. external requesters such as when discovering "logevent.Logging…" which is an externally accessible interface for an Agency).

DRAFT

  - Defining Publication: RFC 3261
- Application Protocol Tag: sips
- Defining Publication: RFC 3261

4.3 NG9 1 1 Computer Aided Dispatch (CAD)

Computer Aided Dispatch (CAD) systems are used to assign and dispatch resources for an Incident and to update Incident state information during the response to an active Incident. A CAD system is an example of a system that incorporates multiple FEs. This document defines a Dispatch FE and an Incident Handling FE (IHFE) that standardize communication between these FEs and other FEs that deal with Incident state. The CAD's Dispatch FE defined in this document manages resources (emergency responders, equipment, etc.) and is responsible for notifications and logging of Incident state changes relative to those resources. The CAD's IHFE is responsible for receiving notifications, and logging of all other Incident state changes handled by the CAD system. Both the Dispatch FE and IHFE implement the EIDO Conveyance interface to communicate Incident state. Of course, assigning resources changes Incident state. See the Dispatch Functional Element section for details. This document does not define a full CAD system; it defines the Incident Handling and Dispatch FEs that CAD MUST support in order to communicate with other NG9 1 1 FEs and Services.

4.4 The Emergency Incident Data Object (EIDO)

NENA-STA-021 [22], the NENA Emergency Incident Data Object (EIDO) standard, defines the EIDO data structure that is used to communicate Incident state changes between FEs. The information contained in an EIDO is what is known to the sender at the time the EIDO is sent and may change during Incident processing.

All FEs that send EIDOs MUST implement the notifier side of the EIDO subscription mechanism specified in the *Conveyance of Emergency Incident Data Objects (EIDOs) between Next Generation (NG9-1-1) Systems and Applications,* NENA-STA-024 [25]. Upon receipt of an EIDO subscription request, the FE consults its EidoConveyance Policy to determine if the requester is permitted to subscribe, and if so, the FE MUST immediately return a notification containing an EIDO complying with that policy. Details on the EIDOConveyance interface are in NENA-STA-024 [25]. If the subscriber is not allowed to subscribe, the FE MUST return the appropriate error response code.

An Incident state change that affects any element of the EIDO is considered a relevant change, and the FE processing the state change MUST:

1. Send an EIDO containing the Incident state to other FEs that are subscribed (subject to policy)

2. Log an EIDO containing the Incident state (even if no EIDOs are being sent).

Incident state changes that affect any component in the EIDO MUST always be logged, even if no EIDO is sent. For example, if the Call Handling FE and the Dispatch FE are implemented in the same system, they might communicate Incident state changes internally. If so, Incident state changes that would affect any component in an EIDO must be logged with the IncidentOrResourceStateChangeLogEvent, even if no EIDO was sent or received.

See NENA-STA-024 [25] for details on standard mechanisms required for exchanging EIDOs. FEs that exchange EIDOs MUST conform to NENA-STA-024 [25].

Commented [MS25]: TODO:
Dan or Guy propose any changes to this text - the IANA registries might not be correctly identified. Create text in IANA section after Dan says these are all we need to register.

Commented [MLS26]: Conveyance 1.1 specifies an EIDOConveyance *interface name* but not an EIDOConveyance *Service Name*. Choose one of these:
- Service Name is CHFE, Dispatch, IDX, or RDS. This allows each of these services to have a separate DRM policy
- Service Name is IHFE. CHFE, Dispatch, IDX, and RDS all have the same policy.

Commented [MS26R2]: ***mls pick up here to continue TODOs

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EIDO data components and elements MUST be filtered by the sender according to policy based on the identity and role of the receiver per NENA-STA-024 [25].

If multiple FEs are bundled in a single system, they all MUST implement and enforce an EIDO data filtering policy applicable to the data they convey.

FEs that exchange EIDOs MUST support sending and/or receiving EIDOs by reference or by value as defined in NENA-STA-024 [25], according to applicable policy. The following data received with a call MAY be provided in an EIDO sent to an outside Agency in accordance with applicable policy:

- additionalDataReference
- agentReference
- callBackReference
- Verstat
- sipIdentity
- locationReference
- personReference

EIDO elements in this list that end in "Reference" contain a reference to another data component present in this EIDO which contains the actual data being shared. These "Reference" elements should not be confused with an external link reference provided by an outside entity that allows an Agency or other entity to retrieve some data.

Commented [MS27]: These are element names in the Call Data Component, so we're stuck with them. This comment is to remind us why.

EIDOs sent to external Agencies that contain a URI to external media MUST contain the original media URI and not a Media Proxy URI.

Deleted: that is associated with an Incident

Other data received in an EIDO by value from another Agency SHALL NOT be provided to an outside entity. If an entity provides a reference to the data, the receiving Agency MAY provide the reference to an outside entity because the original entity can then control access to the data when it is dereferenced.

Agency FEs that receive EIDOs SHALL subscribe to the Intra-Agency IDX if they need to be notified of Incident status changes from other Agency FEs. When any FE needs a coalesced view of an Incident involving multiple Agencies, it MUST obtain such view from the Inter-Agency IDX described in Intra-Agency IDX Interface to the Inter-Agency IDX section below.

4.5 The Intra-Agency Incident Data eXchange (IDX) FE

The Incident Data eXchange (IDX) FE is defined in NENA-STA-010 [7]. Every Public Safety Agency, whether they handle calls or not, MUST have an intra-Agency IDX for exchanging EIDOs with other Agencies. The intra-Agency IDX performs the following major functions:

- SHALL subscribe to all FEs in the Agency that send EIDOs for all Incidents and coalesces intra-Agency EIDOs so it can provide the Agency's view of an Incident
- SHALL act as the EIDO "hub" within the Agency. Agency FEs SHALL subscribe to the Intra-Agency IDX. This way, all subscribed FEs will receive a coalesced view of the Incident.
- The Intra-Agency IDX SHALL be the recipient of EIDO requests for Incident data from outside entities, i.e. the Intra-Agency IDX is the eidoInterfaceUri target in the Agency's record in the S/AL (defined in NENA-STA-010 [7]).
- SHALL process subscription and dereference requests from other Agencies

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The intra-Agency IDX MUST support the EIDO Conveyance interfaces specified in NENA-STA-024 [25].

The intra-Agency IDX SHALL subscribe to EIDOs from all of its Agency FEs that send or receive EIDOs, so it can provide a full view of Incident state as known by the Agency at any given time. This allows the IDX to serve as a “data hub” within the Agency, allowing FEs to subscribe to the intra-Agency IDX for EIDO updates instead of subscribing to all other FEs that send EIDOs.

An intra-Agency IDX SHALL filter the data it sends to another Agency according to the Agency’s policy.

The intra-Agency IDX MUST be used by the Agency to send EIDOs to another Agency.

The intra-Agency IDX MUST be capable of processing Incident subscriptions from other Agencies.

The intra-Agency IDX MUST implement the interfaces and functions specified in the Incident Data eXchange (IDX) section of NENA-STA-010 [7]. The intra-Agency IDX uses these interfaces and functions to interact with FEs and Services inside and outside the Agency.

The intra-Agency IDX MUST implement the Coalescing Policy mechanism described in the Incident Data eXchange section of NENA-STA-010 [7].

The intra-Agency IDX MUST implement the Alternate Values mechanism defined in the Incident Data eXchange section of NENA-STA-010 [7]. The intra-Agency IDX need not provide Alternate Values to an inter-Agency IDX, as the inter-Agency IDX will ignore Alternate Values and not include them in the EIDOs it sends to subscribers.

The intra-Agency IDX MUST implement the Logging functionality described in the Incident Data eXchange section of NENA-STA-010 [7].

The intra-Agency IDX MUST implement the interface to the inter-Agency IDX as described below in the subsection Intra-Agency IDX Interface to the inter-Agency IDX.

The intra-Agency IDX FE instances MUST implement ElementState as defined in the Element State section of NENA-STA-010 [7].

The intra-Agency IDX is a Service and MUST implement ServiceState as defined in the Service State Requirements section of this document.

The Service Name for the intra-Agency IDX is Intra-AgencyIdx and is listed in the “serviceNames” registry section of this document.

4.5.1 Intra-Agency IDX Policy

The intra-Agency IDX MUST implement a DRM Policy called IntraIDX per the Policy section of this document. This document adds “IntraIDX” to the IANA Service Names registry.

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Commented [MS29]: TODO: modify this text as needed to agree with what EIDO-JSON and i3 Architecture decides to do with Alternate values (once they’ve decided - still under discussion).

Commented [MS29R2]: TODO: discuss this - EIDO-JSON says:
If the coalescing algorithm result turns out to not update a value because a source does not include it, the EIDO MAY contain an alternate value Data Component that does not include an “OtherValue” field.
EIDO-JSON allows it, but does not require or recommend it. Do we want to include any text about this?

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Deleted: EIDO Reference Factory interface

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Deleted: EidoReference

Deleted: EIDO Reference interface

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4.5.2 Intra-Agency IDX Interface to the Inter-Agency IDX

The NGCS provides an Inter-Agency IDX that coalesces Incident information on multi-Agency Incidents as defined in NENA-STA-010 **Error! Reference source not found.**. For Incidents that involve more than one Agency, all involved Agencies' intra-AgencyIdxs MUST use the inter-Agency IDX for EIDO updates from other Agencies. The inter-Agency IDX performs the following major functions:

- Subscribes to all Incidents on the Intra-Agency IDX of each of its constituent Agencies
- Processes subscription and dereference requests from involved Agencies
- Provides a coalesced view of the Incident to all subscribers according to a coalescing policy
- Enforces a policy that specifies what can be shared with subscribers via its EIDOConveyance interface.

The Intra-Agency IDX MUST accept subscriptions from the Inter-Agency IDX according to the Agency's policy.

The Intra-Agency IDX MUST subscribe to the Inter-Agency IDX to obtain updates for Incidents being worked on with other Agencies. The Intra-Agency IDX MAY subscribe to the Inter-Agency IDX for situational awareness of other Incidents when allowed to do so by applicable policy.

If two agencies are sharing Incident updates directly under a cooperative agreement, both Agencies MUST also update the Inter-Agency IDX in accordance with their Inter-Agency IDX DRM policy.

4.6 Dispatch Functional Element

The primary functions of the Dispatch Functional Element (Dispatch FE) are:

- Recommend appropriate resources (emergency responders, equipment, etc.) available for an Incident
- Assign emergency responders and other resources to an Incident
- Manage the inventory and status of resources
- Provide and receive relevant resource information for the ECC and responders
- Monitor and log all transactions associated with resources

The Dispatch FE is used to manage emergency resources within its area of responsibility. It tracks the real-time statuses and locations of emergency resources.

The Dispatch FE MUST support exchanging Incident data with other FEs using EIDOs. See the Emergency Incident Data Object section of this document for requirements for communicating with EIDOs.

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Deleted: EIDO Retrieval interface
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Deleted: EidoRequestForService
Deleted: EIDO RequestForService interface
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Deleted: EidoResourceMessage
Deleted: EIDO ResourceMessage interface
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Deleted: EidoRequestAwareness
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Deleted: [7]

Commented [MS31]: If two Agencies have mutual aid agreement, they wouldn't want to share the data they agreed to share with some third party (NGCS Agency) that is not part of the agreement. The solution might be to let them share data with each other, but they also update the Inter-AgencyIDX with the data their policy allows them to share with any other Agency. If a third Agency joins the response, it will get that more limited view from the IDX. The IDX won't have the more extensive data that is covered by the agreement.
I propose this text to cover that situation - if we agree that we need a requirement like this.

The Dispatch FE SHALL provide EIDO status updates to subscribers about an Incident and/or Incident location based on information from emergency responders and other sources, subject to policy.

When assigning a resource to an Incident, the Dispatch FE SHALL notify authorized subscribers by sending an EIDO that identifies the assigned resource and its status.

The Dispatch FE SHALL notify authorized subscribers of resources offered by, or to, another ECC in connection with an Incident, subject to policy.

The Dispatch FE MUST support the RdsPsap interface to the Responder Data Services FE in order to exchange Incident status updates with Responders (see the Responder Data Services section of this document).

4.6.1 Dispatch Policy

Dispatch is a Service that uses a Dispatch FE and an IHFE. The Service Name of the Dispatch Service is “Dispatch”, and it is included in the IANA serviceNames registry.

The Dispatch Service MUST implement a DRM Policy called DispatchPolicy as described the Policy section of this document.

Commented [MLS32]: Hyperlink

EIDO interface names for EIDO DRM rules are specified in the Incident Handling FE (IHFE) section and are inherited by the Dispatch Service. See the Incident Handling FE section for details.

Commented [MS33]: Maybe not. Will ask NENA staff where the EIDO Interface names should be defined.

4.7 Incident Handling FE (IHFE)

An Incident Handling FE collects, displays, updates, and distributes Incident data, including data about incoming and outgoing calls. The IHFE is also responsible for merging, linking, and splitting Incidents as described below. When a call is an NG9 1 1 call, it will be accompanied by a unique Incident Tracking Identifier which is used to track the Incident throughout its lifecycle.

The IHFE utilizes EIDOs to exchange Incident information with other FEs and MUST conform to the specifications in the section titled "The Emergency Incident Data Object”.

An IHFE MAY be bundled into a single system that includes:

- Dispatch FE in a CAD system
- CHFE in a Call Handling system
- Dispatch FE and CHFE in a combined system

If the IHFE is bundled with other FEs that use internal mechanisms to communicate Incident data, the IHFE MUST still provide the standard interface defined above in The Emergency Incident Data Object (EIDO) section for communication with external systems.

The diagram below shows separate CAD and Call Handling systems where an IHFE is bundled into each, and EIDOs are used to communicate Incident data between them:

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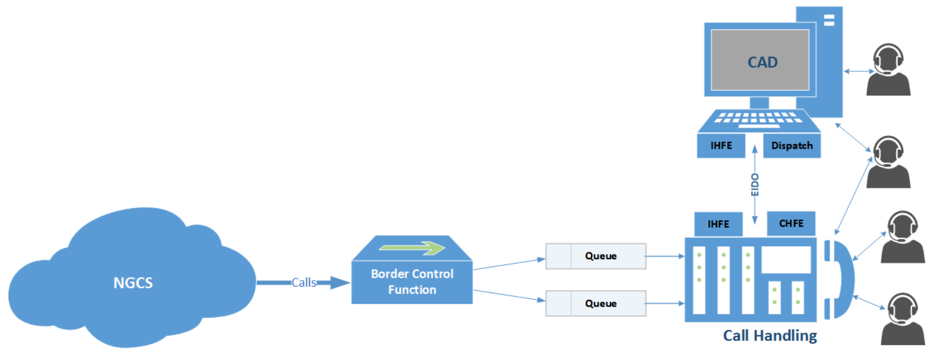

Figure 2 IHFE in CAD and Call Handling Systems

The IHFEs in CAD and Call Handling function identically. An Agent using both CAD and Call Handling could use either system to update Incident data. EIDOs will be sent between these systems to keep both systems updated with Incident state changes as specified in the Emergency Incident Data Object section of this document.

4.7.1 Manual Incident Creation

For any Incident reported via other means, the IHFE MUST:

- Assign a unique Incident Tracking Identifier as defined in the "Identifiers" section of NENA-STA-010 [7]
- Assign an initial Incident type for the new Incident
- Notify EIDO subscribers of the creation of the Incident

The IHFE updates, and if necessary, creates Incident data. The IHFE is also responsible for sending and receiving Requests for Service and Requests for Awareness as specified in NENA-STA-024 [25].

As with all FEs that modify Incident state, the IHFE FE MUST log all EIDOs it sends or receives and MUST log all Incident state changes it causes with the appropriate LogEvent as specified in NENA-STA-010 [7].

4.7.2 Adding, Modifying and Deleting Incident Data

An Agent is allowed to identify the location that came with the call (even if it came by reference) as the Incident location, and the IHFE MAY then use that location when informing EIDO subscribers by copying the location by value to the EIDO Incident data component.

When one or more EIDO data elements need to be added, modified or deleted, the IHFE MUST:

- Generate an IncidentOrResourceStateChangeLogEvent which specifies for each affected data element:
    - The JSON Path of the data element
    - If the data element is being added or modified: populate the "new" element with the updated data element value
    - If the data element is being deleted or modified: populate the "old" element with the previous data element value

Commented [MLS34]: Since an FE must always send an EIDO to the AgencyIDX when any element of a data component changes, all Incident state changes get logged in an EidoLogEvent.
This says the FE also logs this event that tells what elements changed.
You can't tell whether an element is new or changed in this event. Better to have a changeType element with "New", "Changed", or "Deleted" values. Then an elementValue that contains the value from the element, or an empty string if it was deleted.
And is this logged only by the IHFE? Shouldn't other FEs that send EIDOs log this too?

Commented [MS34R2]: TODO: look at this log event and discuss with group.

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- Notify EIDO subscribers of the update to the Incident state

If the IHFE needs to access media through a URI, it MUST use a mediaProxyUri to access the media, thereby ensuring that the media are logged. See the Media Proxy section for details.

If different Agents or witnesses have reported different descriptions for the same thing (person, location, object, etc.), the Agents MAY include multiple data components containing the different descriptions and a Note that explains that these are believed to be different descriptions of the same thing.

Commented [MS35]: Added this proposed text per discussions in EIDO-JSON WG to allow duplicate data components that contain conflicting values from different agents. Remove this if that functionality is removed from EIDO-JSON.

4.7.3 Merging an Incident

If the agent determines that the call is about a previous Incident, the IHFE SHALL merge the current and prior Incidents as described in the Logging Service section of NENA STA 010 [7]. Once a merge has been performed with a prior Incident Tracking Identifier, the prior Incident Tracking identifier SHALL be used from that point forward. This prior Incident Tracking Identifier is called the "surviving" Incident Tracking Identifier. The current Incident Tracking Identifier (the one being merged with the prior one) is called the "replaced" Incident Tracking Identifier. If the agent determines that the Incidents are related but are not the same Incident, the Incidents should be linked as described below in Linking Incidents.

The merge SHALL be communicated to other FEs using the Merge Information data component of an EIDO as specified in the Standard for Emergency Incident Data Object (EIDO), NENA-STA-021 [22]. To inform those subscribing to the surviving Incident, the IHFE MUST:

Commented [MS36]: The yaml already called it "Merge". EIDO-JSON 3.1 fixed it in the text to match the yaml. Did global search/replace in this doc.

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- Put the surviving Incident Tracking Identifier in the header of an EIDO
- Put the replaced Incident Tracking Identifier in the incidentTrackingIdentifier element of the Merge Data component and set the incidentMergeDirectionCode to "REPLACED".

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Subscribers to the replaced Incident Tracking Indentifier (who might not be subscribed to the surviving Incident) must also be notified. To do so, the IHFE MUST:

- Put the replaced Incident Tracking Identifier in the header of another EIDO
- Put the surviving Incident Tracking Identifier in the incidentTrackingIdentifier element of the Merge Data component and set the incidentMergeDirectionCode to REPLACING.

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4.7.4 Splitting an Incident

If it is determined that an incident must be split (cloned), then a copy of the Incident data SHALL be made by the IHFE and a new Incident Tracking Identifier assigned. The IHFE SHALL log the split as described in the Logging Service section of NENA STA 010 [7].

The split SHALL be communicated to other FEs by setting the incidentMergeDirectionCode to "SPLIT" in the Merge Information data component.

Deleted: The split SHALL be communicated to other FEs using the Link data component of the EIDO as specified below in Linking Incidents.

4.7.5 Linking Incidents

If the Incident is determined to be related to an existing Incident (hereafter, Incident A), the current Incident (hereafter, Incident B) SHALL be linked to Incident A by the IHFE as described in the Logging Service section of NENA STA 010 [7].

The link SHALL be communicated to other FEs using the Link data component of an EIDO as specified in the NENA-STA-021 [22].

To inform those subscribing to Incident A, the IHFE MUST:

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- Put the Incident Tracking Identifier of Incident A in the header of an EIDO.
- Put the Incident Tracking Identifier of Incident B in the incidentTrackingIdentifier element of the Link Data component.

Subscribers to Incident B (who might not be subscribed to Incident A) must also be notified. To do so, the IHFE MUST:

- Put the Incident Tracking Identifier of Incident B in the header of another EIDO.
- Put the Incident Tracking Identifier of Incident A in the incidentTrackingIdentifier element of the Link Data component.

The nature of the link is given by the linkDirectionCode:

- “PARENT” The Incident Tracking Identifier contained in the Link data component represents an Incident that caused or triggered the Incident represented by the Incident Tracking Identifier contained in the EIDO header.
- “CHILD” The Incident Tracking Identifier contained in the Link data component represents an Incident that was caused or triggered by the Incident represented by the Incident Tracking Identifier in the EIDO header.
- “RELATED” The Incident Tracking Identifier contained in the Link data component is related to the Incident Tracking Identifier in the EIDO header but is not a parent or child.
- “UNLINK” The Incident Tracking Identifier contained in the Link data component that was previously linked is now unlinked from the Incident Tracking Identifier in the EIDO header. The Unlink is included in only one EIDO to denote the action.

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Deleted: Related

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4.7.6 Incident Handling Functionality

The IHFE SHALL support the exchange of EIDOs with other FEs as specified in the Emergency Incident Data Object section of this document.

The IHFE MUST be able to obtain the initial location for a call and any subsequent location updates. The IHFE SHALL implement both the HELD dereference interface and the SIP Presence Event Package interface to query the LIS or LNG FE for current location as specified in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. The IHFE SHALL also support manually entered locations.

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When the IHFE needs to render a map for which it doesn’t have map data, it MUST use the Mapping Data Service (MDS) to obtain the map, as described in the Mapping Data Service section of this document. To get the desired map data, the IHFE would typically send a findService query to the ECRF for urn:emergency:service:serviceagencylocator.mds with the Incident location, or a polygon that includes the Incident location, then use the S/AL to find the MDS web service interfaces for the Agency or Agencies returned by the ECRF.

The IHFE SHALL support Additional Data included by value in a call, and the Additional Data dereference mechanism specified in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

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The PSAP IHFE SHALL implement the LoST client interfaces to support interaction with the ECRF and LVF FEs. Use of the LoST client interface is defined in the LoST subsection of the Interfaces section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

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Systems that implement the IHFE need to validate civic locations. The IHFE SHALL support the LVF client interface for such validations or use an internal LVF. If the implementation contains an LVF, that LVF is not required to support external requests for location validation.

If the IHFE is using Incident location to determine the responding services that are needed, then it MUST use the ECRF to obtain the services. In cases where location-based routing is not needed, the IHFE MAY determine the services via an internal mechanism, which is not standardized in this document.

The IHFE SHALL implement the client side of the Service/Agency Locator interface as described in the Service/Agency Locator section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. The Service/Agency Locator Service is used to discover an Agency's interface entry points and other important Agency information.

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Commented [MLS39]: TODO: This will change per our U-NAPTR Service Discovery mechanism, so don't accept it yet.

4.7.7 Incident Handling Policy

The IHFE is an integral part of the Call Handling Service, the Dispatch Service, and Responder Data Services (RDS), so the IHFE doesn't have its own Service Name. DRM rules for the IHFE's EIDO interface are the DRM rules specified in the CallHandlingPolicy, DispatchPolicy, or RdsPolicy respectively.

The IHFE MUST enforce the EIDO DRM Policy of the Service that is using the IHFE (CallHandlingPolicy, DispatchPolicy, or RdsPolicy) per the Policy section of this document.

Commented [MS40]: If this doc creates EIDO Interface Names, put them here. Otherwise, reference Conveyance.

4.7.8 Closing an Incident

If the IHFE is used to close an Incident, the IHFE SHALL:

- Initiate logging of a IncidentClearLogEvent to the Logging Service as specified in the Logging Service section of NENA-STA-010 [7].
- Notify EIDO subscribers of the closing of the Incident.

4.7.9 Reopening an Incident

If the IHFE needs to log events on an Incident for which an IncidentClearLogEvent has been logged, the IHFE SHALL:

- Log an IncidentReopenLogEvent.
- Notify EIDO subscribers of the reopening of the Incident.

If the IHFE is subsequenty used to close the reopened Incident, it SHALL follow the procedure specified in "Closing an Incident" above.

4.8 Call Handling Functional Element (CHFE)

The Call Handling Functional Element (CHFE) is responsible for the functionality and interfaces related to the processing of calls. This includes the interface to the NGCS network over which Emergency Calls are received and MAY interface to the PSAP's other communication systems and networks (an administrative PBX, Public Switched Telephone Network (PSTN), etc.) via the External Switching System (ESS) SIP interface described in this section.

To process the underlying Incident and related data being reported by a caller (e.g. the ability to display and update Incident data such as location, etc.) a PSAP must have (in addition to the CHFE) an IHFE. The two FEs each have distinct responsibilities in support of call and incident handling. The IHFE is described in the Incident Handling FE Section.

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4.8.1 Early Media

Early Media is defined as media streams that flow between parties before the session establishment process is complete, that is, before a final response has been provided. It is important the CHFE accepts Early Media that it receives, but because some devices don't work with Early Media, making it inconsistent, this document restricts how it may be used.

A PSAP SHALL NOT offer Early Media in response to inbound call requests.

A PSAP MUST support Early Media offers on outgoing call requests originating from it.

See RFC 3960 [82] for Early Media specifications.

4.8.2 Interfaces

A PSAP compliant CHFE MUST implement the following interfaces.

4.8.2.1 Service State and Security Posture Subscription

The CHFE MUST subscribe to Management Console's PSAP Service State/Security Posture notifications per the specifications in the Service State section of NENA STA 010 [7]. Upon any change to the Service State of the PSAP, the QueueState of any queues the CHFE manages MUST be set to represent the current PSAP Service State with an appropriate value per the mapping specified in the Queue State Subscription section of this document.

4.8.2.2 Element State Subscription

The CHFE MUST implement the notifier side of the Element State subscription mechanism as described in the Element State Requirements section of this document.

4.8.2.3 Emergency Services Routing Proxy (ESRP) Interface

The CHFE MUST implement the SIP call interface as defined in the SIP Call section of NENA-STA-010 [7], including support for PSAP Call Control features as specified in Appendix C of NENA STA 010 [7], in jurisdictions where PSAP Call Control features are desired or required.

4.8.2.4 Queues

The CHFE MUST support multiple Queues, although a small PSAP might use only one Queue. Each Queue MUST have a globally unique name as defined in the Queue Identifier section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.** where the host part must contain in whole or in part the Agency Identifier of the Agency owning or subscribing to the CHFE. For example, "sip:sos@psap.allegheny.pa.us" or "sip:sos@backup.psap.allegheny.pa.us".

4.8.2.5 Dequeue Registration

The CHFE MUST implement the client side of the DequeueRegistration Web Service mechanism as specified in the DequeueRegistration Web Service section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. As stated in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**, a queue that has only a single dequeuer does not use DequeueRegistration. However, the CHFE might be provisioned to dequeue from queues that have multiple dequeuers and will require use of the DequeueRegistration client. For each queue the CHFE can receive calls on, the CHFE MUST register to dequeue as specified in the DequeueRegistration Web Service section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

Commented [41]: Does CHFE need to subscribe to Service State of any other Services? If so, specify it here and point to the General Service State sections.

Commented [MS41R2]: Asked this question on list, no answer yet, will ask again. Should it subscribe to Service State of the ESRP.

Commented [DM42]: This is not unique to the CHFE, it should be moved to the generic FE section of the document

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Commented [43]: Consensus on sideways routing by CHFE:
Sent a note to i3: we want to be able to dynamically respond to an INVITE with a 300 response that specifies a PSAP URI to redirect the call to. We basically want a redirect mechanism that a PSAP manager can control that redirects calls to a mutual aid agency when the manager deems necessary.

Commented [MS43R2]: I3 has changed Dequeue Registration, check that this is no longer an issue.

Moved down [2]: If a call is received on a queue that has a queue state of Standby, the call SHOULD be rejected.[5]¶

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The CHFE MUST renew a registration prior to its expiry. The CHFE SHOULD renew a registration when half of the subscription validity time has passed.

The CHFE MUST unregister when appropriate, for example if the system is about to go out of service or a queue is about to be removed.

A queue's Dequeue Registration state MUST not affect the CHFE's ability to process a call on that queue.

4.8.2.6 Queue State Subscription

The CHFE MUST implement the notifier side of the Queue State subscription mechanism as specified in the QueueState Event Package section of the NENA STA 010 [7]. Upon reception of a SUBSCRIBE request addressed with the Queue Identifier and specifying the emergency-QueueState event package, the CHFE MUST consult the policy (queueState) to determine if the requester is permitted to subscribe. If not, the CHFE MUST return 603 Decline. The CHFE determines whether the specified queue is one managed by the CHFE. If not, the CHFE MUST return 488 Not Acceptable Here. If the request is acceptable, the CHFE MUST return 200 OK. The CHFE MUST implement event rate filters, RFC 6446 [18]. Upon acceptance, the CHFE MUST send an initial NOTIFY message.

NOTIFY messages MUST contain the following information:

- **queueUri**: Queue Identifier of the queue.
- **queueLength**: current number of calls in the queue waiting to be answered.
- **queueMaxLength**: maximum number of calls this queue can hold.
- **state**: current QueueState of the queue, MUST be a value found in the queueState registry.

NOTE: queueLength MUST include only calls waiting to be answered, not calls that are currently being handled by an Agent or an automaton. The number of calls currently being handled by Agents and/or automatons can be tracked by monitoring LogEvents.

Upon any change to the Queue's state (which includes when calls are queued and dequeued) and respecting any rate filter the subscriber may have specified, the CHFE MUST generate a QueueState NOTIFY message containing the updated state information.

When the PSAP ServiceState changes, the CHFE MUST report the QueueState as the value per the following table:

Commented [MS44]: Edits from 6/18/26 call in this paragraph and the table below were merged into Dan's new CHFE text by Michael Smith.

Table 4-1 PSAP ServiceState to QueueState Mapping

PSAP ServiceState	QueueState state element
Unstaffed	Disabled
ScheduledMaintenanceDown	Disabled
Overloaded	ResourceExhausted
GoingDown	Disabled
Down	Disabled

4.8.2.7 ESRP Notifications

The CHFE MUST implement the subscriber side of the ESRP notifications subscription mechanism as specified in the ESRPnotify Event Package section of NENA STA 010 [7]. The CHFE MUST send a SUBSCRIBE request to provisioned ESRP URIs specifying the emergency-ESRPnotify event package.

4.8.2.8 Abandoned Call Notifications

The CHFE MUST implement the subscriber side of the AbandonedCall notifications subscription mechanism as specified in the AbandonedCall Event section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. The CHFE, at startup time, MUST send a SUBSCRIBE request to provisioned ESRP URIs specifying the emergency-AbandonedCall event package and MUST maintain the subscriptions active at all time. The CHFE MUST implement the ability to alert Call Takers of Abandoned Calls and upon reception of an Abandoned Call notification, if enabled by local policy, MUST alert appropriate Call Taker(s) of the abandonment of a call. The CHFE MUST notify EIDO subscribers of the callCancel call state change where the EIDOs are populated as follow:

Incident Identifier: MUST be the identifier found in the notification

- Agency Data Components:
  - There MUST be one Agency Data Components specifying the Agency receiving the notification

- Location Data Components:
  - There MUST be as many Location Data Components as there are locations in the notification
  - locationTypeDescriptionRegistryText MUST be "Caller"
  - If there was a Location by Reference, it was dereferenced and a location was obtained, MUST populate an additional Location Data Component where:
  - locationTypeDescriptionRegistryText MUST be "Caller"
  - locationByValue element MUST contain the dereferenced location and
  - locationDereferencedFromReference MUST point to the Location Data Component which contains the reference used to obtain this value

- Additional Data Components:
  - There MUST be as many Additional Data Components as there are in the notification

- Callback Data Component:
  - This is populated as follow:
  - If one or more P-Asserted-Identify headers are found in the notification, then there MUST be one callBackInformationUri per header else
  - MUST be only one callBackInformationUri containing the From header URI
  - deviceContactHeader MUST contain the Contact URI found in the notification
  - "verstat" MUST contain the verstat parameter value, if provided
  - sipIdentities array MUST contain the contents of all of the Identity headers, if present

- Person Data Component:
  - There MUST be a Person Data Component providing information regarding the caller:
  - "personIncidentRoleRegistryText" array MUST contain "Caller"
  - "callIdentifier" MUST link to the Call Data Component instance in the EIDO

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- "callBackReference" MUST link to the Callback Data Component instance in the EIDO

- Call Data Component:
  - Contains the details of the abandoned call:
    - Identifier of this Data Component MUST be the Call Identifier assigned to the abandoned call in the notification
    - "standardPrimaryCallType" MUST be set to an appropriate value found in the IANA Emergency "LogEvent CallTypes" registry
      - For an emergency call (including abandoned 9-1-1 calls) value MUST be "emergency"
    - "direction" MUST be set to "incoming"
    - "callStartTimestamp" MUST contain the time the "inviteTimestamp" element of the notification
    - "callStateRegistryText" MUST contain "callCancel"
    - "verstat" MUST contain the verstat parameter value, if provided. MUST also be the same value as the "verstat" element in the Callback Data Component.
    - sipIdentity MUST contain the content of the first Identity header, if any
    - sipIdentities array MUST contain the contents of all of the Identity headers, if any (even if only one and it was provided above). MUST be identical to the "sipIdentities" element in the Callback Data Component
    - "callBackReference" MUST link to the Callback Data Component instance in the EIDO
    - "locationReference" array MUST contain links to Location Data Component instances found in the EIDO applicable to the call
  - "additionalDataReference" array MUST contain links to Additional Data Component instances found in the EIDO applicable to the call
    - "personReference" array MUST contain links to Person Data Component instance found in the EIDO applicable to the call

4.8.2.9 Test Calls

The CHFE MUST implement support for Test Calls as specified in the Test Call Functions section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. The CHFE MUST support test calls signaled with both the urn:service:test.sos and urn:emergency:service:test Service URN trees. Upon reception of a Test Call request addressed with a Queue Identifier, the CHFE MUST consult the policy (TBD) to determine if the function is enabled and if the requester is permitted to place such test call.[6] If not, the CHFE returns 603 Decline. Policy must also specify the maximum rate of test calls from a given source and if exceeded, CHFE MUST return a 486 Busy Here response. The CHFE MAY decline a Test Call from a source when it is already processing another Test Call from that source. If the Service URN found in the Request URI is not supported by the CHFE, it MUST return a 404 Not Found. The identity of the source is the P-Asserted-Identity header if present, and From header if not. Returning 486 or 603 responses is intended for legitimate sources; the CHFE SHOULD NOT respond to attempts to use the Test Call function if it appears that a Denial-of-Service attack is underway. If no error, CHFE returns a 200 OK to accept the Test Call with a Body of MIME type "text/plain" with the following:

- The name of the PSAP (as specified in the Service/Agency Locator) followed by CR/LF.

[6] Because Test Call is a critical function, it would be highly unlikely that Test Calls would be disabled.

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Commented [45]: TODO: Need a PSAP policy for Test Calls

Commented [MLS45R2]: TODO: discuss this Policy. There is a TestCalls Policy Type in i3 v3.1 that controls "Which originators may process PRR test calls". Should "PRR" be removed from that use description? How can we control who can send test calls when any person can send a test call? From our DRM section: "Subject attributes can include the id (as an agentId), agency (as an agencyId), role (from the Agent or Agency role registries, Sections 10.27 and 10.28), and agencyType (from the agency locator record)." A private individual has none of those.

Commented [MS45R3]: Dan might have promised a contribution on this, check with him.

- The string "urn:service:test:sos" followed by CR/LF.
- The location provided with the call, dereferencing, if necessary, with the address elements separated by commas (',') followed by CR/LF.
- If the value of the attestation-level in the Identity header does not indicate A-level attestation or the Verstat parameter doesn't indicate that validation passed, the string "Identity Attestation Failed", followed by CR/LF.

The CHFE MAY return other MIME body parts as specified in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

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The CHFE SHOULD accept a media loopback test as defined in *An Extension to the Session Description Protocol (SDP) and Real-time Transport Protocol (RTP) for Media Loopback,* RFC 6849 [27] and SHOULD support the "rtp-pkt-loopback" and "rtp-start-loopback" options. The CHFE is the media attribute of "loopback-mirror". The PSAP MUST loop back no more than three (3) packets of each media type accepted (voice, video, text).

After automatically answering the Test Call and if applicable, looping back the media, the CHFE MUST send a BYE to terminate the Test Call.

4.8.2.10 Location Information Server (LIS) Interface

Commented [46]: TBD: what to do about a new location received in an UPDATE or re-INVITE.

Commented [MS46R2]: Ask Dan about this.

The CHFE MUST be able to receive the calling device's location in a Geolocation header field by either value within an INVITE or MESSAGE body or via a reference. The CHFE MUST support references with schemes of HTTP(S) and SIP(S). If the scheme is HTTP(S), the CHFE MUST utilize the mechanism specified in HELD Dereferencing as defined in *A Location Dereference Protocol Using HTTP-Enabled Location Delivery (HELD)*, RFC 6753 [28] to retrieve the location. If the scheme is SIP(S), the CHFE MUST utilize the mechanism specified in the SIP Presence Event Package defined in *A Presence Event Package for the Session Initiation Protocol (SIP)*, RFC 3856 [29]. For SIP location conveyance, the CHFE MUST be able to control the rate of location update notifications through location filters as defined in *Filtering Location Notifications in the Session Initiation Protocol (SIP),* RFC 6447 [30] and event rate control filters RFC 6446 [18]. The CHFE MUST support the reception of multiple Geolocation: header fields. If multiple Geolocation header fields are provided, the location in the first Geolocation header field SHALL be assumed to be the location used for call routing.

If the incoming call includes one or more HELD reference(s), the PSAP can make an initial dereference to obtain an initial location quickly by setting the 'responseTime' value to 500 ms. Subsequent requests for dispatch location MUST be accomplished by setting the 'responseTime' value to 'emergencyDispatch' per NENA-STA-010.[7]

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4.8.2.11 Emergency Call Routing Function (ECRF) Interface

4.8.2.11.1 Determining Transfer-to Agencies' SIP URI

The Call Handling FE assists the Telecommunicator in selecting the responding services that are available for call transfers. The CHFE SHALL implement the LoST client interface to support interaction with the ECRF as specified in the LoST subsection of the Interfaces section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. The CHFE MUST query the ECRF, when necessary, to permit transfer of a call to any of the services available for the given location.

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[7] Local regulations may specify allowed 'responseTime' values.

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The CHFE SHOULD support the caching of ECRF responses as specified in the <expires> element of the response. The CHFE SHOULD compare any new location or location updates against the <serviceBoundary> of cached responses and MAY use cached mappings in order to accelerate call treatment and reduce load on ECRF servers. The LoST listServicesByLocation function may reveal additional services at a new location, but its use SHOULD be rate limited to avoid taxing the ECRF. An expired cached mapping MAY be used in the case where an authoritative ECRF is not reachable.

4.8.2.12 SIP Interface to External Switching Systems

The CHFE MUST support interfacing with additional SIP call sources/destinations. This interface is used to interface to SIP telephony gateways, an administrative PBX, a SIP Trunk Service Provider, etc. The interface MUST comply to the *SIPconnect 2.0 Technical Recommendation,* [31].

4.8.2.13 Incident Data eXchange (IDX) Interface

The CHFE MUST implement the client side of the EIDO Dereference Factory web service specified in NENA-STA-024 [25]. This is used to obtain a reference from an IDX that can be passed along when transferring a call.

4.8.2.14 Bridge Interface

A CHFE implementation MUST support both the Conference Aware UA and Ad Hoc Bridging methods as specified in section Bridging and Transfers in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.** and MUST be provisioned with the appropriate method as specified by the NGCS Service Provider.

If a call is received specifying it originated from a Bridge, the CHFE MUST use the Bridge to add parties[8] to the call, in order to avoid cascading conferences.

The CHFE MUST support the configuration of URI-List as specified in *Framework and Security Considerations for Session Initiation Protocol (SIP) URI-List Services*, RFC 5363 [32] and use a URI-List to add one or more parties to a call at the same time as defined in *Conference Establishment Using Request-Contained Lists in the Session Initiation Protocol (SIP)*, RFC 5366 [33].

4.8.2.14.1 Conference Aware UA Bridge

The Conference Aware UA Bridge is always in the path of 9-1-1 calls thus the CHFE can add parties to the call without having to first establish a new conference, as per section Route All Calls Via a Conference Aware UA in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

4.8.2.14.2 Ad Hoc Bridge

On an Ad Hoc Bridge, a conference must first be created, and the 9-1-1 call must be moved to the Bridge before any party can be added to the call (as specified in the Ad Hoc description of section Attended Transfers in NENA-STA-010 **Error! Reference source not found.**). In addition, once a call is reduced to two parties (for example at the completion of a transfer) the remaining CHFE SHOULD detect that the call only has two parties and remove the call from the Ad Hoc Bridge.

8 In this document, a party is an entity that can be identified by a SIP or TEL URI.

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Commented [47]: This triggered the discussion of what to do if we get a call from an outside (non-NGCS) bridge.

Commented [MS47R2]: Ask Dan for a response.

Commented [48]: The PSAP needs to know if an NGCS Bridge follows i3 rules or not. Ask i3 to add a parameter to contact that indicates the Bridge is an NGCS Bridge and specifies the i3 Bridge method used. Michael will bring up with i3 group, find thread Brian started on Bridge enhancements.

Commented [MS48R2]: Check if i3 did this.

Commented [49]: Asked i3 to add parameter to contact via email on 5/26/22.

Commented [50]: Changed to MUST. Dan still thinks it should be SHOULD.

Commented [51]: i3 has to decide whether the Bridge is in the path for outgoing calls, or whether the Bridge is created when needed to transfer a call. See the note in the Outgoing Calls section of this document.

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Commented [52]: How is this detection accomplished? What if it is the 9-1-1 caller that drops off, which PSAP is responsible to release the Bridge?

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4.8.2.15 ESRP Notifications

Commented [MLS53]: This section has no text. Presence Server isn't a subsection. I think we need text here. I think we already handled QueueState, SecurityPosture, do other event packages need to be covered here?

4.8.2.16 Presence Server Interface

The CHFE MUST publish Call Taker (Agent) Activity State changes to the Presence Server as specified in section Presence Server. If Agent Activity State is required for the CHFE to distribute calls, then the CHFE MUST subscribe to the Presence Server as specified in section Presence Server to obtain that information. The CHFE MUST be provisioned with the Presence Server selected by the PSAP.

Commented [54]: ToDo: Dan will incorporate text from the AgentStates (ActivityStates).

The CHFE MUST support the Common Reason Code values specified in section Activity States for the Call Handling FE. Use of Common Reason Code values in Activity State is optional.

4.8.2.17 Multimedia Support

The CHFE MUST implement support for RTP-based multimedia communications as specified in the Media section and the Media Mixing section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. The CHFE MUST implement support for Audio, Video and Real-Time Text (RTT) as defined by the *International Telecommunications Union (ITU), Telecommunications Standardization Sector* (ITU-T). The CHFE MUST implement support for the audio codecs G.711µ-law and G.711 A-law, which are defined in *Pulse Code Modulation (PCM) of Voice Frequencies, ITU-T Recommendations,* G.711 [34], and ITU-T *Wideband coding of speech*, G.722.2 [34]. The CHFE SHOULD implement support for Enhanced Voice Service (EVS) as defined in *Codec for Enhanced Voice Services (EVS) - The New 3GPP Codec for Communication,* [36] , G.722 [35] and the *Opus Interactive Audio Codec*, RFC 6716 [37], MAY implement support for *ITU-T Coding of speech,* G.729 [38]. The CHFE MUST identify the active party in a conference call as designated in the Synchronization Source (SSRC) field of the received RTP packets. The CHFE MUST implement support for multi-party RTT mixing to support multimedia conference calls as specified in *RTP-Mixer Formatting of Multiparty Real-Time Text*, RFC 9071 [39].

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Commented [SM55]: The previous reference here to G.729AB but that reference is not found .Annex J of this cross reference does speak to the A and B.

The CHFE MUST implement support for Instant Messaging as specified in section Instant Messaging of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

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In order to comply with the Americans with Disabilities Act requirements, the CHFE MUST implement support for receiving and sending characters encoded as *Baudot-Murray code,* ITA-2 [40] over an audio RTP stream. Sending characters encoded as Baudot Murray code must not occur when a RTT media stream is present in a call.

4.8.3 Call Handling Functional Element Call Processing

This section specifies call processing behavior of the CHFE in the PSAP.[9]

The CHFE MUST log every reception or transmission of call signaling messages using the CallSignalingMessageLogEvent.

Upon reception of an INVITE on any interface defined in this section, if the call does not have a Call Identifier (i.e. a Call-Info header with purpose=emergency-CallId), the CHFE MUST assign one to the call as per section Call Identifier of NENA-STA-010 **Error! Reference source not found.**. If the INVITE does not have an Incident Tracking Identifier (i.e., a Call-Info header

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[9] Please note that this standard does not specify an interface between CHFE servers and workstations. Implementers may select any suitable interface that meets their needs as long as the CHFE implementation complies to this document. .

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with purpose=emergency-IncidentId), the CHFE must assign one as specified in the Incident Tracking Identifier section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

Every queue MUST be provisioned with a Call Priority Identifier, which must be one of: Highest, High, Normal, Low, Lowest. Calls received with a Request-URI of urn:service:sos MUST be assigned the Call Priority Identifier of High or Highest.

If the call was received on a Queue which is provisioned with a Call Priority Identifier of Normal, Low or Lowest, but the Request-URI is an emergency Service URN (i.e. the call is an Emergency Call), the CHFE MUST assign the call a Call Priority Identifier of either High or Highest. Choice between High or Highest priority is per local policy. For example, an Emergency Call may be overflowed to a non-emergency queue but must be treated as an Emergency Call.

If any Emergency Call does not include a location, or contains a default location, the Call Taker MUST be informed.

Please note that this standard does not specify an interface between CHFE servers and workstations. Implementers may select any suitable interface that meets their needs as long as the implementation conforms to the external interfaces defined in this standard.

4.8.3.1 Incoming Interactive Calls

The CHFE MUST implement support for Reliable Provisional Responses as specified in *Reliability of Provisional Responses in the Session Initiation Protocol (SIP)*, RFC 3262 [41]. PRACK MUST be used if the INVITE's Require header or the INVITE's Supported header contains the option tag 100Rel.

The CHFE MUST implement support for the following calling party identifiers:

- A SIP URI.
- A TEL URI.
- A SIP URI with parameter user=phone.
- For the latter two cases, the URI identifies a PSTN party. The CHFE MUST implement support for PSTN identifiers formatted as an E.164 global number as specified in *The tel URI for Telephone Numbers,* RFC 3966 [42]. In addition, the CHFE MUST implement support for the verstat parameter as defined in 3GPP TS 24.229 [xxx] which provides the result of STIR/SHAKEN validation, for example:
    - tel:+18195551212;verstat=TN-Validation-Passed
    - sip:+18195551212;verstat=TN-Validation-Passed@osp.com;user=phone

Note that a P-Asserted-Identity header may contain both a SIP URI and a telephone number URI which indicates the calling party can be reached both by SIP call over an IP network and a PSTN call.

Upon reception of a call INVITE, the CHFE MUST do the following:

- Increase by one the length of the Queue on which the call was received on and generate a NOTIFY to Queue State subscriber(s)
- Log a CallStartLogEvent; the standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a SessionStartLogEvent. The standardPrimaryCallType, "to", and "from" elements MUST be populated.

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Commented [56]: Put all token values in quotes.

Commented [57]: PRACKs apply to MESSAGE, state this requirement in non-interactive calls section too.

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- Log a CallStateChangeLogEvent with state set to "callBegin" If the incoming call includes a Call-Info header with "purpose=emergency-eido" the CHFE MUST dereference the given EIDO URI as specified in section Passing Data to Agencies via Bridging of NENA-STA-010 [xxx]
- Notify authorized EIDO Subscribers of the reception of the new Call where the EIDOs are populated as follow:
  - Incident Identifier: MUST be the identifier for the call
  - Agency Data Components:
    - There MUST be one Agency Data Component specifying the Agency receiving the INVITE
    - If the incoming call included an EIDO specifying other Agencies (such as the transferring Agency), there MUST be additional Agency Data Components specifying the identities of the other Agencies
  - Location Data Components:
    - There MUST be as many Location Data Components as there are locations in the INVITE or in the related Call Data Component of the EIDO received with the call
      - locationTypeDescriptionRegistryText MUST be "Caller"
  - Additional Data Components:
    - There MUST be as many Additional Data Components as there are in the INVITE or in the related Call Data Component of the EIDO received with the call
  - Callback Data Component:
    - This is populated as follow:
      - If the incoming call contained an EIDO:
        - As specified in the related Call Data Component of the EIDO received with the call
      - Else
        - If one or more P-Asserted-Identify headers are found in the INVITE, then there MUST be one callBackInformationUri per header else
        - MUST be only one callBackInformationUri containing the From header URI
        - deviceContactHeader MUST contain the Contact URI found in the INVITE
        - "verstat" MUST contain the verstat parameter value, if provided
        - sipIdentities array MUST contain the contents of all of the Identity headers found in the INVITE, if present
  - Person Data Component:
    - If the incoming call contained an EIDO:
      - As specified in the related Call Data Component of the EIDO received with the call
    - Else
      - There MUST be a Person Data Component providing information regarding the caller:
        - "personIncidentRoleRegistryText" array MUST contain "Caller"
        - "callIdentifier" MUST link to the Call Data Component instance in the EIDO

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n

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            - "callBackReference" MUST link to the Callback Data Component instance in the EIDO
- Call Data Component:
    - Contains the details of the received INVITE:
        - Identifier of this Data Component MUST be the Call Identifier assigned to the call
        - "standardPrimaryCallType" MUST be set to an appropriate value found in the IANA Emergency "LogEvent CallTypes" registry
            - For an emergency call (such as when the Request URI of the INVITE is a Service URN, including abandoned 9-1-1 calls) value MUST be "emergency"
        - "queueIdentifier" MUST be the SIP URI on which the call was received
        - "direction" MUST be set to "incoming"
        - "callStartTimestamp" MUST contain the time the INVITE was received
        - "callStateRegistryText" MUST contain "callBegin"
        - If the incoming call contained an EIDO:
            - "verstat", "sipIdentity" and "sipIdentities" as specified in the related Call Data Component of the EIDO received with the call
        - Else
            - "verstat" MUST contain the verstat parameter value, if provided. MUST also be the same value as the "verstat" element in the Callback Data Component.
            - "sipIdentity" MUST contain the content of the first Identity header, if any
            - "sipIdentities" array MUST contain the contents of all of the Identity headers, if any (even if only one and it was provided above). MUST be identical to the "sipIdentities" element in the Callback Data Component
        - "callBackReference" MUST link to the Callback Data Component instance in the EIDO
        - "locationReference" array MUST contain links to Location Data Component instances found in the EIDO applicable to the call
        - "additionalDataReference" array MUST contain links to Additional Data Component instances found in the EIDO applicable to the call
        - "personReference" array MUST contain links to Person Data Component instance found in the EIDO applicable to the call.

If a call is received on a queue that has a queue state of Inactive, Disabled, Full, ResourceExhausted or Standby, the call SHOULD be rejected with an appropriate response code.[10] The CHFE MAY reject an incoming call for other conditions, but SHOULD immediately

[10] The Standby state is only for a diversion queue and will not receive calls unless it has been set to Active.

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update the Queue State to an appropriate value, and MUST change the Queue State when the condition no longer exists.

When rejecting an incoming call, the CHFE MUST:

- Decrease by one the length of the Queue on which the call was received on and generate a NOTIFY to Queue State subscriber(s)
- Log a CallEndLogEvent
- Log a SessionEndLogEvent
- Log a CallStateChangeLogEvent with state set to callEnd
- Log a SessionStateChangeLogEvent with state set to callEnd
- Notify all authorized EIDO Subscribers of the rejection of the call where the EIDO MUST be composed identically to the EIDO with "callStateRegistryText" containing "callBegin" except now callStateRegistryText MUST be "callEnd".

If no Call Taker is available to answer the incoming call and the call needs to be queued waiting for the next available Call Taker to become available, then the CHFE MAY queue the call to an Interactive Media Response system (IMR) as defined in the Interactive Media Response section of this document. Otherwise, if no announcement is to be played, the CHFE MUST Return a 182 Queued or a 180 Ringing provisional response. The provisional response MUST be retransmitted

When the call is queued, the CHFE MUST:

- Log a CallStateChangeLogEvent with state set to callQueued.
- Log a SessionStateChangeLogEvent with state set to callQueued.
- Notify all EIDO Subscribers of the queuing of the call where the EIDO MUST be composed identically as for the EIDO with callStateRegistryText value is "callBegin" except as follow:
    - callStateRegistryText MUST be "callQueued"

CHFE MUST implement the ability to alert Call Takers of Abandoned Calls and if enabled by local policy, MUST alert appropriate Call Taker(s) of the abandonment of the call. Distribution of the incoming call to one or more Call Takers for answering is not specified in this document. If AgentState is required for proper distribution, the CHFE MUST subscribe to the Presence Server to receive notifications of the appropriate Agent and/or Activity State.

When the call is offered to one or more Call Taker(s), the CHFE MUST notify authorized EIDO Subscribers of the offering of the call, specifying the Call Taker(s) the call is offered to.

If one or more Call Taker(s) need to be alerted of the incoming call, then the CHFE MUST do the following:

- MUST return a 180 Ringing provisional response
- If an announcement needs to be played, the CHFE MUST:
    - Answer the call with a 200 OK.
    - MUST NOT log a CallStateChangeLogEvent with CallAnswered but it MUST log a SessionStateChangeLogEvent with CallAnswered[11].

[11] Operationally, a call is only considered answered when it is answered by a human. The SessionStateChangeLogEvent denotes that the session INVITE has been accepted.

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Deleted: <#>Log the EidoLogEvent.¶

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Commented [58]: How to handle forking when it is forbiden by STA -010?

Commented [MS58R2]: Forking is only forbidden to the NGCS. A PSAP can fork to its workstations. The interface to the workstation is out of scope by consensus. It's not really to hide the topology, it's just to allow any implementation of workstation interface.

Commented [59]: Dan will write some text that says there is one set of signaling per call – you don't send 4 180 Ringing when you ring 4 workstations. CHFE will mask the use of forking from the NGCS.

Deleted: <#>If an announcement needs to be played, the CHFE MUST:¶
Answer the call with a 200 OK. The CHFE MUST NOT offer Early Media in response to inbound call requests.¶

        - Log an AnnouncementStartLogEvent when the announcement starts being transmitted.
        - Log a MediaStartLogEvent for each medium in the announcement (audio, video and/or text).
        - Log an AnnouncementStopLogEvent when the announcement stops being transmitted, all with annoucementType of StandardAnnouncement.
    - Log a CallStateChangeLogEvent with state set to callAlerting, one for each Call Taker the call is offered to.
    - Log a SessionStateChangeLogEvent with state set to callAlerting, one for each Call Taker the call is offered to.
    - Notify all authorized EIDO Subscribers of the alerting of the call, with an EIDO which includes all alerted Call Takers where the EIDO MUST be composed identically as for the EIDO with callStateRegistryText value is "callBegin" except as follow:
        - There MUST be one Agent Call Component for each Agent alerted for the incoming call:
            - Identifier of the Agent Component MUST be the Agent's Agent Identifier
            - agentRoleRegistryText MUST contain at a minimum "Call Taking"
            - agencyReference MUST link to the Agency Data Component in the EIDO
        - For the Call Data Component:
            - callStateRegistryText MUST be "callAlerting"
            - agentReference MUST contain as many links as there are Agent Data Components found in the EIDO

If a CANCEL is received prior to the call being answered by a Call Taker, or in the case the session was answered to play an announcement and a BYE is received during processing prior to being answered by a Call Taker, the CHFE MUST treat it as an Abandoned Call and MUST do the following:

- Decrease by one the length of the Queue on which the call was received on and generate a NOTIFY to Queue State subscriber(s)
- Log a CallStateChangeLogEvent with state set to callCancel.
- Log a SessionStateChangeLogEvent with state set to callCancel if the session was never answered, or callEnd if the session was answered to play an announcement.
- Log a CallEndLogEvent.
- Log a SessionEndLogEvent.
- Notify all EIDO Subscribers of the cancelling of the call where the EIDO MUST be composed identically as for the most recent EIDO with callStateRegistryText value is "callBegin" or "callAlerting" except as follow:
    - callStateRegistryText MUST be "callCancel".

The CHFE MUST implement the ability to answer a call and present it to a Call Taker (a feature sometimes called automatic answer). The CHFE MAY implement the ability to play a greeting if policy enables answering a call on behalf of a Call Taker and policy specifies if a greeting is to be played and if so which greeting. If the call is auto answered, the CHFE MUST implement the ability to notify the Call Taker when answered. The CHFE MUST allow the Call Taker to interrupt an auto-greeting. The greeting MUST be in the form of all media accepted for the call (audio,

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Deleted: When the call is queued, the CHFE MUST:¶
Log a CallStateChangeLogEvent with state set to callQueued.¶
Log a SessionStateChangeLogEvent with state set to callQueued.¶
Notify all EIDO Subscribers of the queuing of the call.¶
If a CANCEL is received prior to the call being answered by a Call Taker, or in the case the session was answered to play an announcement and a BYE is received during processing prior to being answered by a Call Taker, the CHFE MUST treat it as an Abandoned Call and MUST do the following:¶
Log a CallStateChangeLogEvent with state set to callCancel.¶
Log a SessionStateChangeLogEvent with state set to callCancel if the session was never answered, or callEnd if the session was answered to play an announcement.¶
Log a CallEndLogEvent.¶
Log a SessionEndLogEvent.¶
Notify all EIDO Subscribers of the cancelling of the call.¶
CHFE MUST implement the ability to alert Call Takers of Abandoned Calls and if enabled by local policy, MUST alert appropriate Call Taker(s) of the abandonment of the call.¶

Commented [60]: TODO: Write some text in the intro of the document to explain what we mean by implement, and MUST implement and MAY deploy.

video and/or text). The CHFE MUST have the ability to allow a Call Taker to record their own greeting to be played, if permitted by policy.

When a call is answered by a Call Taker, the CHFE MUST do the following:

- Decrease by one the length of the Queue on which the call was received on and generate a NOTIFY to Queue State subscriber(s)
- Log a CallStateChangeLogEvent with the state set to callAnswered.
- Log a SessionStateChangeLogEvent with the state set to callAnswered (unless it was already logged for the session).
- Log a MediaStartLogEvent for each media negotiated (audio, video and/or text) (if not already logged for the session).
- If a greeting is played the CHFE MUST log an AnnouncementStartLogEvent when the announcement starts being transmitted and the CHFE MUST log an AnnouncementStopLogEvent when the announcement stops being transmitted, all with announcementType of AutoAnswerGreeting.
- Notify all EIDO Subscribers of the answering of the call where the EIDO MUST be composed identically as for the most recent EIDO with callStateRegistryText value is "callBegin" or "callAlerting" except as follow:
    - callStateRegistryText MUST be "callAnswered",
    - MUST contain only a single Agent Data Component with the agentId of the Call Taker that answered the call
        - All other Call Takers that were offered the call MUST be removed.

Upon answering a call originating from a Bridge (that is, where the Contact header of the incoming call has a parameter of "isfocus"), the CHFE MUST subscribe to the Bridge specifying the "conference" event package per *A Session Initiation Protocol (SIP) Event Package for Conference State,* RFC 4575 [46]. The CHFE MUST implement out of dialog subscriptions per the *Clarifications for the Use of REFER with RFC 6665,* RFC 7647 [47] section "Dialog re-use is prohibited". All EIDOs sent as the result of being invited to a bridged call MUST specify all of the parties on the call.

4.8.3.2 Emergency Calls Originating from External Switching Systems

The CHFE MUST be able to receive Emergency Calls from External Switching Systems, for example, on a queue dedicated to a telephone number advertised for such a purpose. All calls received on such a queue MUST be assigned a Call Priority Identifier of High or Highest as specified above, no different than if they were received from the ESInet. This includes the CHFE assigning Call and Incident Tracking Identifiers. Because such calls will not have a location upon reception, the CHFE MUST assign a location, either a default location, or optionally, a more appropriate location (from a reverse telephone number database or from previous calls received from the same calling number).[12]

4.8.3.3 Incoming Non-Interactive Calls

The CHFE MUST be able to process Non-Interactive Calls, also called data-only Emergency Calls, as specified in the Non-interactive calls section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

Upon reception of a MESSAGE[13], the CHFE MUST do the following:

[12] Handling of location in this section is not meant to apply to mental health crisis services, such as 988 calls.
[13] CHFE must always log the CallSignalingMessageLogEvent in addition to those listed here.

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Commented [61]: TODO: Need text in i3 that requires Bridge to implement out of dialog subscriptions.

Commented [MLS61R2]: Per Guy, this is not settled in i3. We can't write text that requires it until it is. Defer until then. Use case: whether request to remove party from conference goes in-dialog or out-of-dialog.

Commented [62]: If we allow both in-dialog and out-of-dialog subscriptions, how does the CHFE know which to use with a particular bridge?

Commented [63]: i3 could consider having the client try out-of-dialog first, and if error, try in-dialog. Or vice-versa.

Commented [64]: We need to determine how this is exactly done. EIDOs can specify Agents (once we specify how Agent information is provided to a Bridge).

Commented [MS64R2]: Dan says specify this in i3.

Commented [65]: ToDo: The Bridge would only have the queue URI initially, which doesn't identify the particpant(s). There is an RFC that says after the call is established, you can get the user info (AOR) from the conference package. Associated AORs. Dan, take a look at RFC 6502 and submit modified text that does what we want.

Commented [MLS65R2]: Proposed in i3: for anonymized participants, use the display name in the uri. For others, use CNAME. Dependent upon the XCON contribution from Brian in i3. Dan will write this text after the XCON text is settled.

Commented [MLS65R3]: Still waiting on modified XCON from Brian.

Commented [66]: Maybe we need to add to the call information component a "conference participants" component. Dan will contribute text with a proposed solution to the EIDO issue.

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- Log a CallStartLogEvent. The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a CallEndLogEvent.
- If the MESSAGE request is refused:
    - Return an appropriate final response
- Else if accepted
    - Return a 200 OK regardless of Agent availability.
    - Alert one or more Agents of non-interactive call reception.
    - Notify authorized EIDO Subscribers of the reception of the Non-Interactive Call where the EIDOs are populated as follow:
        - Incident Identifier: MUST be the identifier for the call
        - Agency Data Components:
            - There MUST be one Agency Data Components specifying the Agency receiving the INVITE
        - Location Data Components:
            - There MUST be as many Location Data Components as there are locations in the INVITE
            - locationTypeDescriptionRegistryText MUST be "Caller"
        - Additional Data Components:
            - There MUST be as many Additional Data Components as there are in the MESSAGE, including the alarm and other data found in the MESSAGE body
        - Callback Data Component:
            - This is populated as follow:
            - If one or more P-Asserted-Identify headers are found in the INVITE, then there MUST be one callBackInformationUri per header else
            - MUST be only one callBackInformationUri containing the From header URI
            - deviceContactHeader MUST contain the Contact URI found in the INVITE
            - "verstat" MUST contain the verstat parameter value, if provided
            - sipIdentities array MUST contain the contents of all of the Identity headers found in the INVITE, if present
        - Call Data Component:
            - Contains the details of the received INVITE:
                - Identifier of this Data Component MUST be the Call Identifier assigned to the call
                - "standardPrimaryCallType" MUST be set to an appropriate value found in the IANA Emergency "LogEvent CallTypes" registry
                    - For an emergency call (such as when the Request URI of the INVITE is a Service URN) value MUST be "emergency"
                - "queueIdentifier" MUST be the SIP URI on which the call was received
                - "direction" MUST be set to "incoming"
                - "callStartTimestamp" MUST contain the time the MESSAGE was received

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Deleted: <#>Notify authorized EIDO Subscribers of the reception of the new non-interactive call. ¶
Log the EidoLogEvent.¶

Commented [MS67]: EIDO Call Data Component MUST contain an Additional Data Component that contains the content of the SIP MESSAGE extracted from the CAP.

Commented [MS67R2]: Dan: update this to include CAP MESSAGE content in Additional Data.

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- "callStateRegistryText" MUST contain "callBegin"
- "callBackReference" MUST link to the Callback Data Component instance in the EIDO
- "locationReference" array MUST contain links to Location Data Component instances found in the EIDO applicable to the call
- "additionalDataReference" array MUST contain links to Additional Data Component instances found in the EIDO applicable to the call

How a Non-interactive call is "transferred" to another Agency will be the subject of future work.

4.8.3.4 Outbound Calls

4.8.3.4.1 Placing an Outbound Call

4.8.3.4.1.1 Invoking Privacy

The CHFE MUST be able to request that the identity of the calling PSAP is not to be divulged to the called party when making a Callback Call, a Follow-Up Call or a Non-Emergency Call via OCIF or ESS. Privacy is signaled by setting the From header to the URI "sip:anonymous@anonymous.invalid" and specifying a Privacy header with the value "id". The CHFE MUST also be able to override privacy when the facilities used to make the call have privacy enabled by default by including a Privacy header with value set to "none".

4.8.3.4.1.2 Callback Calls

The CHFE MUST be able to perform Callbacks of emergency calls that have ended, within a configurable amount of time of its reception. This configurable timer is the CHFE Callback Period Timer. The PSAP SHOULD request the timer value from its NGCS operator. The default timer value per the *Public Safety Answering Point (PSAP) Callback*, RFC 7090 [43] is 30 minutes. The CHFE MUST NOT originate a Callback request if the time that has elapsed since the reception of the emergency 9-1-1 call being called back is greater than the configured callback period value. Callbacks MUST always be sent to the NGCS Outbound Call InterFace (OCIF). If the emergency 9-1-1 call being called back was received with both a SIP and a telephone number URI, the SIP URI MUST be used first (if that callback attempt fails, then the telephone number URI is used).

To perform a Callback, the CHFE MUST:

- Craft and send to the OCIF an INVITE request with the following:
  - Request URI specifying the Callback URI
  - From header in one of the two following forms:
    - If privacy is not necessary, the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes (this is necessary in order for the NGCS to properly assert the identity of the PSAP)
    - If privacy is necessary, From header MUST specify URI sip:anonymous@anonymous.invalid
  - Privacy header with one of the two following values:
    - If privacy is not necessary:
      - If facilities used for the call are not provisioned with privacy by default, the CHFE SHOULD include a Privacy header with value "none"

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Deleted: <#>Else return an appropriate final response.¶

Deleted: <#>Incoming Non-Interactive Calls¶

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**Deleted: <#>The CHFE MUST be able to process Non-Interactive Calls, also called data-only Emergency Calls, as specified in the Non-interactive calls section of NENA-STA-010 [7]. ¶
Upon reception of a MESSAGE[14], the CHFE MUST do the following:¶
Log a CallStartLogEvent. The standardPrimaryCallType, "to", and "from" elements MUST be populated.¶
Log a CallEndLogEvent.¶
If the MESSAGE request is accepted:¶
Return a 200 OK regardless of Agent availability.¶
Notify authorized EIDO Subscribers of the reception of the new non-interactive call. ¶
Log the EidoLogEvent.¶
Alert one or more Agents of non-interactive call reception.¶
Notify authorized EIDO Subscribers of the alerting of Agents.¶
Else return an appropriate final response.¶
How a Non-interactive call is "transferred" to another Agency will be the subject of future work.¶**

Commented [68]: In Answer All Calls at a Conference-Aware UA implementation, how does the PSAP place the outgoing call? Dan says put the CA-UA in the path just like on an incoming emergency call, Brian and Terry say that's a security issue because the called party will see it as a call from the CA-UA, not a call from the agent. This must be solved in i3, and we follow those specs. Add this to the email I'll send to i3 per the comment up in the Bridge Interface section.

Commented [MS68R2]: TODO: follow up w/ i3. Has this all been solved?

Commented [MLS69]: Discuss this:
When transferring a callback call to another agency, the locationReference may no longer return the latest location. Per 2024 JCM discussion, we could allow CHFE to provide location by value in the EIDO within the call to the other agency - an exception to ou ... [2]

Commented [MLS69R2]: Agreed to propose text in the Transferring Calls section that allows you to ... [1]

Commented [MS69R3]: TODO: Dan will propose text.

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        - If the facilities are provisioned with privacy by default, the CHFE MUST include a Privacy header with value "none"
      - If privacy is necessary, the CHFE MUST include a Privacy header with the value "id"
    - A first P-Asserted-Identity header in the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes (this is necessary in order for the NGCS to properly assert the identity of the PSAP)
    - A second P-Asserted-Identity specifying the identity of the agent originating the callback, in the form "sip: "Agent name" <Agent Identifier>" (display name optional)
    - An Identity header that contains a PASSporT is defined per *Multiple Language Content Type,* RFC 8255 [44] consisting of the calling TN signed using a certificate traceable to the PCA associated with the agent or agency identified in the second P-Asserted-Identity header (see above). Specifically:
    - the PASSporT header information consisting of:
      - "typ" = "passport"
      - "alg" = "ES256",
      - "x5u" = the URI associated with a certificate traceable to the PCA associated with the agent or agency identified in the second P-Asserted-Identity header
    - PASSporT payload information consisting of:
      - "dest" claim = callback TN from the To header/Request URI line
      - "iat" (Issued At) claim = the date and time of issuance of the token/origination of the call
      - "orig" claim = PSAP calling TN from P-Asserted-Identity header (see #7 above)
    - Priority header value of "psap-callback" as per RFC 7090 [43]
    - Resource-Priority header value of "esnet.0" as specified in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**
    - Route header value set to the OCIF's URI and "lr" parameter
    - A Call-Info header containing a new Call Identifier
    - A Call-Info header containing the Incident Tracking Identifier for the Incident to which the call applies
    - An SDP Offer which includes all supported media choices, subject to operational considerations, which includes the media and codecs negotiated for the emergency 9-1-1 call as preferred (top-most of the Offer)
- Log a CallStartLogEvent where standardPrimaryCallType is set to "callback". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a SessionStartLogEvent where standardPrimaryCallType is set to "callback". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a CallStateChangeLogEvent with state set to callBegin
- Log a SessionStateChangeLogEvent with state set to callBegin
- Notify authorized EIDO Subscribers of the origination of callback where the EIDOs are populated as follow:

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Commented [70]: This document needs to add this to the registry

Commented [MS70R2]: TODO: add "callback" to the Call Type registry.

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- Incident Identifier: MUST be the identifier associated to the Incident to which call getting called back is related
- Agency Data Components:
    - There MUST be one Agency Data Components specifying the Agency placing the callback
- There MUST be one Agent Call Component for the Agent placing the Callback:
    - Identifier of the Agent Component MUST be the Agent's Agent Identifier
    - agentRoleRegistryText MUST contain at a minimum "Call Taking"
    - agencyReference MUST link to the Agency Data Component in the EIDO
- If the call being called back had locations associated with it, there MUST be as many Location Data Components as there was in the EIDO related to that call
    - In addition, if any of the locations were provided by reference, and valid locations were obtained from those references, then for each reference there must be a Location Data Component where:
        - "locationByValue" contains the last location obtained for that reference
        - "locationDereferencedFromReference" contains the reference of the Location Data Component which contains the reference used to obtain this location
        - "locationTypeDescriptionRegistryText" containing the same value as the Location Data Component which contains the reference used to obtain this location
- Additional Data Components:
    - There MUST be as many Additional Data Components as there was for the call being called back, except for Additional Data related to the call itself (such as Service or Provider Info)
- Callback Data Component:
    - This is populated identically to the EIDO associated to the call being called back
- Person Data Component:
    - This is populated identically to the EIDO associated to the call being called back except for "callIdentifier" which MUST link to the Call Data Component instance in this EIDO
    - In addition:
        - "locationReference" MUST link to the Location Data Component instances in the EIDO
        - "additionalDataReference" MUST link to the Additional Data Component instances in the EIDO
- Call Data Component:
    - Contains the details of the Callback INVITE:
        - Identifier of this Data Component MUST be a new Call Identifier assigned to the Callback
        - "standardPrimaryCallType" MUST be "callback"
        - "direction" MUST be set to "outgoing"

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- "callStartTimestamp" MUST contain the time the INVITE was sent
- "callStateRegistryText" MUST contain "callBegin"
- "personReference" array MUST contain links to Person Data Component instance found in the EIDO applicable to the call.

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If an error is received in response to the INVITE of the SIP URI and there is a telephone URI, the CHFE MUST retry the Callback with the telephone URI. If the Calllback attempt(s) fail(s), the CHFE SHOULD retry the call as a Follow-Up call (see below).

4.8.3.4.1.3 Follow-Up Calls

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The CHFE MUST be able to return a call to a previous 9-1-1 caller after the CHFE Callback Period Timer has elapsed and while the Incident is active. These calls are defined as Follow-Up Calls. Follow-Up Calls MUST always be sent to the NGCS Outbound Call Interface Function (OCIF). If the emergency 9-1-1 call being followed-up was received with both a SIP and a TEL callback URI, the SIP URI MUST ALWAYS be used first (if the follow-up call attempt fails, then the TEL URI is used).

To perform a Follow-Up call, the CHFE MUST:

- Craft and send to the OCIF an INVITE request with the following:
  - Request URI specifying the callback number
  - From header in one of the two following forms:
    - If privacy is not necessary, the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes (this is necessary in order for the NGCS to properly assert the identity of the PSAP)
    - If privacy is necessary, From header MUST specify URI sip:anonymous@anonymous.invalid
  - Privacy header with one of the two following values:
    - If privacy is not necessary:
      - If facilities used for the call are not provisioned with privacy by default, the CHFE SHOULD include a Privacy header with value "none"
      - If the facilites are provisioned with privacy by default, the CHFE MUST include a Privacy header with value "none"
    - If privacy is necessary, the CHFE MUST include a Privacy header with the value "id"
  - A first P-Asserted-Identity header in the form "sip:TN@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes (this is necessary in order for the NGCS to properly assert the identity of the PSAP)
  - A second P-Asserted-Identity specifying the identity of the agent originating the callback, in the form "sip: "Agent name" <Agent Identifier>" (display name optional)
  - An Identity header that contains a PASSporT (RFC 8225 [21]) consisting of the calling TN signed using a certificate traceable to the PCA associated with the agent or agency identified in the second P-Asserted-Identity header (see above). Specifically:
  - the PASSporT header information consisting of:
    - "typ" = "passport"

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        - "alg" = "ES256",
        - "x5u" = the URI associated with a certificate traceable to the PCA associated with the agent or agency identified in the second P-Asserted-Identity header
    - PASSporT payload information consisting of:
        - "dest" claim = callback TN from the To header/Request URI line
        - "iat" (Issued At) claim = the date and time of issuance of the token/origination of the call
        - "orig" claim = PSAP calling TN from P-Asserted-Identity header (see #7 above)
    - Resource-Priority header value of "esnet.0" as specified in NENA-STA-010 **Error! Reference source not found.**
    - Route header value set to the OCIF's URI and "lr" parameter
    - Call-Info header specifying the same Incident Tracking Identifiers as the emergency 9-1-1 call being followed up
    - Call-Info header specifying a new Call Identifier
    - An SDP Offer which includes all supported media choices, subject to operational considerations, which includes the media and codecs negotiated for the emergency 9-1-1 call as preferred (top-most of the Offer)
- Log a CallStartLogEvent where standardPrimaryCallType is set to "followUp". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a SessionStartLogEvent where standardPrimaryCallType is set to "followUp". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a CallStateChangeLogEvent with state set to callBegin
- Log a SessionStateChangeLogEvent with state set to callBegin
- Notify authorized EIDO Subscribers of the origination of the Follow-Up call where the EIDOs are populated as follow:
    - Incident Identifier: MUST be the identifier associated to the Incident to which call getting followed-up is related
    - Agency Data Components:
        - There MUST be one Agency Data Components specifying the Agency placing the Follow-Up call
    - There MUST be one Agent Call Component for the Agent placing the Follow-Up call:
        - Identifier of the Agent Component MUST be the Agent's Agent Identifier
        - agentRoleRegistryText MUST contain at a minimum "Call Taking"
        - agencyReference MUST link to the Agency Data Component in the EIDO
    - If the call being followed-up had locations associated with it, there MUST be as many Location Data Components as there was in the EIDO related to that call
        - In addition, if any of the locations were provided by reference, and valid locations were obtained from those references, then for each reference there must be a Location Data Component where:
            - "locationByValue" contains the last location obtained for that reference

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            - "locationDereferencedFromReference" contains the reference of the Location Data Component which contains the reference used to obtain this location
            - "locationTypeDescriptionRegistryText" containing the same value as the Location Data Component which contains the reference used to obtain this location

Commented [DM73]: Should we create a lastKnownLocation type?

    - Additional Data Components:
        - There MUST be as many Additional Data Components as there was for the call being followed-up, except for Additional Data related to the call itself (such as Service or Provider Info)
    - Callback Data Component:
        - This is populated identically to the EIDO associated to the call being followed-up
    - Person Data Component:
            - This is populated identically to the EIDO associated to the call being followed-up except for "callIdentifier" which MUST link to the Call Data Component instance in this EIDO
            - In addition:
                - "locationReference" MUST link to the Location Data Component instances in the EIDO
                - "additionalDataReference" MUST link to the Additional Data Component instances in the EIDO
    - Call Data Component:
        - Contains the details of the Follow-Up call INVITE:
            - Identifier of this Data Component MUST be a new Call Identifier assigned to the Follow-Up call
            - "standardPrimaryCallType" MUST be "followUp"
            - "direction" MUST be set to "outgoing"
            - "callStartTimestamp" MUST contain the time the INVITE was sent
            - "callStateRegistryText" MUST contain "callBegin"
- "personReference" array MUST contain links to Person Data Component instance found in the EIDO applicable to the call.

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If an error is received in response to the INVITE of the SIP URI and there is a telephone URI, the CHFE MUST retry the Callback with the telephone URI.

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4.8.3.4.1.4 Placing a 9-1-1 Call From the CHFE

A CHFE user may have need to call 9-1-1 while logged into the CHFE and may not be able to access another device to make the call due to the nature of the emergency (active shooter, must abandon) or the user's location. The CHFE user may be in the PSAP or remotely connected. The user expects the 9-1-1 call to be processed as normal, and the CHFE MUST ensure that the 9-1-1 call conforms to existing regulations for such calls. If the CHFE implements the ability to place the call through the NGCS, the CHFE MUST follow the specifications in NENA-STA-010 **Error! Reference source not found.** for such calls.

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To perform a 9-1-1 call, the CHFE MUST:

- Craft and send INVITE request with the following:
    - Request URI set to "tel:+1911"

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- From header set to the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes
  - Privacy header set to "none"
  - A first P-Asserted-Identity header in the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes (this is necessary in order for the NGCS to properly assert the identity of the PSAP)
  - A second P-Asserted-Identity specifying the identity of the agent originating the callback, in the form "sip: "Agent name" <Agent Identifier>" (display name optional)
  - Priority header value of "emergency"
  - An SDP Offer which includes all supported media choices
- Log a CallStartLogEvent where standardPrimaryCallType is set to "911". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a SessionStartLogEvent where standardPrimaryCallType is set to "911". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a CallStateChangeLogEvent with state set to callBegin
- Log a SessionStateChangeLogEvent with state set to callBegin

There is no EIDO sent for a 9-1-1 call initiated by a CHFE.

4.8.3.4.1.5 Originating an Emergency Support Call to Another Agency on the ESInet

An Emergency Support Call is a call from an Agency to another Agency associated with an Incident. The CHFE MUST be able to originate such calls to destinations served by the ESInet. An example of such call is when a Dispatcher at an EMS Agency needs to contact a Law Enforcement Agency for paramedics involved in an active Incident. To do so the CHFE MUST:

- Send an INVITE to the OCIF with the following:
  - Request URI specifying a service URN in the appropriate responder tree, for example, "urn:emergency:service:responder.police"
  - A From header specifying the identity of the agent originating the call, in the form "sip: "Agent name" <Agent Identifier>" (display name optional)
  - Two P-Asserted-Identity headers, order is an implementation detail:
    - A P-Asserted-Identity header specifying the identity of the Agent originating the call, in the form "sip: "Agent name" <Agent Identifier>" (display name optional)
    - A P-Asserted-Identity header specifying the identity of the Agency originating the call, in the form "sip: "Agency name" <Agency Identifier>" (display name optional)
  - An Identity header that contains a PASSporT RFC 8255 [44] consisting of the first P-Asserted-Identity header signed using a certificate traceable to the PCA associated with the agent or agency identified in that header. Specifically:
  - the PASSporT header information consisting of:
    - "typ" = "passport"
    - "alg" = "ES256",
    - "x5u" = the URI associated with a certificate traceable to the PCA associated with the agent or agency identified in the second P-Asserted-Identity header

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- PASSporT payload information consisting of:
  - "dest" claim = SIP URI found in the To header
  - "iat" (Issued At) claim = the date and time of issuance of the token/origination of the call
  - "orig" claim = SIP URI from the first P-Asserted-Identity header
- An appropriate Resource-Priority header value in the "esnet" namespace specified in NENA-STA-010 **Error! Reference source not found.**
- A first Route header value set to the OCIF's URI and "lr" parameter
- A second Route header value set to the target's URI and "lr" parameter
- Call-Info header specifying the Incident Tracking Identifier
- Call-Info header specifying the new Call Identifier
- A Call-Info header with purpose "emergency-eido" designating either an EIDO by reference using a HTTPS URI or an EIDO by value found in the body of the INVITE

- Log a CallStartLogEvent where standardPrimaryCallType is set to "emergencySupport". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a SessionStartLogEvent where standardPrimaryCallType is set to "emergencySupport". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a CallStateChangeLogEvent with state set to callBegin
- Log a SessionStateChangeLogEvent with state set to callBegin
- Notify authorized EIDO Subscribers of the origination of the Emergency Support call where the EIDOs are populated as follow:
  - Incident Identifier: MUST be the identifier associated to the Incident to which the Emergency Support Call is related
  - Agency Data Components:
    - There MUST be one Agency Data Components specifying the Agency placing the Emergency Support Call
  - There MUST be one Agent Call Component for the Agent placing the Emergency Support Call:
    - Identifier of the Agent Component MUST be the Agent's Agent Identifier
    - agentRoleRegistryText MUST contain all of the Agent's roles
    - agencyReference MUST link to the Agency Data Component in the EIDO
  - If the Incident has location(s) associated to it, there MUST be equivalent Location Data Component(s)
  - If the Incident has Additional Data associated to it, there MUST be equivalent Additional Data Component(s)
  - If the Incident has person(s) associated to it, and these person(s) has(have) a callback URI, there MUST be equivalent Callback Data Component(s)
  - If the Incident has person(s) associated to it, there MUST be equivalent Person Data Component(s):
    - If applicable, "callBackReference" MUST link to the Callback Data Component instance in the EIDO
  - Call Data Component:
    - Contains the details of the Emergency Support Call INVITE:

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- Identifier of this Data Component MUST be a new Call Identifier assigned to the Emergency Support Call
- "standardPrimaryCallType" MUST be "emergencySupport"
- "direction" MUST be set to "outgoing"
- "callStartTimestamp" MUST contain the time the INVITE was sent
- "callStateRegistryText" MUST contain "callBegin"

  - There MUST be an Incident Data Component containing the Incident details, which MUST include the following:
    - Identifier MUST be the identifier associated to the Incident to which the Emergency Support Call is related
    - "incidentTypeCommonRegistryText" which MUST contain a value from the Incident Type registry
    - "callIdentifier" MUST link to the Call Data Component instance in the EIDO
    - "agentReference" MUST link to the Agent Data Component instance in the EIDO
    - "locationReference" MUST link to the Location Data Component(s) instance in the EIDO (if any)
    - "personReference" MUST link to the Person Data Component(s) instance in the EIDO (if any)

4.8.3.4.1.6 Originating a Non-Emergency Call via OCIF

The CHFE MUST be able to originate non-Emergency Calls to any destination via the OCIF where PSAP and NGCS policy allows. To do so the CHFE MUST:

- Craft and send an INVITE request with the following:
  - Request URI specifying the destination URI
  - Two P-Asserted-Identity headers, order is an implementation detail:
    - A P-Asserted-Identity specifying the identity of the agent originating the callback, in the form "sip: "Agent name" <Agent Identifier>" (display name optional)
    - If the target of the call is on the PSTN:
      - A P-Asserted-Identity header in the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for outbound call purposes (this is necessary in order for the NGCS to properly assert the identity of the PSAP)
    - Else:
      - A P-Asserted-Identity header specifying the identity of the Agency originating the call, in the form "sip: "Agency name" <Agency Identifier>" (display name optional)
  - If one of the P-Asserted-Identity headers contains a URI in the form "sip:TN@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes:
    - The From header MUST be specified in one of the following two forms:

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Commented [79]: Guy says we should define this type of call. Mike says we should have an ActivityState for "on a non-emergency call".

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            - If privacy is not necessary, the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes (this is necessary in order for the NGCS to properly assert the identity of the PSAP)
            - If privacy is necessary, From header MUST specify URI "sip:anonymous@anonymous.invalid"
        - Privacy header with one of the two following values:
            - If privacy is not necessary:
                - If facilities used for the call are not provisioned with privacy by default, the CHFE SHOULD include a Privacy header with value "none"
                - If facilities used for the call are provisioned with privacy by default the CHFE MUST include a Privacy header with value "none"
            - If privacy is necessary, the CHFE MUST include a Privacy header with value "id"
    - If none of the P-Asserted-Identity headers contains a URI in the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying one of the PSAP's URIs provisioned for callback purposes, then the From header can be either of the URIs specified in the P-Asserted-Identity headers
    - An Identity header that contains a PASSporT RFC 8225 [21] consisting of the From header signed using a certificate traceable to the PCA associated with the agent or agency identified in that header. Specifically:
    - the PASSporT header information consisting of:
        - "typ" = "passport"
        - "alg" = "ES256",
        - "x5u" = the URI associated with a certificate traceable to the PCA associated with the agent or agency identified in the second P-Asserted-Identity header
    - PASSporT payload information consisting of:
        - "dest" claim = SIP URI found in the To header
        - "iat" (Issued At) claim = the date and time of issuance of the token/origination of the call
        - "orig" claim = SIP URI from the From header
    - a Route header with the OCIF's URI and "lr" parameter
    - a Call-Info header specifying the new Call Identifier
- Log a CallStartLogEvent where standardPrimaryCallType is set to "nonEmergency". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a SessionStartLogEvent where standardPrimaryCallType is set to "nonEmergency". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a CallStateChangeLogEvent with state set to callBegin
- Log a SessionStateChangeLogEvent with state set to callBegin

4.8.3.4.1.7 Originating a Non-Emergency Call via ESS

The CHFE MUST be able to originate non-Emergency Calls to destinations using the External Switching Systems interface. To do so the CHFE MUST:

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- Craft and send an INVITE request with the following:
  - Request URI specifying the destination URI
  - From header in one of the two following forms:
    - If privacy is not required, the form "sip:<telephone number>@<Agency-Identifier>;user=phone" specifying the PSAP URI appropriate for the facilities being used
    - If privacy is required, From header MUST specify URI "sip:anonymous@anonymous.invalid"
  - Privacy header with one of the two following values:
    - If privacy is not required:
      - If facilities used for the call is not provisioned with privacy by default, the CHFE SHOULD include a Privacy header with value "none"
      - Else the CHFE MUST include a Privacy header with value "none"
    - Else the CHFE MUST include a Privacy header with value "id"
  - A P-Asserted-Identity header in the form "sip: :<telephone number>@<Agency-Identifier>;user=phone" specifying the PSAP URI appropriate for the facilities being used
  - a Call-Info header specifying a new Call Identifier per NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.
- Log a CallStartLogEvent where standardPrimaryCallType is set to "nonEmergency". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a SessionStartLogEvent where standardPrimaryCallType is set to "nonEmergency". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a CallStateChangeLogEvent with state set to callBegin
- Log a SessionStateChangeLogEvent with state set to callBegin

4.8.3.4.1.8 Originating an internal Call

The CHFE MUST be able to originate internal calls to entities within the Agency such as another Agent. How this call is signaled internally is not defined in this document, but the CHFE MUST do the following:

- Log a CallStartLogEvent where standardPrimaryCallType is set to "internal". The standardPrimaryCallType, "to", and "from" elements MUST be populated.
- Log a SessionStartLogEvent where standardPrimaryCallType is set to "internal", "to" and "from" elements MUST be populated.
- Log a CallStateChangeLogEvent with state set to callBegin
- Log a SessionStateChangeLogEvent with state set to callBegin

4.8.3.4.2 Outbound Call Processing

4.8.3.4.2.1 Cancelling an Outgoing Call

If an outgoing call request needs to be cancelled, the CHFE MUST:

- Send an in-dialog CANCEL request
- Log a CallStateChangeLogEvent with state set to callCancel
- Log a SessionStateChangeLogEvent with state set to callCancel
- Log a CallEndLogEvent

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- Log a SessionEndLogEventIf an EIDO was sent out when the outgoing call was initiated, notify all authorized EIDO Subscribers of the cancelling of the call where the EIDO MUST be composed identically as for the most recent EIDO with callStateRegistryText value is "callBegin" except as follow:
  - callStateRegistryText MUST be "callCancelled".

4.8.3.4.2.2 Outgoing Call Provisional Responses

The CHFE MUST support all provisional responses and if a response includes an Early Media answer, the CHFE MUST

- Log a MediaStartLogEvent

4.8.3.4.2.3 Outgoing Call Is Answered

When an outgoing call is answered, the CHFE MUST:

- Log a CallStateChangeLogEvent with the state set to callAnswered
- Log a SessionStateChangeLogEvent with the state set to callAnswered
- If an EIDO was sent out when the outgoing call was initiated, notify authorized EIDO Subscribers of the answering of Follow-Up call where the EIDO MUST be composed identically as for the most recent EIDO with callStateRegistryText value is "callBegin" except as follow:
  - callStateRegistryText MUST be "callAnswered"

4.8.3.5 Active Call Processing

The specifications in this section apply to all active calls, regardless of emergency status or call direction.

4.8.3.5.1 Releasing a Call

When the call is released, regardless of which party released the call, the CHFE MUST do the following:

- If the Agent released the call, send an in-dialog BYE request
- Log a CallEndLogEvent
- Log a SessionEndLogEvent
- Log a CallStateChangeLogEvent with state set to callEnd
- Log a SessionStateChangeLogEvent with state set to callEnd
- Log a MediaEndLogEvent for each medium in use at call end time
- If an EIDO was sent out when the call was received or initiated, notify all authorized EIDO Subscribers of the releasing of the call where the EIDO MUST be composed identically as for the most recent EIDO except as follow:
- callStateRegistryText MUST be "callEnd".

4.8.3.5.2 Changing a Call's Priority Identifier

At any time during the CHFE's processing of a call, an Agent MUST be able to change the call's Priority Identifier, that is, the nature of the event reported by the caller is such that the Call Priority Identifier assigned when the call was received is no longer applicable (for example, a 9-1-1 caller reporting a non-emergency Incident). When the Agent performs such action, the CHFE MUST:

- Assign to the call an updated Call Priority Identifier
- If an EIDO was sent out when the call was received or initiated, notify all authorized EIDO Subscribers of the modification of the Call Priority

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The specifications in this section apply to all active calls, regardless of emergency status or call direction.¶

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Identifier where the EIDO MUST be composed identically as for the most recent EIDO except as follow:

- "callPriorityIdentifier" MUST be set to the new value.

4.8.3.5.3 Adding and Removing Parties to a Call

If a party is to be added to a call (for example in the case of a transfer of an Emergency Call), the CHFE MUST be able to refer the call to a Bridge. The CHFE MUST support the conferencing of all media types, including RTT as specified in the *Long-Term Viability of Protocol Extension Mechanisms*, RFC 9170 [45] and instant messages as specified in section Conference Bridging for MSRP text in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. The CHFE MUST support an NGCS-hosted Bridge and MAY implement a local Bridge. If the CHFE implements a local Bridge, it MUST conform to the Bridging and Transfers section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**.

4.8.3.5.3.1 Creating an Ad Hoc Conference

In the case we need to add a party to call but it does not already include a Conference Focus (i.e. the Contact header of the other party of the call does not specify the parameter isfocus), then the CHFE must create an ad hoc conference, to do so it MUST:

- Send an INVITE to the Conference Factory of the NGCS Bridge to request a conference session with the following:
  - For a call where the Call Priority Identifier is High or Highest, the Request URI must be "urn:service:sos" and Route header with the SIP URI of the Conference Factory
    - If not an emergency call then the Request URI is the Conference Factory URI
  - A From header specifying the identity of the agent originating the call, in the form "sip: "Agent name" <Agent Identifier>" (display name optional)
  - Two P-Asserted-Identity headers, order is an implementation detail:
    - A P-Asserted-Identity header specifying the identity of the Agent originating the call, in the form "sip: "Agent name" <Agent Identifier>" (display name optional)
    - A P-Asserted-Identity header specifying the identity of the Agency originating the call, in the form "sip: "Agency name" <Agency Identifier>" (display name optional)
  - An appropriate Resource-Priority header value in the "esnet" namespace specified in NENA-STA-010 **Error! Reference source not found.**
  - Call-Info header specifying the Incident Tracking Identifier
  - Call-Info header specifying the new Call Identifier
- If the Conference Factory returns a 302 Moved Temporarily, send an INVITE to the newly created Conference URI as specified in the Contact header of the redirect. The INVITE must be formatted identically to the INVITE sent to the Conference Factory

At this point the CHFE has one call leg connected to the Bridge.

4.8.3.5.3.2 Adding a Party to a Conference

To add a party to the call, the CHFE MUST:

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- If the call does not have an Conference Focus, the CHFE MUST follow the instruction in section Creating an Ad Hoc Conference to create conference on the Bridge
    - If successful, the CHFE sends an in-dialog REFER to the Focus
        - Call-Info headers specifying the Call Identifier and Incident Tracking Identifier
        - A Referred-By header specifying the URI of the referring PSAP
        - A Refer-To header specifying the Contact URI of the existing call with parameters the SIP Call Id and, To and From tags of the existing call that needs to be moved to the Bridge
- Send an in-dialog REFER to the Bridge with the following:
    - Call-Info headers specifying the Call Identifier and Incident Tracking Identifier
    - A Referred-By header specifying the URI of the referring PSAP
    - A Refer-To header with the following elements:
        - The SIP or TEL URI of the party to be added
        - If the party to be added is the 9-1-1 caller (because the caller released the call prematurely) then the URI MUST specify that the outbound call from the Bridge will be a Callback and thus the URI MUST include an escaped SIP Priority header with value "psap-callback" per RFC 7090 [43], for example (unescaped):

```
Refer-To: sip:8195551212@osp.com?Priority=psap-callback
```

        - If the party being added to an Emergency Call is an Agency:
            - An escaped Call-Info header containing the HTTPS URI of the EIDO containing the details of the call, with a purpose of "emergency-eido" per NENA-STA-010 **Error! Reference source not found.**
            - If applicable, a suitable Service URN (such as the Service URN used to query the ECRF to determine the party being added) as a "serviceurn" parameter to the Refer-To header field
        - For example (unescaped):

```
Refer-To: <sip:Poison-Control@cnty.st.us?Call-
Info=https://NG911PSAP-A.911Authority-
A.net/eido09245673;purpose=emergency-eido>;
serviceurn=urn:emergency:service:responder.fire
```

- Log a CallStateChangeLogEvent with state set to partyAdd[15]
- Log a SessionStateChangeLogEvent with state set to partyAdd
- If the added party is an Agency, log an AdditionalAgencyLogEvent
- When the added party answers the call, and if an EIDO was sent out when the call was received or initiated send an EIDO to authorized EIDO Subscribers of the addition of the party to the call where the EIDO MUST be composed identically as for the most recent EIDO except as follow:
    - "callStateRegistryText" MUST be "partyAdd"
        - "partyAdd" MUST be reported exactly once per addition of a new party to a call, subsequent EIDOs MUST specify

[15] The Bridge is expected to log the outcome of the add party request.

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Commented [85]: In STA-010, say that the Bridge logs CallStateChangeLogEvent with partyAdd containing the leg callid, if we don't already.

Commented [86]: In STA-010, say that the Bridge logs CallStateChangeLogEvent with partyAdd containing the leg callid, if we don't already.

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"callStateRegistryText" of "partyAnswer" unless the new EIDO specifies a different Call State

- if the added party is an Agency, there MUST be an additional Agency Data Component identifying the newly added Agency.

Upon answering a call originating from a Bridge (that is, where the Contact header of the incoming call has a parameter of "isfocus"), the CHFE MUST subscribe to the Bridge specifying the "conference" event package per *A Session Initiation Protocol (SIP) Event Package for Conference State,* RFC 4575 [46]. The CHFE MUST implement out of dialog subscriptions per the *Clarifications for the Use of REFER with RFC 6665,* RFC 7647 [47] section "Dialog re-use is prohibited". All EIDOs sent as the result of being invited to a bridged call MUST specify all of the parties on the call.

4.8.3.5.3.3 Removing a Party from a Conference

To remove a party from the call, or cancel a pending request to add a party who has not yet answered, the CHFE MUST:

- Send an in-dialog REFER to the Bridge with a Refer-To header specifying the SIP or TEL URI of the party to be removed (as discovered from the participant notifications received from the Bridge) with the parameter Method equal to BYE. For example:

```
Refer-To: sip:Poison-Control@cnty.st.us?Method=BYE
```

- Log a CallStateChangeLogEvent with state set to partyRemove
- Log a SessionStateChangeLogEvent with state set to partyRemove
- When the Bridge confirms the removal of the party from the call, and if an EIDO was sent out when the call was received or initiated, notify authorized EIDO Subscribers of the removal where the EIDO MUST be composed identically as for the most recent EIDO except as follow:
  - callStateRegistryText MUST be "partyRemove"
    - "partyRemove" MUST be reported exactly once per removal of a party from a call, subsequent EIDOs MUST specify "callStateRegistryText" of "partyAnswer" unless the new EIDO specifies a different Call State
- if the removed party is an Agency, that Agency's Agency Data Component MUST no longer be included.

4.8.3.5.4 Transferring a Call

The specifications in this section apply to all calls, regardless of emergency status or call direction.

4.8.3.5.4.1 Attended Transfer

An Attended (also known as warm) Transfer is a transfer facilitated by a Bridge where the caller, transfer-from CHFE and transfer-to entity are all conferenced together in order to allow the Agent at the transfer-from PSAP to ensure a proper hand off of the call as specified above. An Attended Transfer begins with the process of adding a party as specified in the Add a Party to the Call section. The call transfer is completed when the transfer-from CHFE drops off the Bridge.

The CHFE MUST implement support for Attended Transfers as specified in the Attended Transfers section of NENA-STA-010 **Error! Reference source not found.** by doing the following:

- Add a party to the call as specified in the Add a Party to a Call section.

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Commented [87]: TODO: Need text in i3 that requires Bridge to implement out of dialog subscriptions.

Commented [MLS87R2]: Per Guy, this is not settled in i3. We can't write text that requires it until it is. Defer until then. Use case: whether request to remove party from conference goes in-dialog or out-of-dialog.

Commented [88]: If we allow both in-dialog and out-of-dialog subscriptions, how does the CHFE know which to use with a particular bridge?

Commented [89]: i3 could consider having the client try out-of-dialog first, and if error, try in-dialog. Or vice-versa.

Commented [90]: We need to determine how this is exactly done. EIDOs can specify Agents (once we specify how Agent information is provided to a Bridge).

Commented [MS90R2]: Dan says specify this in i3.

Commented [91]: ToDo: The Bridge would only have the queue URI initially, which doesn't identify the particpant(s). There is an RFC that says after the call is established, you can get the user info (AOR) from the conference package. Associated AORs. Dan, take a look at RFC 6502 and submit modified text that does what we want.

Commented [MLS91R2]: Proposed in i3: for anonymized participants, use the display name in the uri. For others, use CNAME. Dependent upon the XCON contribution from Brian in i3. Dan will write this text after the XCON text is settled.

Commented [MLS91R3]: Still waiting on modified XCON from Brian.

Commented [92]: Maybe we need to add to the call information component a "conference participants" component. Dan will contribute text with a proposed solution to the EIDO issue.

Commented [93]: For STA-010: if you send a BYE, the Bridge will send BYE, unless the party didn't answer, in which case it can send a CANCEL. To be specified in i3.

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- When requested by the Agent at the transferring PSAP, its CHFE MUST release the call from the Bridge by following the process specified in section Releasing a Call and in addition MUST log a CallTransferLogEvent when it disconnects from the Bridge.
- If the NGCS Bridge method is Ad Hoc, the transfer-to CHFE MUST release the Bridge as specified in the SIP Ad Hoc Flow section of NENA-STA-010 **Error! Reference source not found.**, if it is alone on the Bridge along with the 9-1-1 caller.

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Commented [94]: i3 must solve this. The CHFE doesn't know the bridge method unless using its own NGCS bridge. Brian brought this up on i3 list on 1/26/23.

The transferring CHFE MUST be able to process a request by an Agent to release the call after the REFER has been sent but prior to receiving a call answered notification. In such a case the transferring CHFE MUST NOT log a CallTransferLogEvent because the outcome of the transfer is unknown.

4.8.3.5.4.2 Blind Transfer

A Blind (also known as cold) Transfer is a transfer where there is no conversation between the transferring PSAP and the transfer-to party. Note that some PSAPs may prohibit or restrict the use of blind transfer for Emergency Calls per local policy. If the bridging method is Ad Hoc, the CHFE MUST NOT create a conference prior to performing the Blind Transfer. To perform a Blind Transfer:

- The transferring CHFE MUST send an in-dialog REFER with the following:
  - Request-URI which contains the URI of the Contact header from dialog-forming INVITE
  - Call-Info headers specifying the Call Identifier and the Incident Tracking Identifier
  - A Referred-By header specifying the URI of the referring PSAP
  - A Refer-To header with the following elements:
    - The SIP or TEL URI of the transfer-to party
    - If the transfer-to party is a PSAP:
  a) An escaped Call-Info header containing the HTTPS URI of the EIDO containing the details of the call, with a purpose of "emergency-eido" per NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**[16]
  b) A suitable Service URN (such as the Service URN used to query the ECRF to determine the party being added) as a "serviceurn" parameter to the Refer-To header field

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For example (unescaped):

```
Refer-To: <sip:Poison-Control@cnty.st.us?Call-Info=
https://NG911PSAP-A.911Authority-
A.net/eido09245673;purpose=emergency-eido>;
serviceurn=urn:emergency:service:responder.fire
```

- If the transfer-to party is an Agency, the transferring CHFE MUST log an AdditionalAgencyLogEvent
- If the transfer-to party is a PSAP, that PSAP's CHFE MUST process the incoming call as specified in the Incoming Interactive Calls section
- When the transferring CHFE receives a NOTIFY specifying the transfer-to party has answered the call it MUST release the call by following the

[16] In the case the NGCS Bridging method is Ad Hoc, the REFER goes to the OSP, and there is no guarantee the transferred party is able to propagate the EIDO reference onto the transferred-to PSAP

process specified in the section Releasing a Call above and then MUST log a CallTransferLogEvent.

The transferring CHFE MUST be able to process a request by an Agent to release the call after the REFER has been sent but prior to receiving a call answered notification. In such a case the transferring CHFE MUST NOT log a CallTransferLogEvent because the outcome of the transfer is unknown.

4.8.3.5.4.3 Transfer of a External Switching Systems to an ESInet Target

The CHFE MUST be able to transfer a External Switching System call (regardless of the direction of the ESS call, it may have been received or originated by the CHFE) to a target on the ESInet (another PSAP for example). To do so the CHFE MUST:

- The CHFE MUST follow the instruction in section Creating an Ad Hoc Conference to create conference on the Bridge Send a second INVITE addressed to the Conference URI obtained from the Conference Factory, this time proxying the call leg to the ESS:
  - If the ESS call was received:
    - Same From and P-Asserted-Identity headers
  - Else:
    - From and P-Asserted-Identity headers identical to the Request URI of the call sent to the ESS
- When this call to the Conference Bridge is answered, this is the call leg conveying the ESS call
- From this point follow the process specified in section Attended Transfer

Because the External Switching System does not have a direct interface to the Bridge, the CHFE MUST remain in the call path for the call leg going to the External Switching System. The CHFE MUST NOT propagate any reINVITE, UPDATE or REFER requests to the ESS and instead MUST accept the request and process any requested signaling or media rehoming. An Agent SHOULD be able to rejoin a transferred call at any time as long as the call remains up. The CHFE MUST be able to tear down a transferred call that has effectively ended without any party sending a BYE (for example upon detection of media or signaling path loss).

4.8.3.5.5 Call Hold and Park

The CHFE MUST implement support for putting an active call in a state to allow the Agent to accomplish other tasks, including answering another call. The Agent, or another Agent (depending on the type of suspension), can then retrieve the call and resume conversation with the caller. The CHFE MUST:

- Implement support for an exclusive Hold (only the Agent who put the call on hold can return back to the call)
  - CHFE MUST log a CallStateChangeLogEvent with state set to callHold
  - Notify authorized EIDO Subscribers of the holding of the call where the EIDO is identical to the previously transmitted EIDO except that callStateRegistryText MUST be "callHold"
- Implement support for a non-exclusive Park (any authorized Agent can retrieve the call)
  - CHFE MUST log a CallStateChangeLogEvent with state set to callPark
  - Notify authorized EIDO Subscribers of the parking of the call where the EIDO is identical to the previously transmitted EIDO except that callStateRegistryText MUST be "callPark"

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Commented [95]: The ESS has an emergency queque for receiving calls from an external system. Need a way to dial a specific agent. i3 could specify the agentId is a sip URI and could be used to contact that agent.

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Deleted: <#>Upon successful conference session creation, move the call to the bridge so the two call legs are bridged (the leg to the External Switching System and the leg to the Agent)
For each call leg:
Send an INVITE to the Bridge targeting the conference session with the following:
For a call where the Call Priority Identifier is High or Highest:
the Request URI must be "urn:service:sos" and Route header with the SIP URI of the conference session
A Call-Info header with the Incident Tracking Identifier
A Call-Info header with the Call Identifier

- When a Held or Parked call is retrieved:
  - CHFE MUST log a CallStateChangeLogEvent with state set to callRetrieve
  - Notify authorized EIDO Subscribers of the retrieval of the call where the EIDO is identical to the previously transmitted EIDO except that callStateRegistryText MUST be "callRetrieve"
- Maintain the SDP "sendrcv" attribute state while the call is on Hold or Parked
- Implement support to inform the caller that the call has been suspended using all media in use for the call (such as specified in *Session Initiation Protocol Service Example -- Music on Hold*, RFC 7088 [52]).

4.8.3.5.6Call Handling Functional Element Management Functions

The specifications in this section describe functions of the CHFE that support common monitoring and management capabilities.

4.8.3.5.6.1 Silent Monitoring of a Call Taker

The CHFE MUST permit authorized Agents to silently monitor the calls of another Agent as permitted by policy.

When an authorized Agent initiates Silent Monitoring, the CHFE MUST log a SilentMonitoringStartLogEvent.

If the monitored Agent is on a call or when the monitored Agent answers or places a call that is monitored, the CHFE MUST log the following:

- A SilentlyMonitoredCallStartLogEvent
- A SilentMonitorMediaStartLogEvent for each medium being monitored.
-
- As media are added and/or removed from a silently monitored call, the CHFE MUST log appropriate SilentMonitorMediaStartLogEvent and/or SilentMonitorMediaStopLogEvent.

When the monitored Agent releases a call, the CHFE MUST log the following:

- A SilentMonitorMediaStopLogEvent for each medium being monitored
- A SilentlyMonitoredCallStopLogEvent.
- When an authorized Agent stops Silent Monitoring, the CHFE MUST log a SilentMonitoringStopLogEvent.

4.8.3.5.6.2 Agent Intervene/Barge In

The CHFE MUST permit an authorized Agent to intervene on an active call as permitted by policy. If authorized to do so the CHFE MUST:

- If the call to be intervened is not on a Bridge, the CHFE MUST follow the instructions in section Creating an Ad Hoc Conference to create conference on the Bridge which will be the call leg of the Agent on the call
  - If successful, the CHFE sends an in-dialog REFER to the Focus
    - Call-Info headers specifying the Call Identifier and Incident Tracking Identifier
    - A Referred-By header specifying the URI of the referring PSAP

Commented [96]: i3 will provide text on how to do silent monitoring using the Bridge.
In i3, there is a callType of "silentMonitoring". Dan says it's not necessarily a call – an FE can fork media.

Commented [97]: How is this supposed to be logged? I have strong reservation logging this using a CallStartLogEvent where standardCallType is silentMonitoring as this is not a call, implementation is purely up to vendor. Is the media of a silently monitored Agent logged? It is already recorded…

Commented [MS97R2]: Discuss when Dan is on the call.

Commented [98]: STA-010 needs to detail how media/events get logged by the bridge – different users can have different streams. What if the Bridge is not the SR ... [5]

Commented [MS98R2]: Raise issue with i3.

Commented [99]: We have partyAdd, partyRemove, and callBargeIn callStateChanges. How about addind party ... [4]

Commented [MS99R2]: Raise issue with i3

Commented [100]: If the CHFE or a logger provides silent monitoring without involving a Bridge or a separate call ... [3]

Commented [101]: The Logger must also log this if it provides monitoring. Brian also proposed standardizing ... [6]

Commented [MS101R2]: TODO: add text to logger section if not already there.

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Commented [103]: Need to specify this new LogEvent in the LogEvent section

Commented [MS103R2]: TODO: add event if not done

Commented [104]: Need to specify this new LogEvent in the LogEvent section

Commented [105]: Need to specify this new LogEvent in the LogEvent section

Commented [106]: Need to specify this new LogEvent in the LogEvent section

Commented [107]: Need to specify this new LogEvent in the LogEvent section

Commented [108]: The Logger must also log this if it provides monitoring. Brian also proposed standardizing ... [7]

Commented [109]: Need to specify this new LogEvent in the LogEvent section

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Commented [110]: Add text about policy in a section that applies to all instances of "Authorized Agent".

Commented [MS110R2]: TODO: study this and propose a solution. ... [8]

Commented [111]: This needs to be specified in STA-010 Bridging FE section. For ad hoc it is simple, CHFE simp ... [9]

Deleted: If the call to be intervened is not on a Bridge, to do so the CHFE MUST

Deleted: Send an INVITE to the Conference Factory of the Bridge to request a confer ... [10]

        - A Refer-To header specifying the Contact URI of the existing call with parameters the SIP Call Id and, To and From tags of the existing call that needs to be moved to the Bridge
- For the intervening Agent’s call leg, the CHFE sends an INVITE to the Conference URI with the following:
    - For a call where the Call Priority Identifier is High or Highest, the Request URI must be “urn:service:sos” and Route header with the SIP URI of the Conference Factory
        - If not an emergency call then the Request URI is the Conference Factory URI
    - A From header specifying the identity of the intervening Agent, in the form “sip: “Agent name” <Agent Identifier>” (display name optional)
    - Two P-Asserted-Identity headers, order is an implementation detail:
        - A P-Asserted-Identity header specifying the identity of the Agent originating the call, in the form “sip: “Agent name” <Agent Identifier>” (display name optional)
        - A P-Asserted-Identity header specifying the identity of the Agency originating the call, in the form “sip: “Agency name” <Agency Identifier>” (display name optional)
    - An appropriate Resource-Priority header value in the “esnet” namespace specified in NENA-STA-010 **Error! Reference source not found.**
    - Call-Info header specifying the Incident Tracking Identifier
    - Call-Info header specifying the new Call Identifier
-

Once the intervening Agent has been conferenced in, the CHFE MUST:

- Log a CallStateChangeLogEvent with state set to callBargeIn
- Log a SessionStateChangeLogEvent with state set to callBargeIn
- Send an EIDO to authorized EIDO Subscribers of the addition of the party to the call where the EIDO is identical to the previously transmitted EIDO except that callStateRegistryText MUST be “callBargeIn” and in addition shows the intervening Agent.

Once the intervening Agent has been conferenced in, the CHFE MUST process the call no differently than any other conference call.

4.8.3.6 Media logging

The CHFE MUST implement support to log media as specified in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. If the CHFE is acting as the Session Recording Client (SRC) it must log media in the following manner:

- All calls with Call Priority Identifier of High or Highest MUST be recorded
- Queue recording: the CHFE MUST support the logging of the media of all calls received on any designated Queue based on local capture policy. For example, a non-emergency queue may be designated as “not recorded”. This includes all early media and media exchanged when a call is parked/put on hold
- Agent recording: the CHFE MUST support the logging of the media of all calls received or placed by a particular Agent, that is media received from

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Deleted: <#>Upon successful conference session creation move the call to be intervened to the bridge so the two call legs are bridged (the leg to the other party and the leg to the Agent)¶
For each Agent (Agent on the call and intervening Agent):¶
Send an INVITE to the Bridge targeting the conference session with the following:¶
For a call where the Call Priority Identifier is High or Highest, the Request URI must be “urn:service:sos” and Route header with the SIP URI of the conference session¶
Place the URI of the Agent in From:¶
A Call-Info header with the Incident Tracking Identifier per NENA-STA-010 [7]¶
A Call-Info header with the Call Identifier per NENA-STA-010 [7]¶
If the other party of the intervened call is on the ESInet:¶
Send a REFER with Replaces to the party where the Refer-To header specifies the conference session¶
If the other party of the intervened call is on the External Switching System interface:¶
For the call leg to the party on the ESS, send an INVITE to the Bridge targeting the conference session with the following:¶
For a call where the Call Priority Identifier is High or Highest, the Request URI must be “urn:service:sos” and Route header with the SIP URI of the conference session¶
A Call-Info header with the Incident Tracking Identifier per NENA-STA-010 [7]¶
A Call-Info header with the Call Identifier per NENA-STA-010 [7]¶
If the call to be intervened is already on a Bridge, the CHFE MUST:¶
For the intervening Agent, send an INVITE to the Bridge targeting the conference session with the following:¶
For a call where the Call Priority Identifier is High or Highest, the Request URI must be “urn:service:sos” and Route header with the SIP URI of the conference session¶
A Call-Info header with the Incident Tracking Identifier per NENA-STA-010 [7]¶
A Call-Info header with the Call Identifier per NENA-STA-010 [7]¶

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the Agent's microphone and media sent to the Agent's earpiece, or text or video sent or received.

- Immediately once a call is placed or received on a monitored interface and media is negotiated, to establish a SIPREC session the CHFE MUST:
    - Send one or more SIPREC INVITE requests to the Logging Service with any desired metadata
    - Once a SIPREC session is accepted, do the following:
        - Log a RecCallStartLogEvent
        - Once media starts flowing on the recorded call (regardless of whether it is early media or not), the CHFE MUST:
            - Log a RecMediaStartLogEvent
            - Send a copy of every exchanged RTP packet to the Logging Service
- When parties are added or removed, or media is renegotiated, the CHFE can send any desired SIPREC metadata updates to the Logging Service
- When the recorded call is released with no remainining parties, the CHFE MUST:
    - Send a BYE for every SIPREC session established for that call
    - Log a RecMediaEndLogEvent
    - Log a RecCallEndLogEvent

4.8.3.7 Instant Recall Recorder

The CHFE MAY use the Logging Service as the media source for Instant Recall playback, subject to local policy. See the Instant Recall Recorder section in NENA-STA-010 **Error! Reference source not found.**,**Error! Reference source not found.**.

4.8.3.8 Interactive Media Response (IMR)

Interactive Media Response (IMR) is an automated function that is commonly implemented to answer calls when a PSAP is receiving more calls than it has call takers to answer, or to interact with the caller before an agent answers. An IMR offers the ability to interact with the caller like an Interactive Voice Response (IVR) system except that it supports all forms of media including audio, video, and text. An IMR may implement advanced features such as Automated Speech Recognition (ASR) and Robotic Process Automation (RPA); standards for these features are out of scope of this document.

IMR is an optional component of the CHFE. If implemented, IMR MUST rely on services of the CHFE:

- Session Recording Client interface defined by Session Recording Protocol (SIPREC), RFC 7866 [48] to log streaming media.
- Reporting changes in its state to subscribed elements via the ElementState event and ServiceState notification packages.
- For calls that are queued for the IMR, the queue of calls is a queue as defined in Section 4.2.1.2 of NENA-STA-010 **Error! Reference source not found.**,**Error! Reference source not found.** and maintains the specified QueueState and DequeRegistration events so that PSAP management can monitor and control the queue as it does all other queues.
- Generation of a MediaStartEvent log event for each media supported during a call

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Commented [MLS112]: TODO: Add to terms table
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- Generation of an AnnouncementStartLogEvent when an announcement begins
- Generation of an AnnouncementStopLogEvent when the announcement stops, with the proper announcementType

IMR components MUST support the standards for basic network media services with SIP as defined in Basic Network Media Services with SIP, RFC 4240 [49] . If implemented, the IMR function MUST support *Voice Browser Call Control: CCXML,* [62] for call control and Voice Extensible Markup Language (VoiceXML), (VoiceXML) [63] for media services and support the SIP standards associated with VoiceXML as defined in *SIP Interface to VoiceXML Media Services,* RFC 5552 [64]. VoiceXML <audio> tags MUST specify multiple MIME types to support the media types received by the CHFE and respond using the appropriate media type.

The IMR MUST implement at least the media specifications listed in Section 3.1.9 of NENA-STA-010 **Error! Reference source not found.**. The IMR SHOULD support the language preference provided in the Session Description Protocol (SDP) as specified in *Negotiating Human Language in Real-Time Communications,* RFC 8373 [53]. IMRs MUST interpret an MSRP, RTT, or other text received consisting of the digits '0-9', '#,' or '*', immediately following a prompt for input as equivalent to DTMF key presses.

IMR MAY be used to provide informational announcements. If automated interaction with the caller is needed, the IMR MUST be used. The IMR MUST always answer the call with a 200 OK irrespective of whether it interacts with the caller or not. The IMR MUST NOT offer Early Media in response to inbound call requests.

If the IMR presents options to the caller and a BYE is received from the caller prematurely, the call MUST be treated as an Abandoned Call. The Abandoned Call MUST be handled as specified in the Incoming Interactive Calls section of this document.

If the IMR is engaged with a caller and cannot progress the call any further, it may, based on local policy, perform a blind transfer as specified in the Blind Transfers section of NENA-STA-010 **Error! Reference source not found.**, leave the call in queue, or terminate the call.

4.8.3.9 Operational Considerations

The text in this document describes the possibility of a local Bridge. While the PSAP needs a Bridge to handle calls that are not Emergency Calls, use of local Bridges for Emergency Calls should be discouraged, and NGCS provided Bridges SHOULD be used in preference to a local Bridge for Emergency Calls. There are a limited number of circumstances where a local Bridge may be needed. One is where the NGCS Bridge is not adequately sized to handle the Non-Emergency Calls the PSAP handles. Another is where the NGCS operator is unable or unwilling to provide bridging for calls it does not handle directly. If a local Bridge is available, it MUST NOT be used in a call that already has a Bridge (indicated by the presence of the "isfocus" parameter), as cascaded bridges provide unacceptable distortion of media. It is essential that local Bridges conform to the Bridge requirements of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**, and particularly the requirements for how Bridge participants in other ESInets are engaged.

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Commented [MS113]: Dan will review this text to make sure it still agrees with STA-010 and propose a fix if it doesn't.

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4.9 Mapping Data Service (MDS)

The MDS is defined in NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**. When answering calls or handling Incidents out of area, the PSAP or ECC (hereafter, ECC) may need to display an appropriate map covering the area in which the caller or Incident is located. All ECCs MUST have an MDS available to hold their data. Multiple ECCs MAY hold their data in a common MDS, or each ECC could have its own MDS. ECCs MUST make their MDS interfaces available to other requesting ECCs via mdsFeatureInterfaceUri and mdsImageInterfaceUri as specified in the Service/Agency Locator section of NENA-STA-010 **Error! Reference source not found.**. The MDS for a PSAP or ECC can be discovered via the S/AL. See the Mapping Data Service section of NENA-STA-010 **Error! Reference source not found.Error! Reference source not found.**, and the Incident Handling section of this document for more details about PSAP use of the MDS.

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When a PSAP needs a map for which it does not have map data, it MUST:

- Query the ECRF using the desired location and the Service URN for the Service/Agency Locator (S/AL) to obtain the URL of the S/AL record of the responsible ECC
- Use the S/AL URL to obtain the S/AL Record of the ECC containing the mdsFeatureInterfaceUri and mdsImageInterfaceUri for that ECC
- Use the mdsFeatureInterfaceUri to request a feature map, if a feature-layer map is desired
- Use the mdsImageInterfaceUri to obtain an image map, if an image map is desired

The bounding rectangle of the requested map may contain fragments of more than one service area. To create a map from such fragments, the ECC MUST request each fragment from the MDS of the responsible ECC and then combine them to create the desired map.

4.9.1 MDS Policy

The MDS is a Service and MUST implement a DRM Policy called “MappingDataService” (defined in NENA-STA-010) per the Policy section of this document. The “MappingDataService” Interface Name defined in NENA-STA-010 represents both MDS interfaces in DRM rules.

4.10 Presence Server FE

The Presence Server FE is an FE within the PSAP that is responsible for receiving activity state information from other FEs, storing it, and distributing agent presence information to subscribing entities based upon Policy rules for the Agency. The Presence Server FE accepts AgencyID, AgentID, and Activity State from other FEs responsible for determining the activity state of agents. At a minimum, functional elements that will provide activity state information to the Presence Server FE include the Call Handling FE, Collaboration FE, and PTT System Services FE.

Commented [MS114]: Is Presence Server FE a good name? Is Presence FE better because Server might imply functionality beyond what we specify?

Commented [MS114R2]: TODO: decide whether to use Presence FE or Presence Server FE, then make a global replacement.

Commented [MS114R3]: ***mls picked up here to continue reviewing comments/changes.

The Presence Server uses the Activity States published from FEs in combination with the operational policy for the Agency maintained in the Policy Store to determine an agent’s availability (Agent State) to receive communications. Some Agencies support the notion that an agent is capable of being involved in multiple communications simultaneously, while other Agencies restrict agents to only handle one communication at a time. As a result, received Activity states from the various FEs must be continually compared with the Agency’s policy to determine the Agent’s availability.

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As an example, an agent that is active with a voice call may be considered unavailable for all other types of communications at one Agency but, at another Agency, may be permitted to accept another communication such as a text call. (Note: This should not be confused with handling a single communications session with multiple media types such as a video communication receiving both video and audio).

The Agent's state may be reported differently depending upon the entity to which the report is being sent. For example, an Agent may be shown as available for internal communications within an Agency, but not for communications with another Agency.

4.10.1 Presence Server Clients

The Presence Server has two distinct types of clients. One set of clients, called "Presentities", provide activity state information to be stored in the Presence Server FE; the other set of clients, called "Watchers", receive presence information derived from the stored activity state and Agency policy as shown below.

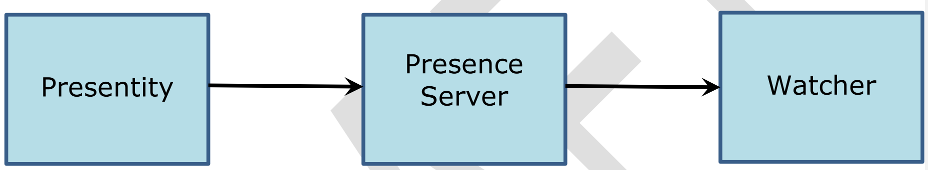

Figure 3 Presentities and Watchers

Presentities include other FEs in the Agency such as the Call Handling FE, Collaboration FE, and PTT Services FE. Watchers include client applications wishing to report on Agent states such as a reader board. FEs may also be Watchers to determine whether the FE is able to accept other communications based on the Agency Policy.

4.10.2 Presence Server FE Interfaces

The Presence Server FE accepts Activity State information for Presentities from the FEs through the use of the SIP PUBLISH or Web method. An FE can use SIP PUBLISH or Web PUT to update the FE Activity State for an agent in the Presence Server. The Presence Server stores the Activity State information for each Presentity. The relevant Activity State values for each FE are described in detail in the Activity State sections for the Call Handling, Collaboration, and PTT System Services FEs.

The Presence Server FE uses the SUBSCRIBE/NOTIFY method to send agent state information to Watchers authorized to receive information about agent activity states. Subscriptions may be requested by clients internal to the Agency or external to the Agency based upon Agency policy defined in the Policy Store. Watchers wishing to receive information about the state of agents MUST support SUBSCRIBE/NOTIFY.

Policies in the Policy Store determine which FEs are allowed to publish to the Presence Server FE, which clients are allowed to subscribe and what information is allowed to be received by Watchers.

The figure below depicts the available interfaces for the Presence Server FE.

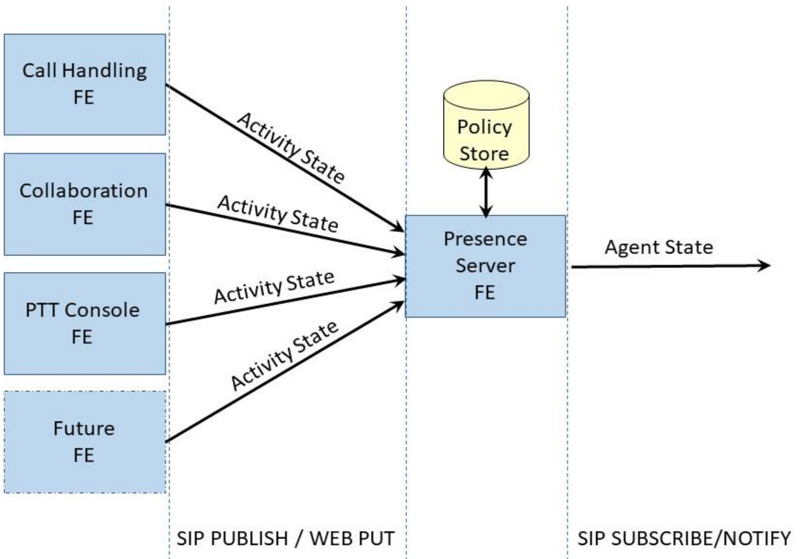

Figure 4 Interfaces for Presence Server FE

The model supports the ability for future FEs or multiple instances of currently defined FEs to provide Activity State information to the Presence Server. The Presence Server MAY implement policy rules that reflect an Agency's policy to determine Agent availability based on current Agent Activity State, for example, whether an Agent is available to take a text call while on an unrelated voice call. A standard mechanism for such policy rules may be provided in a future revision of this document.

4.10.3 Presence Information Document (PIDF)

Presence information is provided in the form of a Presence Information Document (PIDF) which is an XML document defined in *Presence Information Data Format (PIDF),* RFC 3863 [54] that contains the identity and current status of an agent. This document extends PIDF to include the Activity State information defined below in an <activityState> element within the <status> element of a <tuple>.

Per RFC 3863, the <basic> element of the <status> (which only specifies "open" or "closed") is OPTIONAL. Implementations conforming to this standard SHALL include the <basic> element and SHALL specify "open" only if the Agent is available for some type of communication. The <activityState> element carries much more detail about Agent status. The <basic> element is useful to subscribers that don't understand <activityState>. See the section Presentity Activity States for the Presence Server for the definition of the <activityState> extension.

4.10.3.1 Standard PIDF Extensions

In addition to Activity State, the Presence Server SHALL support the following extensions to PIDF:

- *RPID: Rich Presence Extensions to the Presence Information Data Format (PIDF)*, RFC 4480 [50].
- *CIPID: Contact Information for the Presence Information Data Format (PIDF)*, RFC 4482 [51].
- *A Session Initiation Protocol (SIP) Event Notification Extension for Resource Lists*, RFC 4662 [55].
- *An Extensible Markup Language (XML)-Based Format for Event Notification Filtering*, RFC 4661 [56].
- *A Presence-based GEOPRIV Location Object Format*, RFC 4119 [57], as updated by:
  - *Revised Civic Location Format for Presence Information Data Format Location Object (PIDF-LO)*, RFC 5139 [58]
  - *GEOPRIV Presence Information Data Format Location Object (PIDF-LO) Usage Clarification, Considerations, and Recommendations*, RFC 5491 [59]
  - *Representation of Uncertainty and Confidence in the Presence Information Data Format Location Object (PIDF-LO)*, RFC 7459 [60].

4.10.3.2 Presentity Activity States for the Presence Server

The SIP PUBLISH method (described in Session Initiation Protocol (SIP) Extension for Event State Publication, RFC 3903 [61]) or the Web PUT method (described in RFC 2616 [19]) are methods used to report the Activity State for a Presentity (Agent) to the Presence Server FE. Information for the Activity State of an Agent is provided using the Presence Information Document Format (PIDF). PIDF is an XML document that contains the identifier, contact information, and current status of an agent. The Presence Server implements the Notifier side of the SIP "presence" event package, which returns a PIDF document containing the activityState extension element defined below.

Event Package Name: presence
Event Package Parameters: None
PUBLISH Bodies: The PUBLISH body contains a PIDF document that incorporates the <activityState> extension (defined below) within the PIDF's <status> element.

The Presentity URL in the PIDF SHALL be constructed with the Agent Identifier (defined in NENA-STA-010 **Error! Reference source not found.**) of the agent for which presence is being published. This is the content of the "entity" property in the PIDF, for example: entity="pres:agent@agency.gov".

Deleted: [7]

The PIDF MAY include a Location object (making it a PIDF-LO) if the agent's location is needed. Local policy controls whether location is included. 17

All <tuple> elements included in the PIDF MUST include the standard <contact> element and <timestamp> element.

The <contact> element MUST contain the contact address for the Agent relative to the FE for which <activityState> is being provided.

[17] The PIDF is an object used to hold presence information, which may include location. In STA-010, location of a caller or incident is carried in a PIDF containing location (PIDF-LO) but typically does not have any other presence information.

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The <basic> element defined in RFC 3863 [54] MUST be included in the PIDF and MUST indicate whether the Agent is generally available ("open") or unavailable ("closed"). See the section titled "reasonCode PIDF Extension" for optional reason codes that can be used to specify why an Agent is "closed".

4.10.3.2.1 The activityState PIDF Extension

This section specifies a PIDF XML extension element called "activityState". FEs MUST include this <activityState> extension (defined below) within the PIDF's <status> element.

The <activityState> element is a complex element that contains Agent status information. Appendix B contains PIDF examples that contain the <activityState> extension described in this section.

Table 4-2 <activityState> Extension Element

Element	Required	Description
activityState	Yes	Extension to the PIDF <status> element. It is a complex element that provides the Agent's status for given media types, relative to the given FE. The FE is specified by an <fe> element that contains one of CallHandling, Collaboration, or PttConsole).

The <activityState> element MUST be placed within the <status> element of a standard <tuple> element as described in RFC 3863 [54].

The <activityState> element contains the following information for the Agent whose activity is being reported:

Table 4-3 <activityState> Element Content

Sub-Element	Required	Description
supportedIncomingMedia	Yes	A Supported Media Data Component (see below) that describes all the incoming media the FE supports.
supportedOutgoingMedia	Yes	A Supported Media Data Component that describes all the outgoing media the FE supports.
sessions	Yes	An Array containing the list of session data components that describe the sessions/calls the FE is currently active on.
fe	Yes	The functional element the activityState applies to. Valid values are: CallHandling, Collaboration, and PttSystemServices.

A system that implements multiple FEs (Call Handling and Collaboration, for example) MUST report activityState for the Agent in separate <tuple> elements when reporting such activityStates within a single PIDF document, as described in RFC 3863 [54].

4.10.3.2.1.1 Supported Media Data Component

The Supported Media Data Component is a component that contains the media types the FE supports and the Agent's availability for receiving a call with each media type. The media types are represented by the sub-elements in the table below.

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There are two instances of supportedMedia Data Component in <activityState>: supportedIncomingMedia and supportedOutgoingMedia. (see the "<activityState> Element Content" table above):

If a media type is supported by the given FE, then the corresponding media sub-element from the table below MUST be included. If a media type is not supported, the corresponding media sub-element MUST NOT be included. If any media sub-element is not included, that media type SHALL be considered unavailable for the FE specified in the <activityState>.

Table 4-4 <supportedMedia> Data Component Content

Sub-Element	Required	Description
voiceMedia	Conditional	Supported Media Data Component that describes the Agent's availability for voice media. MUST be included if the media type is supported.
videoMedia	Conditional	Supported Media Data Component that describes the Agent's availability for video media. MUST be included if the media type is supported.
imMedia	Conditional	Supported Media Data Component that describes the Agent's availability for instant messaging media. MUST be included if the media type is supported.
rttMedia	Conditional	Supported Media Data Component that describes the Agent's availability for real-time text media. MUST be included if the media type is supported.

Each included sub-element above contains the following two sub-elements that describe whether the Agent is currently available for emergency communications or for non-emergency communications with the given media type, as reported to or by the Presence Server. The isAvailable and isAvailableForEmergency sub-elements contain either "true" or "false".

Table 4-5 <isAvailable> and <isAvailableForEmergency> Sub-Elements

Sub-Element	Required	Description
isAvailable	Yes	Boolean value that describes the availability of the functional element to use the media type.
isAvailableForEmergency	Yes	Boolean value that describes the availability of the functional element to use the media type for emergencies.

4.10.3.2.1.2 Sessions Data Component

The <activityState> component also contains a <sessions> data component that is an array of individual <session> data components that each describe a session the Agent is currently in. An agent can be on zero, one, or more sessions simultaneously. Each session is reported in a <session> data component.

A <session> data component can describe a SIP session, a Collaboration session, or a PTT session. A <session> data component contains three sub-elements:

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Table 4-6 <session> Data Component Sub-Elements

Sub-Element	Required	Description
incomingMedia	Yes	Array of zero or more media types currently being received by the FE for the session, in <mediaType> tags.
outgoingMedia	Yes	Array of zero or more media types currently being sent by the FE for the session, in <mediaType> tags.
isEmergency	Yes	Boolean value indicating if the session is an emergency session.

A <mediaType> tag contains one of the following media type values:

Table 4-7 <mediaType> Possible Values

Media Type	Description
Voice	Voice stream
Video	Video stream
IM	Instant Message text block
RTT	Real-Time Text stream

Each <mediaType> tag indicates a medium in the session. For example, a session with incoming Voice and Video media would contain:

```
<incomingMedia>
   <mediaType>Voice</mediaType>
   <mediaType>Video</mediaType>
</incomingMedia>
```

The <mediaType> tag represents the fact that the <session> contains that media type, regardless of the number of streams of the media type.

If a session has only outgoing media, such as in a PTT transmission, the <incomingMedia> sub-element MUST be empty.

The isEmergency sub-element contains either "true" or "false" to indicate whether the session is an emergency call.

If a given media type ends during the session, the state change MUST be reported by sending a new PIDF with that media type removed.

Appendix B contains PIDF examples showing the use of <mediaType> in context.

4.10.3.2.2 The <reasonCode> PIDF Extension

The <basic> element defined in RFC 3863 [54] MUST be included in activityState. It is used to indicate whether the Agent is generally available ("open") or unavailable ("closed"), like when the Agent is at lunch or on a break. The FE MAY include an OPTIONAL <reasonCode> element within the PIDF's <status> element to indicate why the Agent is currently "closed". Values for <reasonCode> (Work/Wrap-Up, Break, Lunch, etc.) are from the Agent Reason Codes registry defined in the IANA Registry Actions section of this document. An example PIDF containing <reasonCode> elements can be found in Appendix B.

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4.10.3.2.3 The <agentRoles> PIDF Extension

An Agent is typically acting in one or more of the Agent Roles defined in NENA-STA-010, for example, "Call Taking", "Dispatching", and "Supervisor". The Agent's roles are specified in the "role" substring from the Subject Alternate Name field of the Agent's PCA certificate.

Commented [MS115]: TODO: reference the IANA Agent Roles registry instead of STA-010.

Commented [MS115R2]: TODO: probably remove

This section extends the PIDF's <status> element by adding an <agentRoles> string element whose value is the entire "role" substring from the Agent's PCA certificate. Any PIDF that includes the <activityState> extension element MUST also contain an <agentRoles> extension element within the PIDF's <status> element. The <agentRoles> extension element MUST contain the entire "role" substring from the Agent's PCA certificate. PIDF examples that include the <agentRoles> extension element can be found in Appendix B.

4.10.3.3 PUBLISH Response Bodies

The response to a PUBLISH request indicates whether the request was successful or not. The following subset of responses may be returned from the Presence Server for a PUBLISH request. Refer to RFC 3903 [61] for a complete description of required responses. Other standard response codes may be returned and MUST be handled as specified in the applicable standard.

Table 4-8 PUBLISH Response Status Codes

Code	Response	Description
200	OK	Request processed successfully
403	Unauthorized	Publisher not authorized to publish
404	Not Found	Resource (agent) not supported
412	Conditional Request Failed	Precondition given for the request has failed
415	Unsupported Media Type	Improper content type
423	Interval Too Brief	Expiration interval expired
489	Bad Event	Incorrect Event Header

User agent processing of PUBLISH Request: The Publisher (e.g. User Agent FE) publishes Activity State information for an agent to the Presence Server when an agent logs into the FE or when the activity state changes for one of the defined communication methods.

Presence Server FE processing of PUBLISH Request: The Presence Server FE is the User Agent Server (UAS) that processes and responds to PUBLISH requests.

The Presence Server FE processes the published event state (Activity State) contained in the body of the PUBLISH request. If the content type of the request does not match the event package or is not understood by the Presence Server, the Presence Server FE MUST reject the request with an appropriate response, such as 415 (Unsupported Media Type) and skip the remainder of the steps.

If the publisher is not authorized to publish to the Presence Server, the Presence Server SHOULD reject the request and return 401 (Unauthorized).

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4.10.3.4 Presence Server Web Service

The Presence Server SHALL provide a SetAgentActivityState Web Service that allows FEs that do not support SIP to publish the Activity State of their agents to the Presence Server FE. The SetAgentActivityState Web Service is defined in the Web Services section of this document.

4.10.3.5 Watcher Agent State Information

Agent State information is made available to authorized Watchers using the SUBSCRIBE/NOTIFY model as described in *Session Initiation Protocol (SIP)-Specific Event Notification,* RFC 3265 [66]. Watchers wishing to subscribe to the Presence Server for Agent State information must issue a SUBSCRIBE request to the Presence Server specifying the Presentity (Agent) for which state information is desired. Agent State information for multiple agents MAY be specified using a resource list, identified by a URI, representing one or more URIs of agents as described in RFC 4662 [55].

The SIP event framework defines a SIP extension for subscribing to and receiving notifications of events. The event package, Presence, is used to provide presence information for an agent to which the Watcher has subscribed.

Notifier Processing of SUBSCRIBE Requests: Per RFC 3856 [29]. In addition, the Notifier (i.e., the Presence Server) consults the policy (presenceState) to determine if the requester is permitted to subscribe. If not, the Presence Server declines the request as described in RFC 3265 [66]. If the request is acceptable, the Notifier returns 202 Accepted.

Notifier Generation of NOTIFY Requests: Per 3856 [29]. In addition, when state of the agent changes (e.g. activity state changes for the agent for any of the FEs), a new NOTIFY may be generated adhering to the filter requests and policy.

<u>Message Flow</u>

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The following depicts a typical message flow for the Presence Server FE.

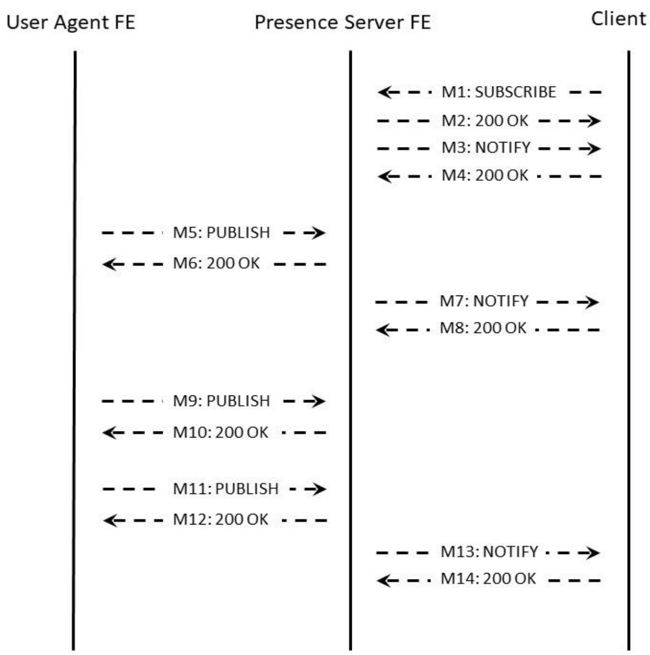

Figure 5 Message Flow for Presence Server FE

M1: The client initiates a new subscription to the Presence Server FE

M2: The Presence Server FE validates the client is permitted to subscribe per the policy of the Presence Server (from the Policy Store), processes the subscription request, and creates a new subscription

M3: The Presence Server FE sends the client a NOTIFY with the current presence state of the presentity (agent)

M4: The client confirms receipt of the NOTIFY.

M5: The User Agent FE representing a presentity (agent) initiates a PUBLISH request to the Presence Server FE in order to update it with the new Activity State information

M6: The Presence Server FE receives the Activity State for the presentity (agent), validates with its policy that the User Agent FE is authorized to publish the Activity State, and accepts the publication. The published state is incorporated into the presentity's state information.

M7: The Presence Server FE determines that a reportable state change has been made to the presentity's state information for a subscriber and sends the new state information (Agent State with detailed Activity states) to the authorized client.

M8: The client confirms receipt of the NOTIFY.

M9: The User Agent FE representing a presentity (agent) initiates a PUBLISH request to the Presence Server FE in order to update it with the new Activity State information.

M10: The Presence Server FE receives the Activity State for the presentity (agent), validates with its policy that the User Agent FE is authorized to publish the Activity State, and accepts the publication. The published state is incorporated into the presentity's state information. In this example, no Watcher will be notified either because a filter on the subscription determines that no notification is wanted, the presentity's state is only refreshed, not changed, or the policy of the Presence Server for this Watcher results in no changed state.

M11: The User Agent FE representing a presentity (agent) detects a change in the Activity State and initiates a PUBLISH request to the Presence Server FE in order to update it with the new Activity State information

M12: The Presence Server FE receives the Activity State for the presentity (agent), validates with its policy that the User Agent FE is authorized to publish the Activity State, and accepts the publication. The published state is incorporated into the presentity's state information.

M13: The Presence Server FE determines that a reportable state change has been made to the presentity's state information for a subscriber and sends the updated state information via NOTIFY (Agent State with detailed Activity states) to the authorized client. The Presence Server consults its policy to determine what information the subscriber will receive in the NOTIFY.

M14: The client confirms receipt of the NOTIFY.

4.10.4 PIDF Extension for Agent Role

Each <tuple> that reports <activityState> for an FE MUST include the extension defined in this section, which contains the entire "role" substring from the Subject Alternate Name of the Agent's certificate. The role string contains the Agent's roles and any role modifiers, which come from the IANA Agent Roles registry under the "emergency" namespace.

Commented [MS116]: Add a reference to PCA's CP. Might reference NG-SEC, Brandon will let us know.

Commented [MS116R2]: Look at NG-SEC, if the text works with future versions, then reference that. Otherwise, reference the published CP.

Table 4-9 PIDF Extension for Agent Role

Extension Element	Required	Description
agentRole	Yes	String element that contains the entire "role" substring from the Subject Alternate Name of the Agent's certificate.

The roles in <agentRole> only apply to the FE in which the agentRole appears. The agentRole MAY contain one or more roles that have no meaning in the context of this FE (for example, a Dispatching role in the "CallHandling" <agentRole>). In this case, a subscriber SHOULD ignore the role. Appendix B contains PIDF examples with <activityState> and <agentRole> elements.

4.10.5 Presence Server Policy

Commented [117]: Dan: if there is a rule that controls access to agentId, that same rule should apply in EIDOs and SIP transactions – it does no good to obfuscate the identity in one case and release it in another. An attribute like agentId that is common to multiple transactions might be better implemented as a General DRM Rule that is enforced by Presence Server, Call Handling, and IDX. We need to discuss and possibly implement something in a General Policy Store that all 3 FEs would read and enforce.

Commented [MS117R2]: Add for consideration in future work. Check whether we already added it.

The Presence Server utilizes rules stored in its provisioned Policy Store to determine the FEs that are allowed to PUBLISH or PUT activity state information to the Presence Server and Watchers that are permitted to subscribe for Agent State information from the Presence Server.

The Presence Server is a Service and MUST implement a DRM Policy called "PresenceServerPolicy" per the Policy section of this document. This document adds "PresenceServerPolicy" to the IANA Policy Types Registry. This document adds the following

Presence Server Interface Names to the [IANA Policy Types Registry]() for use in PresenceServerPolicy DRM rules:

Table 4-10 Presence Server Interface Names

Interface Name	Description
AgentPresencePublish	The PUBLISH interface of the Presence Server for SIP elements to report Activity State for an Agent
AgentPresencePut	The SetAgentActivityState Web Service interface of the Presence Server for non-SIP elements to report Activity State for an Agent
AgentPresenceSubscribe	The SUBSCRIBE interface of the Presence Server to retrieve AgentActivityState

For example, the AgentPresencePut interface would be specified in a DRM rule as <serviceName>.<resourceName>, i.e. "PresenceServer.AgentPresencePut". Data Rights Management (DRM) rules for the Presence Server use the mechanisms described in the Authorization and Data Rights Management section of this document to define the permissions for entities that utilize the Presence Server.

Data items are named by the name of the object (activityState in this case) and the element tag (if access is controlled by element) separated by a period. For example, "presence.tuple.activityState.voiceMedia". If the object name is specified (e.g. "presence") without a member name, then access to the entire object is controlled by the rule.

4.11 Media Proxy Functional Element

Commented [MS118]: Make sure we say that the requesting entity's credentials MUST be used to retrieve the media. Critical!

Commented [MS118R2]: Add this to the DRM section if not already there.

The Media Proxy FE provides external one-way media to a requesting Agent via a mediaProxyUri it creates and returns from the MediaProxyUriRequest web service, acting as a proxy for the external media source. The mediaProxyUri MUST be used by all Agents to access the external media and the Media Proxy MUST log all Media it provides. Otherwise, media retrieved directly by a Responder would not be logged.

Retrieved media could contain malware and MUST be treated by the Media Proxy as specified in the *Security for Next Generation 9-1-1 Standard (NG-SEC)*, NENA-STA-040 [12].

One-way media comes from three sources:

- Responders via Responder Data Services (body and dash cameras for example)
- Public or private sources of streaming media (surveillance cameras, smartphone video stream, etc.)
- Public or private sources of files accessible via a given URI (pictures, audio or video clips, documents, etc.)

In all cases, media can be delivered to an Agent in a PSAP, to an Agent in another Agency, or to an Agent who is a responder in the field. The Media Proxy does not render the media, it just delivers the media to the requesting Agent, who is expected to have a device or application capable of rendering the media. The Media Proxy handles URIs for streaming media and file media as described below.

Media consumed by any entity, including responders, MUST be logged. Some external media sources may not have access to the NG9-1-1 logging service if a responder accesses the media

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directly. Also, policy may govern which entities are allowed access to certain media sources. The Media Proxy FE fulfills these needs.

When an FE wishes to send an EIDO referencing one-way media, it MUST send the original URI, not a Media Proxy URI. The entity that wishes to access the media MUST obtain a Media Proxy URI via the Media Proxy Uri Request Web Service.

Commented [MS119]: Make sure IHFE text agrees with this.

Media Proxy URIs MUST be RTSP or HTTP URIs, and the Media Proxy MUST support original URIs that are RTSP or HTTP URIs. The Media Proxy MAY support other protocols, but such implementations are not standardized herein. The Media Proxy maintains a mapping between the original URI and the Media Proxy URI it returns.

The Media Proxy MUST ensure that any mediaProxyUri returned by the MediaProxyUriRequest web service remains valid until the expirationTimestamp provided in the response has passed. Subsequent requests for the media MUST receive a "410 Gone" response and the requestor MUST determine whether the expirationTimestamp has passed, thus requiring it to obtain a new mediaProxyUri to retrieve the media. In all other cases, the Media Proxy returns the status code it received from the original URI.

Commented [MS120]: TODO: Check v3f to see if i3 added any status codes to its registry that are standard codes, but just provided a different description.

Commented [MS120R2]: Guy said the Omnibus RFC creates a status codes registry under emergency namespace.

Commented [MS120R3]: I don't see a status codes registry, only the "EIDO-IncidentStatus-Common" Registry. No problem for us, since we now require the client to check whether the URI has expired before deciding how to handle "410 Gone".

The Media Proxy FE MUST log each MediaProxyUriRequest and MediaProxyUriResponse with the MediaProxyUriRequestLogEvent and MediaProxyUriResponseLogEvent respectively, as defined in the Logging Service section of this document.

4.11.1 Streaming Media

One-way streaming media is handled, in all cases, by a URI to the media, which the consumer uses by resolving the URI to an RTSP media stream. The Media Proxy FE retrieves the media from the original source and streams it to the consumer. See the Media Proxy URI Request section for details on how a local Media Proxy URI to the media is retrieved by the consumer.

The Media Proxy FE MUST support *Real Time Streaming Protocol (RTSP) 2.0,* RFC 7826 [67], for both media sources and consumers. *Real Time Streaming Protocol (RTSP) RTSP 1.0,* RFC 2326 [68] MAY be supported through a gateway, but such implementations are not standardized in this document.

The Media Proxy URI Request web service returns a secure RTSPS URI for retrieving streaming media even if the original URI is not secure. The Media Proxy MUST allow access to the media only through this secure URI using TLS.

Commented [MS121]: There is no reason to allow non-TLS connections, is there? RTSP over TLS is specified in RFC 7826.

When an Agent desires to retrieve the media, it uses the Media Proxy URI, with the SRTSP protocol and the Agent's credentials. The Media Proxy consults its policy to determine if access to the media stream should be granted. If allowed, the Media Proxy uses the mapping between the Media Proxy URI and the original URI to retrieve the media, forwards the media to the consumer and logs it using the Media Recording Interface to the Logging Service defined in NENA-STA-010 **Error! Reference source not found.**. If the original stream uses a protocol other than RTSP, and the Media Proxy supports that protocol, the Media Proxy MUST interwork that protocol to RTSP. The Media Proxy MUST support multiple Agents retrieving the same media; therefore, the Media Proxy MUST support multiple output streams from a single input stream.

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The Media Proxy MUST support the Session Recording Client (SRC) functionality described in the Logging Service section of NENA-STA-010 **Error! Reference source not found.** and MUST log all streaming media that it proxies via this SRC interface. If either incidentId or callId were

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included in the MediaProxyUriRequest, the Media Proxy MUST include them in the SIPREC INVITE when logging streaming media.

The Media Proxy MUST log all RTSP commands and responses with the MediaProxySignalingMessage LogEvent (see the LogEvents section for details). If a protocol other than RTSP is used, text commands and responses MAY be logged with the MediaProxySignalingMessage LogEvent. All LogEvents generated by the Media Proxy associated with a media session (whether from the original source or relayed to a responder) MUST populate the LogEvent's sessionId field with a unique session identifier. When the RTSP protocol is used, the sessionId field MUST contain the RTSP session identifier.

4.11.2 File Media

File media available via an HTTP URI is retrieved using a mediaProxyUri obtained through a MediaProxyUriRequest as described above and MUST consist of a MIME block containing one of the Media Types from the IANA Media Types registry [81]. The Media Proxy MUST support the Media Types that are listed in Appendix C, and MAY support others.

The Media Proxy MAY refuse to proxy specific Media Types prohibited by local policy, such as those that may contain executable content.

The Media Proxy retrieves the file from the original source and sends the data to the requesting agent, caching the file locally so that it can be logged. When the retrieval operation is done, the Media Proxy logs the file in a RetrievedFileMediaLogEvent and specifies the Media Type and subtype of the retrieved file. If either incidentId or callId were included in the MediaProxyUriRequest, the Media Proxy MUST include them in the RetrievedFileMediaLogEvent when logging file media. If the retrieve operation was aborted before completion, the Media Proxy MUST log any partial data retrieved in a RetrievedFileMediaLogEvent and set the "abortedRetrieval" member to TRUE. See the LogEvents section of this document for details of the RetrievedFileMediaLogEvent.

The Media Proxy URI Request web service returns a secure HTTPS URI for retrieving streaming media even if the original URI is not secure. The Media Proxy MUST allow access to the media only through this secure URI using TLS.

> **Commented [MS122]:** Why would we allow fallback to HTTP when retrieving file media from the Media Proxy?

The Media Proxy caches file media locally until the time given by the expirationTimestamp returned by the MediaProxyUriRequest web service and MUST remove the file from its cache after that time. The MediaProxy MAY expire the file earlier than expirationTimestamp when it deems necessary, e.g., if the file is corrupted, infected with malware, etc., in which case it MUST return If the URI has expired, a client that needs access to the source URI again MUST make a new MediaProxyUriRequest.

4.11.3 Media Proxy Policy

The Media Proxy is a Service and MUST implement a DRM Policy called "MediaProxyPolicy" per the Policy section of this document. The Media Proxy has a MediaProxyUri web service interface, an HTTP GET interface, and an RTSP interface. DRM rules for the MediaProxy specify the MediaProxy Service Name as the Resource and apply to all three interfaces of the MediaProxy.

4.12 Management Console

The Management Console FE (MCFE) supports management functions for the PSAP and implements required event mechanisms specified in NENA-STA-010 **Error! Reference source**

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not found., including PSAP Service State and Security Posture. The MCFE implements web services that control or influence key PSAP state and service-related properties.

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The Service Name for the Management Console is "ManagementConsole". This document requests IANA to add this to the "serviceNames" Registry. See the serviceNames registry section for details.

The MCFE MUST implement web services that control or influence key PSAP state and service-related properties. These include the Discrepancy Report Proxy Web Service, Set Queue State Web Service, and the Set Agent Activity State Web Service. See the Web Services section of this document for details of these web service interfaces.

4.12.1 PSAP Service State

The MCFE is responsible for representing the PSAP's Service State and Security Posture and therefore MUST implement the notifier side of the emergency-ServiceStatesubscription mechanism as specified in the Service State section of the NENA-STA-010 **Error! Reference source not found.**. Upon reception of a SUBSCRIBE request addressed to the Service Identifier of the PSAP (the URI specified in the Service/Agency Locator for the Agency) and specifying the emergency-ServiceState event package, the MCFE consults the ServiceState policy for the Service Identifier of the PSAP to determine if the requester is permitted to subscribe. It returns 603 (Decline) if not acceptable. If the request is acceptable, it returns 200 OK. The MCFE MUST implement event rate filters, RFC 6446 [18]. Upon acceptance, the MCFE MUST send an initial NOTIFY message containing the following information:

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- service.name: MUST be "PSAP"
- service.serviceId: Service Identifier
- service.domain: Service Identifier[18]
- serviceState.state: MUST be a value found in the serviceState registry
- serviceState.reason: text containing reason for change
- securityPosture.posture: MUST be a value found in the securityPosture registry
- securityPosture.reason: text containing reason for change

Upon any change to the Service State or to the Security Posture of the PSAP and respecting any rate filter the subscriber may have specified, the MCFE MUST generate a NOTIFY message that includes the updated information.

Changes in PSAP Service State/Security Posture can affect the state of call queues managed by the CHFE. The CHFE subscribes to PSAP Service State/Security Posture and is responsible for changing QueueState as appropriate.

MCFE subscribes to Service State of all Services in the Agency. A change in Service State of critical PSAP Services can cause the MCFE to change the PSAP Service State. The MCFE MUST support setting the PSAP Service State manually.

4.12.2 Requesting a Change in QueueState

The MCFE may request a change in the state of a specific queue. Examples include when a manager wishes to change a diversion queue from Standby to Active, or to temporarily disable a queue for some reason.

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[18] Due to an error in the original NENA-STA-010.3 schema [7], "domain" was defined instead of "serviceId". To maintain compatibility with early implementations, "serviceId" has been added to the schema. A future version of NENA-STA-010 [7] will deprecate "domain" and make "serviceId" mandatory.

The MCFE MUST implement the client side of the CHFE's Set Queue State web service to request the CHFE to set the state of the given queue to Active, Disabled, or Standby as appropriate. Other QueueState values are not allowed by the Set Queue State web service. See the Set Queue State Web Service section for details.

4.12.3 Discrepancy Reporting

The MCFE MUST implement the server side of the Discrepancy Reporting web service defined in the Discrepancy Reporting section of NENA-STA-010 **Error! Reference source not found.** to receive Discrepancy Reports (DRs) filed against the PSAP by outside entities.

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The MCFE MUST handle DRs from FEs or Services in the PSAP filed against other FEs or Services in its own PSAP.

The MCFE MUST be able to respond to Status Update Requests for Discrepancy Reports that were filed against the PSAP or FEs and Services within the PSAP.

The MCFE SHALL implement the client side of the Discrepancy Reporting web service to support sending Discrepancy Reports to outside entities on behalf of the PSAP.

The MCFE SHALL send Status Update Requests for Discrepancy Reports that were filed against entities outside the PSAP.

The MCFE MUST implement the Discrepancy Report Proxy web service defined in the Web Services section of this document, which accepts DRs from PSAP FEs and Services to be filed. Upon receiving a proxy request, the MCFE MUST:

- Populate the resolutionUri in the received DR with the URI where responses MUST be sent.
- Use the serviceAgencyId provided in the proxy request to search the S/AL for the dscRptSvc URI for that Agency and submit the DR to the Discrepancy Reporting web service at that URI
- Set the discrepancyReportSubmittalTimeStamp to the time stamp provided by the submitting FE, rather than the time the MCFE is submitting the DR to the outside entity

The MCFE MUST NOT modify other fields in the proxied DR.

4.12.4 Management Console Policy

The Management Console is a Service and MUST implement a DRM Policy called "ManagementConsolePolicy" per the Policy section of this document. This document requests IANA to add "DrProxy" to the Interface Names registry. The standard DiscrepancyReport, ServiceState, and ElementState interfaces are implemented by the Management Console, and MUST have DRM rules that control access to them as described in the Authorization and Data Rights Management section of this document.

The MCFE MAY implement a Policy Editor that interfaces to a Policy Store Web Service as described in the Policy section of this document.

4.12.5 Management Console PSAP Monitoring Functions

The MCFE SHALL implement the LogEvent web service interface defined in NENA-STA-010 **Error! Reference source not found.** to receive copies of LogEvents from the PSAP's LogEvent Replicator. This allows the MCFE to display real-time call statistics to the user.

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The MCFE MUST subscribe to ServiceState/SecurityPosture of all Services in the PSAP.

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The MCFE MUST subscribe to ElementState for every FE in the PSAP

The MCFE MUST be capable of subscribing to ServiceState/SecurityPosture of a provisioned list of Services outside the PSAP.

The MCFE MUST subscribe to the provisioned ESRPs' ESRPNotify event so it knows if calls are being diverted.

The MCFE MUST subscribe to Agent Activity State notifications provided by the Presence Server.

The MCFE MUST subscribe to the QueueState of every queue managed by the PSAP's CHFE.

4.13 Responder Data Services Functional Element

The Responder Data Services Functional Element (RDS FE) is an optional FE that can be used to transfer data (including links to streaming media) between an ECC and a Field Responder Client (FRC). Responders may be an individual agent, identified with an Agent ID, or may be a unit, identified with a Unit ID-Common, from the EIDO-JSON standard, NENA-STA-021 [22]. Whether the Responder is an individual Agent or a unit of some type, it is typically the Responder device that reports status, such as location or device state. Responder devices have identifiers similar to Element Identifiers as defined in NENA-STA-010 **Error! Reference source not found.** – a fully qualified hostname and have credentials traceable to the PCA that are used to authenticate the device. In the case of a device belonging to a private Agency (tow truck service, ambulance service, etc.) credentials from a trusted Certificate Authority acceptable to the ECC MAY be allowed.

The RDS FE uses EIDOs to transfer data between an Agency and Responders. Like the CHFE and Dispatch FE, the RDS MUST have an IHFE to exchange EIDOs between Responders and the ECC using the mechanisms described in the Incident Handing FE section of this document. See the Emergency Incident Data Object section of this document for requirements for communicating with EIDOs.

The RDS FE SHALL provide EIDO status updates to subscribers about an Incident and/or Incident location based on information from emergency responders and other sources through its IHFE, subject to policy.

The RDS FE has an EidoConveyance interface that is used by the ECC and Responders to exchange EIDOs. While the RDS has mechanisms that allow it to provide data from a single responder, in many cases, the RDS reports the state of the set of Responders (e.g., a unit) assigned to an incident, as well as the state of Responders not currently assigned to incidents. To do so, it sends EIDOs with multiple Agent and/or Emergency Resource data components, one for each Responder.

Commented [MS123]: TODO: group should decide. We don't define Responder. It's not in the glossary, but "First Responder" is (which includes calltakers and dispatchers). We use Responder a lot, and we always mean field responder, AFAIK.
Should we define Responder in this document as specifically a "field responder"? Or should we define Field Responder and use that longer term throughout.

Commented [MS123R2]: TODO: Change Responder to Field Responder throughout.

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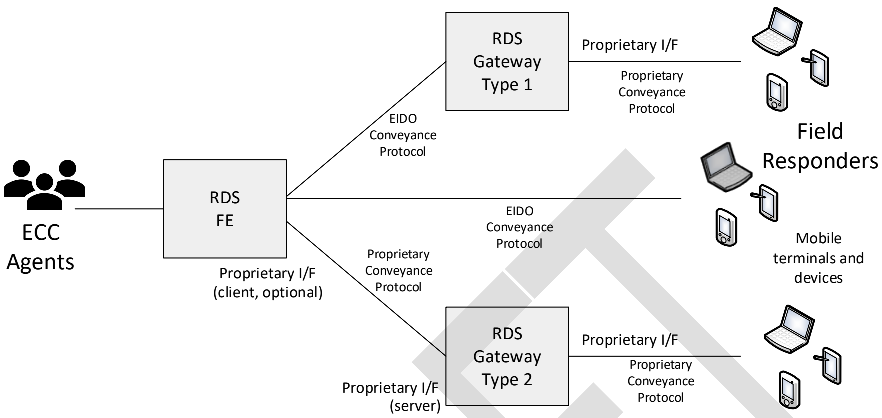
Figure 6 Responder Data Services (RDS) Concept of Operations



The RDS uses the EIDO as the primary data object and uses the EIDO Conveyance Protocol (specified in NENA-STA-024 [25]) as the transport mechanism. Some Responder data sources that communicate with the RDS FE provide their data via the Additional Data component in an EIDO. Adding a new data source may require creation of a new "block" of Additional Data but otherwise does not require any other new mechanism to be implemented on either side of the RDS. High-volume data sources might be accessed through a URL provided in an Additional Data component.

The EIDO Conveyance protocol may go through an optional Gateway as shown in Figure 1 above to convert it to a proprietary protocol. The Gateways are not specified in this document, but MAY be used to connect the RDS to mobile terminals and devices that do not support the EIDO Conveyance protocol.

When the data source provides streaming media, the EIDO will contain an RTSP URI. When the data source provides file media (images, video clips, documents, etc.), the EIDO will contain an HTTP URI. EIDOs can contain other URIs.

Reporting the location of responders is a key requirement of the RDS. The RDS MUST obtain location updates from all responder devices connected to the RDS that provide updated location information. A location reported by the RDS in an EIDO MAY be a location by reference, and the RDS will need to obtain updates of location from the connected responder device. The RDS MAY be permitted, following Agency policy, to act as a repeater for location updates for responder location. Normally, location received by reference must be passed by reference to upstream (towards the ESInet or other Agency FEs) entities.

Deleted: The RDS is permitted to send location received by reference from a responder to other permitted entities. The RDS MUST provide responder location by reference, via either HELD for query, or SIP for subscriptions, including the filter functionality specified for those interfaces on a Location Information Server per NENA STA-010 [7], Section Location Information Server. When location updates are received from the responder, the RDS, following Agency policy sends location updates to subscribers. A location reference in an EIDO sent by the RDS MUST have a lifetime of a minimum of fifteen minutes.

4.13.1 RDS Use of EIDO Subscribe Conveyance

Updates to Incidents, either from other FEs to responders or responders to other entities use the Subscribe mechanism. The RDS MUST subscribe to responders on any incident the responder is assigned to, and MAY subscribe to responders that are not assigned to an Incident in order to receive resource status.

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The RDS MUST support subscription requests on its EidoConveyance interface from authorized entities that wish to receive Incident updates from responders and MAY support subscription requests to receive updates from responders that are not assigned to an Incident in order to receive resource status. Agency policy determines which entities are permitted to subscribe to the RDS.

Due to issues in coalescing input from multiple responders, the Intra-Agency IDX SHOULD be the only internal ECC FE that subscribes to the RDS EidoConveyance interface; other FEs would direct their subscriptions to the Intra-Agency IDX. The RDS SHOULD be the only FE that subscribes to the FRCs. Responders provide status updates to the RDS but subscribe to the IDX to receive updates from other FEs and Agencies.

The normal subscription from an FE to the RDS causes the subscriber to receive aggregate updates for all responders assigned to the incident managed by the RDS. The RDS MUST support an extension in the SUBSCRIBE request that includes a "Responder" parameter containing an agent ID or Unit ID-Common of a responder. A subscriber that provided a Responder parameter in the subscription receives EIDOs containing only information from the identified responder. Agency Policy determines which entities may subscribe to the RDS for a specific responder.

Commented [MS125]: TODO: This extension was envisioned by the author (Brian), but no one ever specified it. Should we specify it? Or remove this text for version 1 and tag for possible future development?

4.13.2 RDS Logging

The RDS provides logging of data to and from responders. FRCs MAY log data to an i3 Logging Service. The RDS MUST log EIDOs it sends or receives to/from responders to its Logging Service(s).

4.13.3 RDS Policy

The RDS is a Service that incorporates an IHFE that implements the EidoConveyance interface for exchanging EIDOs. The "IhfePolicy" has DRM rules that control access to its EIDO interfaces. See the IHFE Policy section for details.

Deleted: The RDS has an RdsHeld interface and an RdsSubscribe interface defined in the IANA Actions section of this document, both of which are used to provide Responder location updates.

4.14 The Logging Service

Commented [MLS128]: Stopped Policy/DRM edits here on 4/2/25. Logging Service needs a Policy/DRM section and some edits to the LogEvents defined herein.

The Logging Service defined in NENA-STA-010 **Error! Reference source not found.** provides critical logging and recording services to the ECC. Every ECC MUST have a Logging Service. An ECC MAY have its own Logging Service, or it MAY use a Logging Service provided in the NGCS. This document sometimes refers to the Logging Service as simply "the logger". The Logging Service implementation MUST be designed with enough built-in redundancy to meet the ECC's reliability requirements for event logging, media recording, legal record production, and troubleshooting activities. Use of an NGCS Logging Service by an ECC, alone or in combination with a local ECC Logging Service, is one way to achieve an ECC's required Logging Service reliability. It is possible to implement Logging Service redundancy completely within the ECC, but when redundant instance(s) are physically co-located, the implementation is more vulnerable to physical disasters. The required level of Logging Service reliability, and actions to take if the Logging Service becomes unavailable, are typically driven by policy and/or regulation. Per NENA-STA-010 **Error! Reference source not found.**: "Clients to the logging service MUST support logging to at least two loggers for redundancy purposes, with support for three (3) or more DESIRABLE, and some jurisdictions might require four or more." ECCs may implement Logging Services in any configuration that meets or exceeds these requirements.

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Commented [129]: Make sure this document specifies the MULTI-TENANT requirements from REQ-001 (FEs shared by multiple agencies). Perhaps an introductory section in the document that covers what an FE is, and the difference between FEs located in a PSAP, and those hosted and/or shared with other agencies.

Commented [130]: Look at our existing "Quality and Reliability" section. Make sure it is adequate not only for Logging Services, but for other PSAP FEs. It doesn't mention geo-diversity currently.

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The NGCS's Logging Service logs all significant call processing steps that occur as the call is being routed to the ECC. The ECC MUST log all significant call and Incident processing steps until the call and any associated Incidents end. The IANA LogEvent registry defines many events that MUST be logged by ECC FEs, including, but not limited to, the CHFE, Dispatch, and FEs that send or receive EIDOs. Call and Incident processing events are logged as LogEvents (see NENA-STA-010 **Error! Reference source not found.**). ECC FEs MUST log all required LogEvents they generate from the time the call hits the first ECC FE until both of the following conditions have occurred:

Deleted: [7]

- the call in question has ended, and
- the associated Incident has been closed.

Note that multiple calls may be received about the same Incident, and logging continues for those calls and that Incident until call and Incident handling is completed. LogEvents may be divided between multiple Logging Services (the NGCS Logging Service, and the Logging Services of ECCs that handled the Incident).

Media recording MUST begin before the call is answered if Early Media are available (see NENA-STA-010 **Error! Reference source not found.**); otherwise, media recording MUST begin when the call is answered. When a call is transferred to another ECC, that portion of the call will be logged in the other ECC's Logging Service. Therefore, querying more than one Logging Service is sometimes necessary to retrieve all media that was logged for a call. Note that an AdditionalAgencyLogEvent is logged when an ECC becomes aware that another Agency is involved in the Incident. The presence of the AdditionalAgencyLogEvent tells the querying Agency that part of the media on this call or Incident may exist in another Agency's logger. See the LogEvent section of NENA-STA-010 **Error! Reference source not found.** for details.

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4.14.1 LogEvents

This document defines the following additional LogEvents to be added to the LogEvent Registry:

DrmLogEvent – when a Service allows or denies access to an interface or a data object available on an interface based on DRM policy as described in the Policy section of this document, it MUST log the DrmLogEvent. The Service Identifier of the Service that evaluated the access request is included in a "serviceId" member. The PolicyId of the Policy that was evaluated is logged in a "policyId" member. The owner of the policy is logged in a "policyOwner" member. The URI of the user's request (either the URI of the Resource's interface or the full URI of a data object available on the Resource's interface, which is the target in the XACML rule) is logged in a "target" member. A "requesterCredentials" member contains two members taken from the requester's certificate: a "commonName" member and a "subjectAltName" member, which contain the data from those elements of the certificate. The "requesterCredentials" member MUST be included if the transaction was over TLS; otherwise, it MUST be omitted. The outcome of the evaluation is included in a "policyDecision" member, which has a value of "granted" or "denied". If "granted", the permissions granted to the subject are given in a "permissionsGranted" member. The "permissionsGranted" member is a string that contains the permissions granted to the subject. The "permissionsGranted" string contains one or more of Read, Create, Update, Delete, or Execute, separated by commas if more than one permission is specified. If "policyDecision" is "denied" then the "permissionsGranted" member MUST be omitted.

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Table 4-11 DrmLogEvent

Name	Required	Description
serviceId	Yes	The Service Identifier of the Service that evaluated the access request
policyId	Yes	The PolicyId of the policy that was evaluated
policyOwner	Yes	The Agency Identifier of the Agency responsible for the Policy that was evaluated
target	Yes	The URI of the user's request, either the URI of the Resource's interface or the full URI of a data object available on the Resource's interface
requesterCredentials	Conditional	A complex object containing Common Name and Subject Alternate Name from the requester's certificate and the content of the From: member for SIP requests. See condition for this member above.
policyDecision	Yes	Either "granted" or "denied"
permissionsGranted	Conditional	One or more of "Read", "Create", "Update", "Delete", and "Execute". See condition for this member above.

IncidentOrResourceStateChangeLogEvent – changes to the state of an Incident or existing resource are logged with the IncidentOrResourceStateChangeLogEvent if not already logged pursuant to an EIDO being sent or received by the FE. For example, when a unit is assigned, arrives on scene, a note is typed, resource goes off duty, etc. The current state of the Incident or resource, as known by the FE logging the event, is logged in an EIDO, in an "EidoBody" member. This LogEvent is generated when the FE didn't send an EIDO because there were no external subscribers to notify, such as when a system contains the other FEs that would normally be notified but uses an internal mechanism to convey Incident state changes.

Table 4-12 IncidentOrResourceStateChangeLogEvent

Name	Required	Description
EidoBody	Yes	An EIDO containing the current Incident state as known to the FE logging the event

MediaProxyUriRequestLogEvent – logs a request to the Media Proxy FE for a local URI that can be used to retrieve media through the Media Proxy from an external URI. A requestBody member contains the body of the request sent to the MediaProxyUriRequest web service, which is a JSON object. See the Media Proxy URI Request section for content of the JSON object. A "queryId" member is provided by the element logging the event, and contains a unique identifier for this request, used to match the request LogEvent to the response LogEvent.

Commented [MS131]: When we get here, review the changes I made in the Media Proxy URI Request web service section per previous group discussion. These LogEvents were then modified to agree with the web service changes.

The Media Proxy MUST log this event. The requesting client MAY log this event.

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Table 4-13 MediaProxyUriRequestLogEvent

Name	Required	Description
requestBody	Yes	The body of the request, which contains a JSON object as specified in the Media Proxy URI Request section of this document
queryId	Yes	Unique identifier provided by the FE logging this LogEvent, used to match the request LogEvent to the response LogEvent

The queryId is generated by the FE logging the event, MUST be globally unique, and it is suggested that it be of the form: "urn:emergency:uid:queryid:'globally unique id'".

MediaProxyUriResponseLogEvent – logs a MediaProxyUriResponse returned to a requesting client from a Media Proxy FE. A mediaProxyResponseBody member contains the body of the response sent to the client (see the Media Proxy URI Request section for details).The Status Code from the response is included in a "statusCode" member.

The Media Proxy MUST log this event. The requesting client MAY log this event.

Table 4-14 MediaProxyUriResponseLogEvent

Name	Required	Description
responseBody	Yes	The body of the response received from the Media Proxy
statusCode	Yes	The status code from the response sent back by the MediaProxy. If the request timed out, the requesting client SHOULD provide 408 Request Timeout as the status code.
queryId	Yes	Unique identifier provided by the FE in the MediaProxyUriRequestLogEvent, used to match the response LogEvent to the request LogEvent

MediaProxySignalingMessageLogEvent – media session dialog messages such as RTSP commands and responses are logged by the Media Proxy FE via the MediaProxySignalingMessageLogEvent. The entire message is included in a "mediaProxySignalingMessageText" member. The RTSP session identifier is included in the sessionId field of the common prologue of the LogEvent and is used to match the signaling messages to the Media Proxy streaming session being logged. A "mediaProxySignalingProtocol" member indicates the protocol (i.e., "rtsp"). Absence of the "mediaProxySignalingProtocol" member defaults to "rtsp". This LogEvent is used to log both the RTSP commands from the user to the Media Proxy and the Media Proxy commands, regardless of protocol, toward the media source.

Commented [MS132]: TODO: see what we did in STA-010, defined these as arrays instead of JSON objects.

Commented [MS133]: Same form used in LostQueryLogEvent in i3.

Commented [MS134]: See if we can modify this to use for logging RTSP use for IRR and playback. Change the name, maybe change the content.

Commented [MS134R2]: Dan: we could log this stuff with siprec metadata if we defined the objects.

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Table 4-15 MediaProxySignalingMessageLogEvent

Name	Required	Description
mediaProxySignaling MessageText	Yes	The text of a media session dialog message (e.g., an RTSP command) sent through the MediaProxy to the server providing the media
mediaProxySignaling Protocol	Conditional	The protocol used to deliver the media. Defaults to "rtsp" if not present.

FileMediaRequestLogEvent – The MediaProxy FE MUST use the FileMediaRequestLogEvent to log a request for file media via a Media Proxy URI. A "request" member contains the entire HTTP request including the path and headers. A "queryId" member is provided by the element logging the event, and contains a unique identifier for this request, used to match the request LogEvent to the response LogEvent.

The FileMediaRequestLogEvent MUST be logged by the Media Proxy and MAY be logged by the FE requesting the media.

Table 4-16 FileMediaRequestLogEvent

Name	Required	Description
request	Yes	The entire HTTP request including the path and headers
queryId	Yes	Unique identifier provided by the FE logging this LogEvent, used to match the request LogEvent to the response LogEvent

FileMediaResponseLogEvent – The MediaProxy FE MUST use the FileMediaResponseLogEvent to log file media that is retrieved using a Media Proxy URI. A "media" member contains the file media, or portion thereof, that was retrieved, Base64 encoded. A "mediaType" member contains the Media Type of the file media, which MUST be a value from the IANA Media Types registry [81]. A "statusCode" member contains the HTTP status code that was returned by the server. If the request timed out, the FE logging this event MUST set the value to "408" (to denote a Request Timeout). A "mediaProxyUri" member contains the mediaProxyUri used to retrieve the media. See the Media Proxy URI Request Web Service section for information about the mediaProxyUri.[19]

The FileMediaResponseLogEvent MUST be logged by the Media Proxy and MAY be logged by the FE retrieving the media.

Table 4-17 FileMediaResponseLogEvent

Name	Required	Description
media	Yes	The file media, or portion thereof, that was retrieved, Base64 encoded

19 This LogEvent is only used to log File Media. Streaming media is logged using SIPREC as described in the Streaming Media subsection of the Media Proxy section.

Commented [MS135]: New proposed LogEvent to log the request for file media. Dan proposed this while we were discussing the FileMediaResponseLogEvent below.

Commented [MS135R2]: Maybe use a PUT to the logger, create a new Logger web service to receive binary files. It needs to be associated with these LogEvents. Figure that out.
Need a way to log that the PUT failed in the response (?) LogEvent. Figure that out.

Commented [MS136]: We have to Base64 encode binary data to include it in a JSON LogEvent. Just add the words "Base64 encoded per RFC 4648" to this sentence if the group agrees that is how we should handle this issue.

Commented [MS136R2]: Dan suggested that we simply include the entire HTTP response, Base64 encoded. Should we do that instead this? If so, we only need a "response" member and the queryId.

Commented [MS136R3]: Pick up here when Dan and Etienne are on.
TODO: group accepted this with some members not present. Review when Dan, Etienne and other SMEs are on the call.

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Name	Required	Description
mediaType	Yes	The Media Type from the IANA Media Types registry
statusCode	Yes	The HTTP status code returned by the server
mediaProxyUri	Yes	The mediaProxyUri used to retrieve the media
queryId	Yes	Unique identifier that was provided by the FE in the FileMediaRequestLogEvent, used to match the response LogEvent to the request LogEvent

PttTransmissionLogEvent – the PTT System Services FE logs transmission-related metadata with the PttTransmissionLogEvent when an individual transmission starts, and anytime there is new metadata available during or at the end of a transmission being recorded. The LogEvent has a "callMetadata" member that contains a PTT system-specific object which in turn contains the metadata for the transmission. The callIdSIP member in the common prologue of the LogEvent MUST contain the Call-ID of the associated SIPREC session.

For P25 PTT systems, the system-specific callMetadata object is the PTT-P25CallMetadata object defined in the PTT System Services FE section.

Table 4-18 PttTransmissionLogEvent

Name	Required	Description
callMetadata	Yes	An object containing system-specific metadata associated with a single PTT transmission (e.g., PTT-P25CallMetadata)

4.14.2 Instant Recall Recorder (IRR) Policy

The Instant Recall Recorder function is described in NENA-STA-010 **Error! Reference source not found.** and uses the Logging Service's standard interfaces to provide the ability to quickly review current or recent emergency communications content. DRM Policy controls who is allowed to use the IRR interface. The policy name of the IRR policy is "IrrPolicy". The name of the IRR interface is "Irr" and is contained in the Interface Names registry section of this document.

A typical IRR policy allows some roles, such as supervisors, managers, and trainers, to recall any media. Agents are typically allowed to only retrieve media from their own calls, often with a restriction on how far back in time they can retrieve media. An example of such an IRR policy is included in the DRM policy examples section in Appendix A – DRM Policy Examples.

4.15 LogEventReplicator FE

The LogEventReplicator is defined in NENA-STA-010 **Error! Reference source not found.**. It sends copies of LogEvents it receives from the Logging Service to applications that use the events for various purposes like displaying status of calls, agents, and services, or for

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Commented [MS137]: Defined in this document. "IRR" was removed from i3 and defined herein as "Irr". The Service Name is Logging Service.

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monitoring and analysis purposes, thereby relieving the FEs that log the events of the additional burden of sending events to such applications. An NG9-1-1 PSAP or ECC MUST have a LogEventReplicator that MUST send copies of LogEvents to the MCFE. The LogEventReplicator MAY send copies of LogEvents to a Management Information System (MIS) and other applications if desired. The list of clients that will receive LogEvent copies from the LogEventReplicator is provisioned.

4.16 Management Information System (MIS) FE

The MIS Functional Element provides reporting services based on data collected from LogEvents. The types of collected data can include:

- Communications processing data generated by FEs.
- Call and Incident state change information.
- Authorization and Data Rights Management events.

Other data that could be used by an MIS system are not standardized in this document.

The reports generated are used to analyze statistics and patterns for management purposes. This document does not mandate specific report types or content. The LogEvents defined in the Logging Service section of NENA-STA-010 **Error! Reference source not found.** provide details about NG9-1-1 processing. The MIS FE SHALL support the LogEvent client interface described in NENA-STA-010 **Error! Reference source not found.** in order to retrieve LogEvents from Logging Services. If the MIS FE wishes to receive LogEvents in real-time, it SHALL implement the LogEvent server interface described in NENA-STA-010 **Error! Reference source not found.**. In this case, the MIS FE SHOULD receive LogEvents from a LogEventReplicator (see NENA-STA-010 **Error! Reference source not found.**) in order to avoid putting additional burden on the FEs sending LogEvents.

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MIS FEs that receive LogEvents in real-time MAY drive real-time status and statistical displays. An MIS FE that provides data to such displays SHALL use the standardized NG9-1-1 metrics defined in *NG9-1-1 Call Processing Metrics Standard,* NENA-STA-019 [69] in cases where the MIS FE wishes to drive display of values based on those specific measurements.

The MIS FE requires information about authorization and Data Rights Management events for reporting purposes. This document defines a new DrmLogEvent for authorization events that the MIS FE MAY use for these purposes. See the Logging Service section of this document for details.

4.17 Push-To-Talk (PTT) System Services FE

The PTT capabilities traditionally offered by various radio system technologies, such as Land Mobile Radio (LMR), provide the main link between Responders and Dispatchers. PTT technologies continue to evolve, including an emerging standard by the 3rd Generation Partner Project (3GPP), *Mission Critical Push To Talk (MCPTT),* [70] included by in its evolving Long-Term Evolution (LTE) standards. There are many legacy PTT technologies in use, and while there is not one protocol that allows all of these technologies to interoperate, gateways are available that provide interoperability between most of them. For the purposes of this document, a "PTT call" can be a single transmission from one user (as is typical with LMR), or it can be a communications session that includes multiple transmissions from multiple users. A PTT call may include multiple media types. The functionality described herein is intended to support either call model. Most PTT systems generate metadata that provides details about

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each PTT transmission, and about any communications session that encompasses the transmission.

The Push-To-Talk (PTT) System Services FE is an OPTIONAL FE that MAY be implemented in any system that provides PTT media and metadata services. The PTT System Services FE MUST implement interfaces for logging PTT media and associated metadata to a Logging Service (defined in NENA-STA-010 **Error! Reference source not found.**), as specified in the Logging Service section of this document. This document defines a new LogEvent for logging metadata associated with an individual PTT transmission and web services for retrieving the details of PTT communications resources from the PTT system. See the Logging Service section of this document for details.

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The PTT System Services FE utilizes a "resourceId", defined by the PTT system, that MUST uniquely identify a shared resource like a talkgroup or conventional radio channel on the PTT system. The Logging Service uses resourceId to request the PTT system to log media and metadata on shared resources as described in the PTT Logging Interface section.

4.17.1 PTT Logging Interface

The PTT Logging Interface is implemented by the PTT System Services FE using the RFC 7866 SIPREC protocol [48] Session Recording Client (SRC) interface for logging PTT media. Associated metadata is logged to the Logging Service. These Logging Service interface specifications are defined in the Logging Service section of NENA-STA-010 **Error! Reference source not found.**. The Incident Tracking Identifier (Call-Info header field, see NENA-STA-010) MUST be provided in the INVITE sent to the Logging Service if available and attributable to the PTT transmission being recorded.

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The SIPREC protocol supports persistent recording sessions where more than one communication session is recorded within a single SIPREC session, along with its associated metadata. The Logging Service supports both persistent recording sessions containing multiple transmissions and sessions that contain only one transmission. The PTT System Services FE can use the method that is most appropriate for the PTT system that is being recorded. A transmission within a PTT system will appear in the SIPREC session as a Media Stream element for the transmitting participant. The PTT System Services FE MUST log the Participant-Stream Association and Media Stream elements from RFC 7865 [71] in the PttTransmissionLogEvent defined in this document. See Session Initiation Protocol (SIP) Recording Metadata, RFC 7865 [71] for details on the Media Stream class. This document defines the PttTransmissionLogEvent for logging system-specific metadata associated with a single PTT transmission, and defines a PTT-P25CallMetadata JSON object for the PTT system to provide transmission-related metadata. The PttTransmissionLogEvent MUST be logged when transmission starts and anytime there is new metadata available during or at the end of the transmission. See the LogEvents section for details of the PttTransmissionLogEvent.

Deleted: The SIPREC metadata stream_id and session_id Media Stream attributes are used to match the Media Stream to the PTT metadata associated with the PTT call. …

The SharedCommunicationResource JSON object is defined in the EIDO-JSON Standard NENA-STA-021 [22] to represent a talkpath on a PTT system. SharedCommunicationResources on a PTT system are discovered by querying the PTT system via the ListSharedCommunicationResourceChannelIds function of the SharedCommunicationResource Information Web Service defined in the Web Services section of this document. The SharedCommunicationResource Information Web Service MUST be implemented by the PTT System Services FE if the PTT System Services FE is implemented by a PTT system. An individual SharedCommunicationResource object is retrieved using the GetSharedCommunicationResourceObject function. The Logging Service will then register

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interest in a SharedCommunicationResource using the SharedCommunicationResourceRegisterInterest function, which requests that the PTT system log media and metadata for transmissions over that SharedCommunicationResource. See the Web Services section of this document for details of these web services. The steps taken by the Logging Service to accomplish registering interest in a SharedCommunicationResource are:

- Get list of SharedCommunicationResources via ListSharedCommunicationResources.
- Retrieve each SharedCommunicationResource of interest via GetSharedCommunicationResourceObject.
- For a P25 resource, register interest in the SharedCommunicationResource via PTT-P25ResourceRegisterInterest.
- The PTT system will then send media (via SIPREC) and metadata (via LogEvent) for each transmission on the given SharedCommunicationResource.

See the Web Services section of this document for details of these web services.

4.17.1.1 Logging Project 25 (P25) PTT Systems

The PTT-P25 SharedCommunicationResource JSON object is defined in the EIDO-JSON Standard NENA-STA-021 [21] to represent a talkpath, which can be a talkgroup or a conventional radio channel on a P25 system; this document uses the PTT-P25 SharedCommunicationResource for the same purpose and defines a PTT-P25CallMetadata JSON object that represents metadata about an individual PTT transmission over a PTT-P25 SharedCommunicationResource.

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The PTT-P25CallMetadata JSON object below represents known metadata associated with an individual transmission over a specific PTT-P25 SharedCommunicationResource. The PTT-P25CallMetadata object MUST be logged with the PttTransmissionLogEvent defined in the Logging Service section of this document for each transmission on a registered resource.

4.17.1.1.1 PTT-P25CallMetadata

This object is implemented by the PTT System Services FE in a P25 radio system and MUST be logged by the PTT-P25 System Services FE in a PttTransmissionLogEvent (defined in the LogEvents section of this document) for each transmission on the resource given by the resourceId member. If the pttMediaAvailability member of the PTT-P25CallMetadata object indicates that media was not available for the transmission, the PttTransmissionLogEvent MUST still be logged by the PTT system.

The PTT-P25CallMetadata object identifies the talkgroup or conventional radio channel that carried the transmission along with the applicable metadata for the transmission. Any members shown as OPTIONAL SHOULD be provided if known and supported by the PTT system being monitored:

Table 4-19 PTT-P25CallMetadata

Name	Required	Description
resourceId	YES	The unique resource identifier of this shared communication resource within the originating system
timestamp	YES	The time the transmission began or ended as specified in the Timestamps and Date/Time Values section of this document

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Name	Required	Description
messageType	YES	Denotes whether this LogEvent is associated with the beginning or end of a PTT call, or if it is an update during the call; values are Begin, End, and Update.
systemId	No	The system ID for a trunked radio system. This with the wacn uniquely identifies a trunked radio system
wacn	No	The Wide Area Communications Network code for the system. This with the systemId uniquely identifies a trunked radio system
channelId	YES	The channelId from the SharedCommunicationResource, can be a talkgroup or conventional radio channel identifier
pttCallType	NO	Type of call, if this is a trunked resource: Group or Individual
groupMode	YES	Type of group resource if this is a group resource. One of: Conventional or Trunked
emergencyStatus	YES	Whether the user has signaled this as an Emergency Call. One of: Emergency or Normal
pttChannelAlias	NO	An OPTIONAL Alias (descriptive name for the talkpath or radio channel such as "LTAC-1").
pttSessionId	NO	The SIPREC Communication Session session_id, if this object is associated with a SIPREC session
pttMediaStreamId	NO	The stream_id of the SIPREC MediaStream, per RFC 7865 [71], if this object is associated with a SIPREC session
pttMediaStreamLabel	NO	The label of the MediaStream, per RFC 7865 [71], if this object is associated with a SIPREC session
pttZoneId	NO	The ID of the zone where the current call was started for systems that implement a zone-based architecture
pttSiteId	NO	The ID of the site where the current call was started
pttUplinkFrequency	NO	Frequency on which the call is transmitted
pttNetworkAccessCode	NO	A special 3-digit Hexadecimal access code used for this transmission for conventional resources.
pttEncryptedCall	YES	True if encrypted, False if not
pttEncryptionAlgorithmId	NO	ID of the Algorithm used to encrypt the call
pttEncryptionKeyId	NO	ID of the Key used to encrypt the call
pttSrcUnitId	YES	ID of the unit (e.g., radio, device, application, console) transmitting
pttSrcUserId	NO	ID of the user transmitting
pttSrcUserName	NO	Name (alias) of the user transmitting, if applicable
pttDstUnitId	NO	For Individual call, ID of the destination unit (e.g., radio, device, application, console ID)
pttDstUserId	NO	For Individual call, ID of destination user
pttDstUserName	NO	For Individual call, Name (alias) of destination user, if applicable

Commented [MS139]: This is the metadata session_id of a CS within the RS. Keep it.
TODO: add the SIPREC Call-Id to associate this with the SIPREC session

Commented [MS139R2]: Should I add participant-stream association back? Would it be useful in any case? Does it provide any info not already covered by unitId and channel?

Name	Required	Description
pttMediaAvailability	YES	Whether media is available for this transmission. Values: Available NotAvailableProhibited NotAvailableKeyNotProvided NotAvailableNoBandwidth NotAvailableUnspecified

For each transmission that is recorded, the PTT System Services FE MUST populate the PTT-P25CallMetadata object and log it in the PttTransmissionLogEvent defined in the Logging Service section of this document.

The resourceId is the unique identifier of the talkgroup or radio channel that carried the transmission, which is also used by the web services that allow the Logging Service to register interest in a PTT-P25 SharedCommunicationResource.

P25 trunked radio systems support two different types of calls, referred to as Individual and Group calls. Individual calls are made by one user directly to another user. Group calls are made by one user to a group of other users who have selected the "talk group" represented by a group identifier or who are monitoring a particular radio channel. Some metadata items are only available on certain types of calls. For example, PTTDstUnitId and PTTDstUserId only apply to Individual calls.

On P25 systems, Emergency Calls are specially marked PTT calls made by a user who is requesting urgent assistance. The emergencyStatus member MUST always be present and MUST have a value of either Emergency or Normal.

For P25 trunked resources, the type of call MUST be provided in the pttCallType member and MUST be one of the following values:

- Group
- Individual

The timestamp is the time the transmission began and MUST be present.

The messageType MUST be provided on all types of calls; it denotes whether the PTT-P25CallMetadata is associated with the beginning or end of a call, or whether it is an update provided during the call.

The systemId, WACN, and siteId MUST always be present. These members identify the system, network, and site on which the transmission started.

The channelId MUST be provided on all types of calls; it gives the talkgroup ID for a trunked call, or the ID of the conventional radio channel over which the call occurred.

The groupMode (Conventional or Trunked) MUST always be provided and indicates whether the resource used for this transmission is a trunked or conventional resource.

If the call is an Individual Call, pttDstUnitId MUST be provided.

If available, pttPresenceInformation SHOULD be provided, and SHOULD include the transmitting device's location if location is available.

If a SIPREC session is associated with this transmission, then pttSessionId MUST contain the SIP Call-ID from the SIPREC session.

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In some cases, such as severe network congestion, the media of a call might not be provided to the Logging Service, but PTT-P25CallMetadata MUST always be logged. If media are not available, pttMediaAvailability MUST be provided and MUST specify the reason. This allows logging of some details about the call even if call media are not available.

The pttSessionId, pttMediaStreamId, and pttMediaStreamLabel from the SIPREC session MUST be provided if there is a SIPREC session associated with this transmission. NOTE: where pttMediaAvailability has a value other than "Available", there may not be an associated SIPREC session.

The PTT System SHOULD provide members of PTT-P25CallMetadata that are marked as not required, if available and allowed by policy.

4.17.2 PTT System Services FE Presence Server Interface

The PTT System Services FE MUST implement Agent activity notifications as described in Section Presence Server FE Interfaces. The PTT System Services FE MUST report activity state to subscribers as specified in the Presence Server FE section, utilizing either the SIP PUBLISH method or the Web PUT method described in that section. A <session> element for the PTT transmission session MUST be included as specified in the Sessions Data Component section if the Agent is actively transmitting.

4.18 Use of NGCS Element State and Service State

An Agency is dependent upon NGCS FEs and Services. A PSAP FE SHOULD subscribe to Element or Service state for NGCS FEs and Services it is dependent upon so that it is notified of state changes. An FE or Service that is dependent upon a Service SHOULD subscribe to Service State rather than to Element State for the individual elements that make up the Service. Use of Element and Service State within an Agency is specified in the General specifications section in this document.

4.19 Time Server

The NGCS provides an NTP time service as defined in the Time Server section of NENA-STA-010 **Error! Reference source not found.**. All PSAP elements MUST synchronize their clocks to a Time Server. An Agency MAY use the Time Server provided by the NGCS, or MAY have its own Time Server, provided that it meets the requirements stated in the NENA-STA-010 **Error! Reference source not found.** Time Server section. An Agency that has its own Time Server SHOULD use the NGCS Time Server for redundancy.

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4.20 Service/Agency Locator (S/AL)

NENA-STA-010 **Error! Reference source not found.** defines a Service/Agency Locator web service that PSAP FEs use to discover interface and contact information for other Agencies. Examples of FEs that use this web service include IHFE, CHFE, Dispatch, and Management Console. PSAP FEs that need the information provided by the Service/Agency Locator web service MUST be provisioned with the S/AL's URL. Agency FEs and Services that use the S/AL MUST support both search by location and search by name.

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4.21 Use of Discrepancy Reporting

NENA-STA-010 **Error! Reference source not found.** defines a Discrepancy Reporting web service that is used to report, inquire, and follow up on errors in call processing. FEs and Services in an Agency MUST file their Discrepancy Reports via the Management Console's Discrepancy Report Proxy Web Service, which will forward the DRs to the specified FE or Service. See the Management Console section for more information.

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5 REGISTRIES

NOTE to document editor: This section is optional. If this section is not needed, delete the section title and all text.

NENA utilizes two registry systems in its standards: The NENA Registry System (NRS) and the Internet Assigned Numbers Authority (IANA) protocol parameter registries.

This section defines one registry to be created and three registries to be modified in the IANA Protocol Registries.

Deleted: This section defines [number] [registry or registries] to be [created or modified] in the [IANA Protocol Registries].

NOTE to document editor: Whenever a standard has a list of items, especially where the list is used in a data structure, and the list is expected to change over time, the list should be maintained in a "registry." A registry is just a table of data, with rows and columns. The registry is established by a standard that defines the columns and what they are used for. Each entry in the registry is a row and has values for the specified columns. The standard that creates or modifies a registry usually defines the initial values (row and column content). It also specifies how a new value is added via the appropriate management policy.

Registries can be hierarchical (i.e., a registry contains sub-registries, nested as needed) if you have a group of registries that are related.

NENA registries are maintained by the NRS, which operates according to NENA-STA-008.2, available at https://www.nena.org/page/standards. The existing registries, with all of the content of the registry, are available in stable locations on the NENA website. The intent of storing the registries at stable URLs is that implementers of standards that use registries can automatically include current values in their implementations. Registries will only modify registries according to the management policy specified for that registry.

Protocol parameter registries represent the authoritative record of many of the codes and numbers contained in a variety of Internet protocols. IANA maintains these records in compliance with the associated technical standards and allocation policies and provides this service in coordination with the Internet Engineering Task Force (IETF).

Registry data is available for bulk retrieval via Rsync (recommended) or FTP.

To obtain a registration in an existing registry or to modify existing registrations, consult the relevant application form. For information on creating or modifying IANA registries, please see RFC 8126, available at https://www.rfc-editor.org/rfc/rfc8126.html, for guidance. For NRS registration, the WG Co-Chairs, upon publication of the document, shall submit the document to the NRS registration email. This section describes what the appropriate registry should do: create a new registry, modify a registry design, add a new value to an existing registry, etc. Each distinct action should be in a separate subsection, although a group of related actions, such as a set of new values for a single registry, should be in a single section.

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This section is written as a set of instructions to the organization that manages the registry. Indicate in a clear and unambiguous way, exactly what you want them to do. Do not make assumptions. They may not read your document in its entirety, so this section has to stand alone. It can reference other parts of the document. For example, you may have a table in your document that has what you want the initial content of the registry to contain. That is acceptable, as long as the table specifies exactly what goes in each "cell" of the rows and columns.

5.1 IANA Registry Actions

The registries mentioned below are all within the "emergency" namespace.

5.1.1 New "Agent Reason Codes" registry

IANA is requested to create a new "Agent Reason Codes" subregistry under the "emergency" namespace. This registry will contain codes the represent the reason an Agent is unavailable.

Commented [MS141]: What other subregistries do we have to create?

5.1.1.1 Registry Title/Name

The name of this registry is "Agent Reason Codes".

5.1.1.2 Parent Registry

This registry is added under the "emergency" namespace

5.1.1.3 Information Required to Create a New Value

The name of Agent Reason Code token value (e.g., "Lunch"), a description of the code, and a reference to the document that specifies the code.

5.1.1.4 Registration Policy

Specification Required [IETF RFC 8126], which requires expert review. The expert should confirm that each entry represents a new type of Agent Reason Code.

5.1.1.5 Content

This registry contains:

- An ASCII "Name" of the code token value (e.g., "Lunch")
- An UTF-8 "Description" that describes what the code denotes
- A "Reference", URI to the document that defines the code

5.1.1.6 Initial Values

Table 5-1 Agent Reason Codes

Reason Code	Description	Reference
WrapUp	The agent is completing work from a previous communication	This Document
Break	The agent is on scheduled break	This Document
Lunch	The agent is taking scheduled meal break	This Document
Consulting	The agent is consulting with a supervisor	This Document

Commented [MS142]: Replace with actual document URI when known.

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Reason Code	Description	Reference
Training	The agent is in a training session	This Document

5.1.2 New Values for "Service Names" registry

IANA is requested to add the following values to the existing registry, "serviceNames":

Table 5-2 New serviceNames

Service Name	Service	Reference
ManagementConsole	Management Console Service	This document
IntraIDX	Intra-Agency Incident Data eXchange Service	This document
CallHandling	Call Handling Service	This document
Dispatch	Dispatch Service	This document
ResponderData	Responder Data Service	This document
MediaProxy	Media Proxy Service	This document
PresenceServer	Presence Server Service	This document
ManagementInformation	Management Information Service	This document
PushToTalk	Push-To-Talk System Service	This document

5.1.3 New Values for "Interface Names" registry

IANA is requested to add the following values to the existing registry, "Interface Names":

Table 5-3 New Interface Names

Interface Name	Reference
DrProxy	This document
SetQueueState	This document
MediaProxyUriRequest	This document
Irr	This document
RdsPsap	This document
RdsResponder	This document
AgentPresencePublish	This document
AgentPresencePut	This document
AgentPresenceSubscribe	This document

5.1.4 New Values for "Policy Types" Registry

IANA is requested to add the following values to the existing registry, "Policy Types":

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Commented [MS144]: Add our interface names, ask i3 to remove their policy type and interface name

Deleted: DRProxy

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Commented [MS145]: From STA-010 section 3.3.1 Policy Store Web Service:
"Policies are identified by the policy type, the identity of the agency that owns the policy and, for Policy Routing Rules, a policyQueueName or an Id."'
From 3.3.1.2 Policies:
"Policy Types come from the Policy Type Registry (section 10.33) and consist of service names (which are policies that describe the access rights for the service), and types specified by this document or other NENA standards."

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Table 5-4 New Policy Types

Policy Type	Format	Use	Reference
ManagementConsolePolicy	XACML	DRM	This document
IntraAgencyIdxPolicy	XACML	DRM	This document
CallHandlingPolicy	XACML	DRM	This document
DispatchPolicy	XACML	DRM	This document
RdsPolicy	XACML	DRM	This document
MediaProxyPolicy	XACML	DRM	This document
PresenceServerPolicy	XACML	DRM	This document
MisPolicy	XACML	DRM	This document
RmsPolicy	XACML	DRM	This document
P25PushToTalkPolicy	XACML	DRM	This document
LogEventReplicatorPolicy	XACML	DRM	This document
IrrPolicy	XACML	DRM	This document
EidoConveyancePolicy	XACML	DRM	This document
IhfePolicy	XACML	DRM	This document
AgentPresencePolicy			
AgentPresencePublishPolicy	XACML	DRM	This document
AgentPresencePutPolicy	XACML	DRM	This document
AgentPresenceSubscribePolicy	XACML	DRM	This document

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Commented [MS146R2]: I will ask Brandon/CRM whether to ask Conveyance to reconvene to add Interface Names for individual conveyance interfaces (see list in IHFE policy section).

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6 DEPENDENCIES

Commented [SD148]: Needs to be completed.

Commented [MS148R2]: TODO:

Note to document editor: This section is optional and is used for documenting dependencies on specific versions of data and interface schema that the document depends on. Remove this section if the document does not reference any data schema or interfaces. For example, a Version 1 document may create or update a registry or NENA Artifact. Version 2 of the document will not mention these actions as they will have been completed in the previous version. Future versions of a document should reference dependencies from previous versions.

This document has the following dependencies on registries or NENA Artifacts, as defined in the NENA Knowledge Base (NENAkb) Glossary [1] and NENA-ADM-012 [4]:

Name	Description	Reference
Logging Service_v_XYZ	Shorthand description	https://github.com/nena911/xyz
EIDO_v_XYZ	Shorthand description	https://github.com/nena911/xyz
[IANA] XYZ Registry	Shorthand description	https://www.iana.org/assignments/xyz/xyz.xhtml
[NENA] ABC Registry	Shorthand description	http://technet.nena.org/nrs/registry/abc.xml

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7 IMPACTS AND CONSIDERATIONS

NOTE to document editor: This section is optional. If this section is not needed, delete the section title and the text.

Use this section to convey any additional information that the Working Group believes is important for the reader to understand. Examples of impacts and considerations include the following:

- operational
- technical
- security-related
- level of effort
- time required to implement
- cost or cost-recovery
- any other information the group believes the reader should understand when using this document

The organization of this section and any subsections is at the discretion of the editor and the Working Group.

8 RECOMMENDATIONS FOR ADDITIONAL DEVELOPMENT WORK

Commented [SD149]: There are two ways to manage future work items. Based on what existed in the document, I chose the table option below. You can choose to remove the table and use the other method instead. If the table is retained, the other method wording can be deleted.

NOTE to document editor: This section is optional. If this section is not needed, delete the section title and the text.

Recommendations for additional development work generally include information that should be considered in a future version of the document or is intended to be referred to another NDG Committee, Working Group, or Standards Development Organization. In general, additional development work can be considered as a list of important topics that the Working Group recommends for further development work.

The following list is intended to help the Working Group decide what to include as additional development work.

- Information that is valuable to the topic, although deemed out-of-scope for the current version's Charter. These items can be considered in a future version of the document, and many times will support the planning of the document's next version.
- Information or a topic that the Working Group recommends be considered by a different NDG Committee or Working Group, or Standards Development Organization (i.e., IETF, ATIS, APCO). If known, include the NDG Committee or Working Group, or the Standards Development Organization in the item description.
- Any information added to this section should be as complete as possible due to the future nature of the efforts. The current Working Group should not assume that it will be the Working Group assigned to support the future work.

Additional development work, whether as a revision to the document or a referral, is supported by the Issue Submission Form (ISF) process for consideration of the NDG membership and direction by the DSC. More information on the ISF process can be found by visiting https://www.nena.org/page/Submit_IssueNDG.

Upon publication, the Working Group Co-Chairs will review this section with their Committee Co-Chairs to determine the next step for each item listed in this section. This would include sending recommendations to external organizations or generating ISFs for NDG consideration.

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This document references existing NENA [Standards, Requirements, Information]. Additional work may be required as noted below.

NOTE to document editor: The section can be structured with numbered section headings, or the table below can be used to document future work. Choose one format and remove the other from this section.

Suggested Table format:

Table 8-1 Future Work for Consideration

Recommended Assignment	Future Work
Future Version of this Document	**Collaboration FE** This FE, described in NENA-REQ-001.1.2-2018, NG9-1-1 PSAP Requirements Document, provides "multimedia chat" functionality that supports text, audio, and video communications and file-sharing capabilities for Agents in the same or different Agencies. A future revision of this document should define a standardized FE that supports these functions.
[Recommended Assignment]	**Outgoing Alerts** NENA-REQ-001.1.2-2018 describes an Outgoing Alert FE that provides interfaces that allow an Agency to provide some information to emergency services personnel or entities, or to the public at large.
[Recommended Assignment]	**Callback Timers** Different OSPs have different callback timer values. A mechanism could be designed to handle this variation if the NGCS had a way for a PSAP to obtain the callback timer of the OSP that originated the Emergency Call.
[Recommended Assignment]	**Presence Server Watcher Improvement** A "Watcher" could discover the agents who are currently logged in to a particular FE. The Presence Server knows when agents log in or out. It would be possible to define an event package for Watchers to get notification of logged in agents via NOTIFY.
[Recommended Assignment]	**Media Proxy Improvement** Support pan/tilt/zoom on video camera sources. There is a protocol called ONVIF that supports this. Consider support for camera systems that include a speaker, where the Agent could talk to someone who is being surveilled.
[Recommended Assignment]	**Transfer of Non-interactive Calls** Specify how a Non-interactive call is "transferred" to another Agency.
Future Version of this Document	**Reliability**

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[Future Work Item]
NOTE to document editor: Provide a distinct name for the future work item.
[Description]
NOTE to document editor: Describe the future work item, its background, a justification for why it should be done, and if applicable, the document section number.
[Recommended Assignment]
NOTE to document editor: Include a recommended assignment such as a future version of this document, NDG Committee, Working Group, a new Working Group, or external Standards Development Organization (i.e., IETF, ATIS, APCO). If unknown, work with the Committee Co-Chairs to identify an appropriate assignment.

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Recommended Assignment	Future Work
	Define multiple levels of reliability for of PSAPs in Calltaking and Dispatch Service Availability section, then in the Physical Considerations section, specify physical characteristics for those levels.
[Recommended Assignment]	**Generic EIDO Endpoint Exchange FE** Define a Generic EIDO Endpoint Exchange FE that would be implemented by outside entities, to include private Ambulance and Tow Truck Companies. It would allow them to exchange EIDOs with cooperating ECCs.
	DRM Rules for Access to Identity Information If there is a rule that controls access to Agent ID, that same rule should apply in EIDOs and SIP transactions – it does no good to obfuscate the identity in one case and release it in another. An attribute like Agent ID that is common to multiple transactions might be better implemented as a General DRM Rule.

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9 ABBREVIATIONS, TERMS, AND DEFINITIONS

Use this table to identify all terms that are unique to NENA or the 9-1-1 system or are important to this document. All abbreviations used in this document are to be included in this table. If only used once within the document, consider fully spelling out the term without its abbreviation so it does not need to be added to this table. Creating abbreviations that are not subsequently used in the document is not recommended.

Compare your definitions with the current NENAkb Glossary to avoid unnecessarily creating conflicting descriptions of the same term. **Review terms just before each approval ballot since the NENAkb Glossary is continually updated and definitions may have changed**.

Please add terms and abbreviations to the table using the following format:

- Place term or abbreviation alphabetically in the **Term or Abbreviation (Expansion)** column. For abbreviations, add the expansion (the spelled-out words) in parentheses, e.g., FCC (Federal Communications Commission).
- Provide a brief definition or description for the abbreviation or term.
- In the **Definition** column, you may also identify other Relevant NENA Documents or External References that provide greater detail about the term or abbreviation. URLs added to cited references will not be included in the NENAkb Glossary.

Instructions for Recommending Changes to the NENAkb (NENA Knowledge Base) Glossary

Use the third column to recommend changes to the NENAkb Glossary or confirm acceptance of the Glossary's definition in this document. The CRM team will remove the third column before the approved document is published. These options are available in the third column.

Select Recommendation drop-down box:

- **In Glossary—**The term and the definition in the document are the same as in the NENAkb Glossary. Please use the definition in the NENAkb Glossary.
- **Add—**The term does not exist in the NENAkb Glossary and meets the criteria for inclusion. You may also identify other Relevant NENA Documents that provide greater detail about the term or abbreviation.
- **Delete—**The term exists in the NENAkb Glossary but is no longer needed or does not meet the criteria for inclusion.
- **Don't Add—**The term is not in the NENAkb Glossary and does not meet the criteria for inclusion **OR** the WG needs a unique definition in this document other than what is in the NENAkb Glossary.
- **Modify—**Update the NENAkb Glossary with this term and definition. If the term originated in a different NENA document, consult with that document's Working Group (or its Committee if not currently active) to develop a single, agreed-upon definition.
- **Other—**Use this option to suggest other actions such as adding this term as a "Related Term" or "Also Known As" term to the definition of another term in the Glossary.
- **Insert Explanation—**If "Other" is selected OR a definition was modified in collaboration with another WG, OR any other explanation is needed, enter the details in the Insert Explanation text box (3rd column). Or else, delete the Insert Explanation text box.

See the NENA Knowledge Base (NENAkb) Glossary [1] for a list of terms and abbreviations used in NENA documents. Abbreviations and terms used in this document are listed below with their definitions as they apply to this version of the document. Please be aware that terms and definitions in the NENAkb Glossary may be different than what is shown in this document.

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
Abandoned Call	An Abandoned Call is an emergency Call in which the caller disconnects before the Call can be answered by an Agent in the PSAP (Public Safety Answering Point) or Agency .	Modify (add as related term to Call)
Activity State	Agent state data that includes the types of communications supported by an FE being used by an Agent, along with the communication types the Agent is currently engaged in on that FE.	Don't Add
Ad Hoc Bridge	One of two bridge methods defined in NENA-STA-010 **Error! Reference source not found.**. The Ad Hoc Bridge method is characterized by using the bridge only when it is needed during transferring or conferencing more than two parties.	Add (add as related term to Bridge)
Agent	In NG9-1-1, 9-1-1 operations, and public safety response, an Agent is an authorized person — employee, contractor, or volunteer — who has one or more authorized functions in, or as directed by, an Agency. In NG9-1-1, an Agent is validated by its parent Agency. The fact that the Agent's Certificate Signing Request is signed by the parent Agency's Agency Certificate constitutes validation of that Agent. A validated Agent will have a unique Agent Identifier and will be issued an Agent Certificate. An Agent in NG9-1-1 can also be an automaton in some circumstances (e.g., an IMR answering a call). The Agent Certificate will enumerate the Role(s) of the Agent.	In Glossary
Agency	An Agency is an entity with a valid public safety purpose under a single discrete recognized administration. In 9 1 1 and public safety operations, an Agency is a governmental entity, or a non-governmental entity under the direction of a governmental entity, responsible for all or some part of 9 1 1 system provisioning, call processing, and field response.	Select

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
	In NG9 1 1, a governmental Agency is recognized through validating an enabling statute, ordinance, municipal incorporation, joint powers agreement, or similar. A private entity is recognized through validating articles of incorporation, business registration, or similar. A validated Agency will have a unique Agency Identifier and will be issued an Agency Certificate. The Agency Certificate can validate Agents that are members of that Agency. The Agency provisioned in the NG9 1 1 environment that holds an Agency Certificate may not necessarily be the same Agency recognized in 9 1 1 and public safety operations, depending on how the NG9 1 1 system is designed and configured.	
Agent Identifier	An Agent Identifier is a username, using the syntax for "Dot-string" in *Simple Mail Transfer Protocol,* RFC 5321 [74] (that is, the user part of an email address, without the possibility of a "Quoted-String").	Add
Agency Identifier	Agency Identifier is a domain name for an agency used as a globally unique identifier.	In Glossary
Attended Transfer	An Attended (also known as warm) Transfer is a transfer facilitated by a Bridge where the caller, transfer-from entity and transfer-to entity are all conferenced together in order to allow the Agent at the transfer-from entity to ensure a proper hand off of the call as specified above.	Add (add as related term to Transfer)
BCF (Border Control Function)	BCF (Border Control Function) provides a secure entry into the ESInet for Emergency Calls presented to the network. The BCF incorporates firewall, admission control, and may include anchoring of session and media as well as other security mechanisms to prevent deliberate or malicious attacks on PSAPs or other entities connected to the ESInet.	In Glossary
Blind Transfer	A Blind (also known as cold) Transfer is a transfer where there is no conversation between the transferring entity and the transfer-to entity.	Add (add as related term to Transfer)

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April 18, 2024 at 10:01 AM
Ask i3 if they will add this term in 3.1

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April 18, 2024 at 10:07 AM
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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
Bridge	In NG9-1-1, a Bridge is used to transfer calls and conduct conferences. The Bridge has a SIP signaling interface conformant to NENA-STA-010 Error! Reference source not found. used to create and maintain conferences and has media mixing capability. Bridges in NG9-1-1 are multimedia capable (voice, video, text).	Add
CAD (Computer Aided Dispatch)	CAD (Computer Aided Dispatch) is a computer-based system which aids PSAP Telecommunicators by automating selected dispatching and record keeping activities.	In Glossary
Call	Call is a generic term referring to any request for public safety assistance, regardless of the media used to make that request. This term may appear in conjunction with specific media, such as "voice Call", "video Call", "text Call", or "data-only Call" when the specific media is of importance. The term "non-interactive Call" refers to an emergency Call that is initiated automatically, carries data, does not establish a two-way interactive media session, and typically does not involve a human at the "initiating" end.	In Glossary
Call Identifier	A Call Identifier is a globally unique identifier assigned by the first element in the first ESInet which handles a call.	In Glossary
Call Priority Identifier	A priority level that is initially assigned to a call based on the type of queue it was received on (emergency or non-emergency), that influences subsequent call treatment. The priority level can change if the call is promoted to, or demoted from, an emergency call. The defined priority levels are Highest, High, Normal, Low, and Lowest.	Add
Call Taker	A Call Taker is an Agent that answers Calls (voice, video, or text) and interacts with the caller. A Call Taker can be an automaton.	Add
Callback Call	A Callback is the action of reconnecting to an Emergency Caller after the Emergency Call has ended where the attempt to reconnect occurs within a configurable amount of time of its reception. Callback calls have special signaling.	Modify (add as related term to Call)

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
CHFE Callback Period Timer	A configurable timer within the CHFE that represents the time period after a call ended during which a Callback can be made.	Add
CHFE (Call Handling Functional Element)	Call Handling is a Functional Element concerned with the details of the management of calls. It handles all communication with the caller. It includes the interfaces, devices and applications utilized by the Agents to handle the call.	Modify (replace Call Handling in glossary)
CJIS (Criminal Justice Information Services)	CJIS (Criminal Justice Information Services) serves as the focal point and central repository for criminal justice information services in the FBI. Programs initially consolidated under the CJIS Division included the NCIC (National Crime Information Center), UCR (Uniform Crime Reporting), and Fingerprint Identification. In addition, responsibility for several ongoing technological initiatives was transferred to the CJIS Division, including the IAFIA (Integrated Automated Fingerprint Identification System), NCIC 2000, and the NIBRS (National Incident-Based Reporting System).	In Glossary
Route All Calls Via a Conference Aware UA	One of two bridge methods defined in NENA-STA-010 **Error! Reference source not found.**, referred to as "Route All Calls Via a Conference Aware UA". In this mechanism, at least the signaling portion of a bridge is always in the call path, and transfer is accomplished by adding and deleting parties to the bridge.	Add (add as related term to Bridge)
CPE (Customer Premises Equipment)	CPE (Customer Premises Equipment) consists of communications or terminal equipment located in the customer's facilities – Terminal equipment at a PSAP.	In Glossary
DNS (Domain Name System)	DNS (Domain Name System) is a global, distributed, delegated database of records about domain names. Each type of record holds a specific type of information. DNS is queried with a domain name and desired record type. The response is the record data associated with the queried data. Some records contain an IP address for a domain name, other record types contain email or SIP server information for a domain name, etc.	In Glossary

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
DR (Discrepancy Report)	DR (Discrepancy Report) is a function that exists in NG9-1-1 to notify agencies and services (including the BCF, ESRP, ECRF, Policy Store, and LVF) when any discrepancy in a database is found. The discrepancy reporting audience is anyone who is using the data and finds a problem.	In Glossary
DRM (Data Rights Management)	The management of permissions to access NG9-1-1 interfaces and data according to rules (a policy) written in the eXtensible Access Control Markup Language (XACML) [17].	Add (Taken from STA-010.3.1)
Early Media	Early Media is defined as media streams that flow between parties before the session establishment process is complete, that is, before a final response has been provided.	Add
ECC (Emergency Communications Center)	A facility designated to receive and process requests for emergency assistance, which may include 9-1-1 calls, determine the appropriate emergency response based on available resources, and coordinate the emergency response according to a specific operational policy.	In Glossary
ECRF (Emergency Call Routing Function)	ECRF (Emergency Call Routing Function) is a functional element in NGCS (Next Generation Core Services) which is a LoST (Location-to-Service Translation) protocol server where location information (either civic address or geo-coordinates) and a Service URN serve as input to a mapping function that returns a URI used to route an Emergency Call toward the appropriate PSAP for the caller’s location or towards a responder agency.	In Glossary
EIDO (Emergency Incident Data Object)	An EIDO (Emergency Incident Data Object) is a JSON-based (JavaScript Object Notation) object that is used to share emergency incident information between and among authorized entities and systems. NENA has adopted the JSON-based EIDO (Emergency Incident Data Object) for sharing incident information among authorized NG9 1 1 entities and systems.	In Glossary

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
Element State	An event package defined in NENA-STA-010 **Error! Reference source not found.** that provides the state of an element either determined automatically or as determined by management. A registry (ElementState) of allowed values is also defined in NENA-STA-010 **Error! Reference source not found.**.	Add
Emergency Call	A call for assistance to a designated emergency number (e.g., 911, 112) initiated by the public requesting immediate response from an Agency.	Add (add as Related term to Call)
ERS (Emergency Response Services)	The Agencies, Services, and Functions involved in the response to a reported Incident, including those specified in this document, and other public or private entities and services that support them, including all entities that use EIDOs to exchange Incident data.	Add
Emergency Support Call	An Emergency Support Call is a call from an Agency to another Agency associated with an Incident.	Add (add as Related term to Call)
ESRP (Emergency Service Routing Proxy)	ESRP (Emergency Service Routing Proxy) is an i3 functional element which is a SIP proxy server that selects the next-hop routing within the ESInet based on location and policy. There is an ESRP on the edge of the ESInet. There is usually an ESRP at the entrance to an NG9-1-1 PSAP. There may be one or more intermediate ESRPs between them.	In Glossary
ESS (External Switching Systems)	ESS (External Switching Systems) are the administrative, non-emergency communications systems within the PSAP that serve as external switching systems for telephony and other administrative communications (voice mail, email, instant messaging, etc.). The ESS equipment might be represented by a traditional PBX or other communications gateway.	In Glossary
EVS (Enhanced Voice Service)	EVS (Enhanced Voice Service) is an audio codec supporting Narrowband, Fullband, Wideband and Superwideband profiles. The EVS standard is found in *Codec for Enhanced Voice Services (EVS)* [75]	Add

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April 18, 2024 at 10:47 AM
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April 18, 2024 at 11:06M
Ask i3 if they will add this term in 3.1

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
FE (Functional Element)	An FE (Functional Element) is an abstract building block that consists of a set of interfaces and operations on those interfaces to accomplish a task. Mapping between functional elements and physical implementations may be one-to-one, one-to-many, or many-to-one.	In Glossary
FRC (Field Responder Client)	A device or software installed on a device, such as a computer, tablet, or smartphone, used by field responders to access public safety systems, such as to receive dispatch instructions or perform database lookups. FRCs are often colloquially referred to as "MDT (Mobile Data Terminal)."	In Glossary
Follow-up Call	A call to a previous 9 1 1 caller after the CHFE Callback Period Timer has elapsed and while the Incident is active. See the Follow-Up Calls section for details.	Add (add as related term to Call)
HTTP (Hypertext Transfer Protocol)	HTTP (Hypertext Transfer Protocol) is typically used between a web client and a web server that transports HTML and/or XML.	In Glossary
PSAP	An NG9-1-1 Public Safety Answering Point (PSAP) that conforms to the specifications in NENA-STA-023 is referred to as an "PSAP" and will interoperate with the NG9-1-1 Core Services (NGCS) defined in NENA-STA-010 **Error! Reference source not found.**	Add
IANA (Internet Assigned Numbers Authority)	IANA (Internet Assigned Numbers Authority) is the departmental entity within ICANN (Internet Corporation for Assigned Names and Numbers) that oversees coordination of global IP address allocation, DNS root zone management, protocol name and number registries, and other Internet protocol assignments. Some NENA documents may use IANA Protocol Registries following the processes described in RFC 8126 [73].	Add
IDX (Incident Data eXchange)	IDX (Incident Data eXchange) is a Functional Element that facilitates the exchange of Emergency Incident Data Objects (EIDOs) among other Functional Elements both within and external to an agency. (Previously called "IDE".)	In Glossary
IHFE (Incident Handling Functional Element)	An FE that creates and updates Incidents and utilizes EIDOs to exchange Incident information.	Add

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
IMR (Interactive Media Response)	IMR (Interactive Media Response) is an automated service used to play announcements, record responses, and interact with callers using any or all of audio, video, and text.	In Glossary
Incident Tracking Identifier	Incident Tracking Identifier is an identifier assigned by the first element in the first ESInet that handles an Emergency Call or declares an incident. Incident Tracking Identifiers are globally unique.	In Glossary
IP (Internet Protocol)	IP (Internet Protocol) is the method by which data is sent from one computer to another on the Internet or other networks.	In Glossary
JSON (JavaScript Object Notation)	JSON (JavaScript Object Notation) is a lightweight data-interchange format based on a subset of the JavaScript Programming Language Standard ECMA-262.	In Glossary
Law Enforcement Online	Law Enforcement Online is a secure, Internet-based information sharing system for agencies around the world that are involved in law enforcement, first response, criminal justice, anti-terrorism, and intelligence. With LEO, members can access or share sensitive but unclassified information anytime and anywhere.	Don't Add
LIS (Location Information Server)	LIS (Location Information Server) is a Functional Element that provides locations of endpoints. A LIS can provide Location-by-Reference, or Location-by-Value, and, if the latter, in geodetic or civic forms. A LIS can be queried by an endpoint for its own location, or by another entity for the location of an endpoint. In either case, the LIS receives a unique identifier that represents the endpoint, for example an IP address, circuit-ID, or Media Access Control (MAC) address, and returns the location (value or reference) associated with that identifier. The LIS is also the entity that provides the dereferencing service, exchanging a location reference for a location value.	In Glossary
LNG (Legacy Network Gateway)	LNG (Legacy Network Gateway) is an NG9 1 1 Functional Element that provides an interface between a non-IP originating network and a Next Generation Core Services (NGCS) enabled network.	In Glossary

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
LogEvent	LogEvent is a standardized JSON object containing information about a processing event that is stored in and retrieved from the Logging Service.	In Glossary
LogEventReplicator	The LogEventReplicator is an FE defined in NENA-STA-010 **Error! Reference source not found.** that sends copies of LogEvents it receives to applications that use the events for various purposes like displaying status of calls, agents, and services, or for monitoring and analysis purposes, thereby relieving the FEs that log the events of the additional burden of sending events to such applications.	Add
Logging Service	The Logging Service is a Service defined in NENA-STA-010 **Error! Reference source not found.** that provides critical logging and recording services to the the NGCS and/or to a PSAP. The NGCS and PSAP may have separate Logging Services or may use the same Logging Service.	Add
LPG (Legacy PSAP Gateway)	LPG (Legacy PSAP Gateway) is a signaling and media interconnection point between an ESInet and a legacy PSAP. It plays a role in the delivery of Emergency Calls that traverse an i3 ESInet to get to a legacy PSAP, as well as in the transfer and alternate routing of Emergency Calls between legacy PSAPs and NG9-1-1 PSAPs. The Legacy PSAP Gateway supports an IP (i.e., SIP) interface towards the ESInet on one side, and a traditional MF or Enhanced MF interface (comparable to the interface between a traditional Selective Router and a legacy PSAP) on the other.	In Glossary
LVF (Location Validation Function)	LVF (Location Validation Function) is a Functional Element in an NGCS (Next Generation 9 1 1 Core Services) that is a LoST protocol server where civic location information is validated against the authoritative GIS database information. A civic address is considered valid if it can be located within the database uniquely, is suitable to provide an accurate route for an Emergency Call, and adequate and specific enough to direct responders to the right location.	In Glossary

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
MDS (Mapping Data Service)	A MDS (Mapping Data Service) provides a PSAP call taker with information showing the location of an out-of-area caller.	In Glossary
MCFE (Management Console Functional Element)	An FE that supports general management functions for the PSAP. It is responsible for the PSAP's Service State and Security Posture and subscribes to important PSAP/ECC notifications. It also sends and receives Discrepancy Reports on behalf of the PSAP and implements the LogEvent web service to receive LogEvents for statistical and analysis purposes.	Modify (include abbr.)
Media Proxy	An FE that provides external one-way non-interactive media to an Agent by acting as a proxy for the external media source.	Add
MIS (Management Information System)	MIS (Management Information System) is a computer system that collects, stores and correlates data from multiple systems and allows users to readily access data support to operational and strategic decision making on performance, trends, traffic capacities, etc.	In Glossary
NAT (Network Address Translation)	Network Address Translation is a way to map multiple private addresses inside a local network to a public IP address before transferring information to or from the internet.	Add
NCIC (National Crime Information Center)	NCIC (National Crime Information Center) is an electronic clearinghouse of crime data that can be tapped into by virtually every criminal justice agency nationwide, 24 hours a day, 365 days a year. It helps criminal justice professionals apprehend fugitives, locate missing persons, recover stolen property, and identify terrorists. It also assists law enforcement officers in performing their duties more safely and provides information necessary to protect the public.	In Glossary
NGCS (Next Generation 9 1 1 Core Services)	NGCS (Next Generation 9 1 1 Core Services) is the set of services needed to process a 9 1 1 call on an ESInet. It includes, but is not limited to, the ESRP, ECRF, LVF, BCF, Bridge, Policy Store, Logging Services, and typical IP services such as DNS and DHCP. The term NG9 1 1 Core Services includes the services and not the network on	In Glossary

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
	which they operate. See Emergency Services IP Network.	
Non-Emergency Call	A call to an Agency that does not pertain to an emergency, nor does it require immediate law enforcement, fire rescue, or EMS service dispatch of resources. Non-Emergency Calls do not contain any specialized signaling for priority communications network treatment (e.g., like a 911 Emergency Call that contains location data). Non-Emergency Calls typically use a ten-digit telephone dialing procedure for a public switched network established in the local service area or country.	In Glossary (add as related term to Call)
NRS (NENA Registry System)	NRS (NENA Registry System) is the entity provided by NENA to manage registries.	In Glossary
NWS (National Weather Service)	The National Weather Service (NWS) is an agency of the U.S. government that provides weather forecasts, warnings, and other weather-related information to the public.	Don't Add
OCIF (Outbound Call Interface Function)	OCIF (Outbound Call Interface Function) is part of the NGCS responsible for handling calls originating from i3-PSAPs over their serving ESInet/NGCS.	In Glossary
Opus	Opus is a lossy audio format developed by the Xiph.org Foundation and standardized by the Internet Engineering Task Force in RFC 6716 [37].	Add
PAI (P-Asserted-Identity)	A header field in a SIP message containing a URI that the originating network asserts is the correct identity of the caller.	In Glossary
PASSporT (Personal Assertion Token)	PASSporT (Personal Assertion Token) is a cryptographically signed token to protect the integrity of the identity of the originator and to verify the assertion of the identity information at the destination.	In Glossary
PIDF (Presence Information Data Format)	The PIDF (Presence Information Data Format) is specified in IETF RFC 3863 [54]. It provides a common presence data format for Presence protocols, and also defines a new media type. A presence protocol is a protocol for providing a presence service over the Internet or any IP network.	In Glossary

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
Presence Server	An FE within the PSAP/ECC that is responsible for receiving activity state information from other FEs, storing it, and distributing agent presence information to subscribing entities.	Add
Presentity	An entity that has presence information associated with it.	Add
PSAP (Public Safety Answering Point)	PSAP (Public Safety Answering Point) is a physical or virtual entity where 9-1-1 calls are delivered by the 9 1 1 Service Provider.	In Glossary
PSTN (Public Switched Telephone Network)	PSTN (Public Switched Telephone Network) is the network of equipment, lines, and controls assembled to establish communication paths between calling and called parties in North America.	In Glossary
QoS (Quality of Service)	As related to data transmission, Quality of Service is a measurement of latency, packet loss, and jitter.	In Glossary
Queue	A Queue is a stored arrangement of calls or data waiting to be processed.	In Glossary
RDS (Responder Data Services)	RDS (Responder Data Services) is a Functional Element that enables near real time wireless data transmissions between PSAPs and emergency responder devices. This includes transmitting dispatch information, creating new Incidents, updating active and closed Incidents. The Responder Data Services FE can support the transmission and receipt of media, or a reference (i.e., URL) to that media.	In Glossary
RTT (Real Time Text)	RTT (Real Time Text) is a text transmission that is character at a time, as in TTY. Technology that allows consumers to send and receive Internet Engineering Task Force (IETF) RFC 4103 [76] text characters, as they are typed, as well as audio simultaneously.	In Glossary
Security Posture	Security Posture is an event, part of Service State, which represents a downstream entity’s current security state (Green for normal, Yellow for suspicious activity, Orange for fraudulent events, Red for under active attack).	In Glossary
Service	Represents a set of FE instances, addressable with a single Service Identifier, that provides the functionality and interfaces specified for the FE in this document.	Add

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
Service State	Service state is an event package defined in NENA-STA-010 **Error! Reference source not found.**, that indicates the state of a service either automatically determined, or as determined by management. It encompasses the state of the designated service, inclusive of its security posture when supported.	Add
Service/Agency Locator	The Service/Agency Locator is a directory ("white pages" and "yellow pages") of agencies, together with key information about the service or agency. The Service/Agency Locator is a distributed database.	Add
SIP (Session Initiation Protocol)	SIP (Session Initiation Protocol) is a protocol specified by the IETF RFC 3261 [77] that defines a method for establishing multimedia sessions over the Internet. Used as the call signaling protocol in VoIP, NENA i2, and NENA i3.	In Glossary
SIP URI	Identifies a communications resource and contains sufficient information to initiate and maintain a SIP communication session with the resource. Its general form is sip:user:password@host:port;uri-parameters?headers	Add (add to URI)
SLA (Service Level Agreement)	An agreement between the provider of a service and a user of that service specifying the level of service that will be provided.	Add
SRV (Service)	SRV (Service) is a specification of data in the Domain Name System defining the location, i.e., the hostname and port number, of servers for specified services.	In Glossary
TEL URI	A TEL URI is a telephone number represented as a SIP URI, for example: tel:+1-213-666-6422	Add (add to URI)
Test Call	An Emergency Call initiated by a system into the 911 ecosystem that serves to monitor the health of the 911 NGCS and CHFE. Test Calls are differentiated from actual Emergency Calls by specific network signaling (e.g., test calls are signaled with both the urn:service:test.sos and urn:emergency:service:test Service URN trees).	Add (add as Related Term to Call)
Time Server	A Time Server is a Functional Element that provides NTP (Network Time Protocol) time services to other Functional Elements.	In Glossary
UA (User Agent)	As defined for SIP in IETF RFC 3261 [77], the UA (User Agent) represents an endpoint in the IP domain, a logical entity that can act as both a	In Glossary

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Commented [MLS168]: Ask i3 if they will add this

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Commented [MLS170]: Ask i3 if they will add this.

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Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
	UAC (User Agent Client) that sends requests, and as UAS (User Agent Server) responding to requests.	
UAC (User Agent Client)	Refer to IETF RFC 3261 [77] for the following definition. "A user agent client is a logical entity that creates a new request, and then uses the client transaction state machinery to send it. The role of UAC lasts only for the duration of that transaction. In other words, if a piece of software initiates a request, it acts as a UAC for the duration of that transaction. If it receives a request later, it assumes the role of a user agent server for the processing of that transaction."	In Glossary
UAS (User Agent Server)	Refer to IETF RFC 3261 [77] for the following definition. "A user agent server is a logical entity that generates a response to a SIP request. The response accepts, rejects, or redirects the request. This role lasts only for the duration of that transaction. In other words, if a piece of software responds to a request, it acts as a UAS for the duration of that transaction. If it generates a request later, it assumes the role of a user agent client for the processing of that transaction."	In Glossary
URI (Uniform Resource Identifier)	URI (Uniform Resource Identifier) is an identifier consisting of a sequence of characters matching the syntax rule that is named <URI> in RFC 3986 [78]. It enables uniform identification of resources via a set of naming schemes. A URI can be further classified as a locator, a name, or both. The term "Uniform Resource Locator" (URL) refers to the subset of URIs that, in addition to identifying a resource, provides a means of locating the resource by describing its primary access mechanism (e.g., its network "location"). The term "Uniform Resource Name" (URN) has been used historically to refer to both URIs under the "urn" scheme (RFC 2141 [79]), which are required to remain globally unique and persistent even when the resource ceases to exist or becomes unavailable, and to any other URI with the properties of a name. An example of a URI that is neither a URL nor a URN is sip:psap@example.com.	In Glossary
URL (Uniform Resource Locator)	A type of URI, specifically used for describing and navigating to a resource (e.g., https://www.nena.org).	In Glossary

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
Virtual PSAP	An operational model directly enabled through NG9 1 1 features and/or network hosted PSAP equipment in which telecommunicators are geographically dispersed, rather than working from the same physical location. Remote access to the PSAP applications by the dispersed telecommunicators requires the appropriate network connections, security, and work station equipment at the remote location. The virtual work place may be a logical combination of physical PSAPs, or an alternate work environment such as a satellite facility, or any combination of the above. Workers are connected and interoperate via IP connectivity.	In Glossary
VLAN (Virtual Local Area Network)	A broadcast domain that is partitioned and isolated within a network at the data link layer. A single physical local area network (LAN) can be logically partitioned into multiple, independent VLANs; a group of devices on one or more physical LANs can be configured to communicate within the same VLAN, as if they were attached to the same physical LAN.	Don't Add
VPN (Virtual Private Network)	A network implemented on top of another network, and private from it, providing transparent services between networks or devices and networks. VPNs often use some form of cryptographic security to provide this separation.	In Glossary
Web Service	A Web service is a self-contained, self-describing, modular application that can be published, located, and invoked across the Web. Web services perform functions that can be anything from simple requests to complicated business processes.	In Glossary
XML (eXtensible Markup Language)	XML (eXtensible Markup Language) is an internet specification for web documents that enables tags to be used that provide functionality beyond that in Hyper Text Markup Language (HTML). In contrast to HTML, XML has the ability to allow information of indeterminate length to be transmitted to a PSAP call taker or dispatcher versus the current restriction that requires information to fit the parameters of pre-defined fields.	In Glossary
YAML (YAML Ain't Markup Language)	YAML (YAML Ain't Markup Language) is a human-readable data-serialization language; commonly used for configuration files and in applications where data is being stored or transmitted.	In Glossary

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
		Select
		Select
		Select

10 REFERENCES

NOTE to document editor: Remove reference [4], [5], and [6] if the Describing Artifacts section was removed from the beginning of the document and it is not referenced elsewhere in the document.

[1] National Emergency Number Association. "NENA Knowledge Base Glossary." Updated [insert the date indicated at "Approved updates to NENA Knowledge Base Glossary last made" as found at the cited link in the format Month DD, YYYY]. https://kb.nena.org/wiki/Category:Glossary.

[2] Internet Engineering Task Force. *Key words for use in RFCs to Indicate Requirement Levels*. S. Bradner. RFC 2119, March 1997.

[3] Internet Engineering Task Force. *Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words*. B. Leiba. RFC 8174, May 2017.

[4] National Emergency Number Association. *NENA Rules for Use of Version Control Repositories*. NENA-ADM-012.1-2022. Alexandria, VA: NENA, approved November 8, 2022.

[5] OpenAPI Initiative. *OpenAPI Specification 3.1.0 Released*. https://www.openapis.org/blog/2021/02/18/openapi-specification-3-1-released, February 18, 2021.

[6] World Wide Web Consortium (W3C). *W3C XML Schema Definition Language (XSD) 1.1 Part 1: Structures*. S. Gao, C. M. Sperberg-McQueen, and H. Thompson. https://www.w3.org/TR/xmlschema11-1/, April 5, 2012.

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When citing a NENA document, do not include an errata version letter in the document number and use the NENA Standards page URL (https://www.nena.org/page/standards) as the link to the document. Links to material outside of this document should be limited to the web page where the referenced document may be downloaded, as opposed to a link directly to the specific document. Questions regarding formatting references should be directed to the Committee Resource Management team at crm@nena.org.

[7] National Emergency Number Association. *NENA i3 Standard for Next Generation 9-1-1*. NENA-STA-010.3-2021. Alexandria, VA: NENA, approved July 12, 2021.

DRAFT

[8] National Emergency Number Association. *Emergency Services IP Network Design Information Document*. NENA-INF-016.2-2018. Alexandria, VA: NENA, approved April 5, 2018.

[9] Association of Public Safety Communications Officials, Intl. and the National Emergency Number Association. *APCO/NENA Emergency Communications Center (ECC) Service Capability Criteria Rating Scale*. ANS APCO/NENA 1.102.3-2020. Daytona Beach, FL: APCO, approved 2020.

[10] National Emergency Number Association. *NENA E9-1-1 PSAP Equipment Standards*. NENA-STA-027.3.1-2025. Alexandria, VA: NENA, approved December 8, 2025.

[11] National Emergency Number Association. *NENA PSAP Site Characteristics Information Document*. NENA-INF-024.2-2018. Alexandria, VA: NENA, approved February 14, 2018.

[12] National Emergency Number Association. *NENA Security for Next Generation 9-1-1 Standard (NG-SEC)*. NENA-STA-040.2-2024. Alexandria, VA: NENA, approved October 28, 2024.

[13] U.S. Department of Justice, Federal Bureau of Investigation. *Criminal Justice Information Services (CJIS) Security Policy*. CJISD-ITS-DOC-08140-5.8. Washington, DC: USDOJ December 27, 2024

[14] Internet Engineering Task Force. *An Architecture for Differentiated Services*. S. Blake, et. al. RFC 2475, December 1998.

[15] Internet Engineering Task Force. *RTP: A Transport Protocol for Real-Time Applications (RTCP)*. H. Schulzrinne, et. al. RFC 3550, July 2003.

[16] Internet Engineering Task Force. *Secure Real-time Transport Control Protocol (SRTCP)*. M Baugher, et. al. RFC 3711, March 2004.

[17] OASIS Open. *eXtensible Access Control Markup Language (XACML) Version 3.0*. OASIS eXtensible Access Control Markup Language (XACML) Technical Committee. XACML-V3.0. January 22, 2013.

[18] Internet Engineering Task Force. *Session Initiation Protocol (SIP) Event Notification Extension for Notification Rate Control.* A. Niemi, et. al. RFC 6446, January 2012.

[19] Internet Engineering Task Force. *Hypertext Transfer Protocol - HTTP/1.1*. R. Fielding, et. al. RFC 2616, June 1999.

[20] Internet Engineering Task Force. *HyperText Transfer Protocol Over TLS (HTTPS)*. E. Rescorla. RFC 2818, May 2000.

[21] Internet Engineering Task Force. *PASSporT: Personal Assertion Token.* C. Wendt, et. al. RFC 8225, February 2018

[22] National Emergency Number Association. *NENA Standard for Emergency Incident Data Objects (EIDO)*. NENA-STA-021.1-2021. Alexandria, VA: NENA, approved October 14, 2021.

[23] Internet Engineering Task Force. *DNS Resource Record (RR) for specifying the location of services (DNS SRV).* A. Gulbrandsen, et. al. RFC 2782, February 2000.

[24] Internet Assigned Numbers Authority. *Internet Assigned Numbers Authority (IANA) Procedures for the Management of the Service Name and Transport Protocol Port Number Registry*. M. Cotton, et. al. RFC 6335, August 2011.

[25] National Emergency Number Association. *NENA Conveyance of Emergency Incident Data Objects (EIDOs) between Next Generation (NG9-1-1) Systems and Applications*. NENA-STA-024.1.1-2025. Alexandria, VA: NENA, approved January 16, 2025.

[26] National Emergency Number Association. *NENA Request for Proposal (RFP) Considerations Information Document*. NENA-INF-021.1-2020. Alexandria, VA: NENA, approved January 19, 2020.

[27] Internet Engineering Task Force. *An Extension to the Session Description Protocol (SDP) and Real-time Transport Protocol (RTP) for Media Loopback*. H. Kaplan, Ed., et. al. RFC 6849, February 2013.

[28] Internet Engineering Task Force. *A Location Dereference Protocol Using HTTP-Enabled Location Delivery (HELD)*. J. Winterbottom, et. al. RFC 6753, October 2012.

[29] Internet Engineering Task Force. *A Presence Event Package for the Session Initiation Protocol (SIP)*. J. Rosenberg. RFC 3856, August 2004.

[30] Internet Engineering Task Force. *Filtering Location Notifications in the Session Initiation Protocol (SIP)*. R. Mahy, B. Rosen. H. Tschofenig. RFC 6447, January 2012.

[31] SIPconnect SIP Forum. *SIP-PBX / Service Provider Interoperability 2.0 Technical Recommendation*. A. Hutton, G. Salguerio. SIP Forum Document Number: TWG-11, 2016.

[32] Internet Engineering Task Force. *Framework and Security Considerations for Session Initiation Protocol (SIP) URI-List Services*. A. Roach, G. Camarillo. RFC 5363, October 14, 2015.

[33] Internet Engineering Task Force. *Conference Establishment Using Request-Contained Lists in the Session Initiation Protocol (SIP)*. G. Camarillo, A. Johnston. RFC 5366, October 2008.

[34] International Telecommunication Union. *G.711: Pulse Code Modulation (PCM) of Voice Frequencies, ITU-T Recommendations*. https://www.itu.int/rec/T-REC-G.711/en, 2010.

[35] International Telecommunication Union. *Wideband coding of speech at around 16 kbit/s using Adaptive Multi-Rate Wideband (AMR-WB) G.722.2*. https://www.itu.int/rec/T-REC-G.722.2/en, July, 2003.

[36] European Telecommunications Standards Institute. *Universal Mobile Telecommunications System (UMTS); LTE; Codec for Enhanced Voice Services (EVS); Performance characterization (3GPP TR 26.952 version 18.0.0 Release 18).* https://www.etsi.org/deliver/etsi_tr/126900_126999/126952/18.00.00_60/tr_126952v180000p.pdf, 2024.

[37] Internet Engineering Task Force. *Definition of the Opus Audio Codec*. J.M. Valin, et. al. RFC 6716, September 2012.

[38] International Telecommunication Union. *ITU-T Coding of speech at 8 kbit/s using conjugate-structure algebraic-code-excited linear prediction (CS-ACELP), Annex J.* G.729, 2006.

[39] Internet Engineering Task Force. *RTP-Mixer Formatting of Multiparty Real-Time Text*. G. Hellstrom. RFC 9071, July 2021.

[40] Crypto Museum Baudot-Murray code, ITA-2, https://www.cryptomuseum.com/ref/ita2/index.htm.

[41] Internet Engineering Task Force. *Reliability of Provisional Responses in the Session Initiation Protocol (SIP)*. J. Rosenberg, H. Schulzrinne. RFC 3262, June 2002.

[42] Internet Engineering Task Force. *The tel URI for Telephone Numbers*. H. Schulzrinne. RFC 3966, section 5.1.4, December 2004.

[43] Internet Engineering Task Force. *Public Safety Answering Point (PSAP) Callback*. H. Schulzrinne, et. al. RFC 7090, April 2014.

[44] Internet Engineering Task Force. *Multiple Language Content Type*. N. Tomkinson, Dr. N. Borenstein. RFC 8255, October 10, 2017.

[45] Internet Engineering Task Force. *Long-Term Viability of Protocol Extension Mechanisms*. M. Thomson, T. Pauly. RFC 9170, December 2021.

[46] Internet Engineering Task Force. *A Session Initiation Protocol (SIP) Event Package for Conference State*. J. Rosenberg, et. al. RFC 4575, August 2006.

[47] Internet Engineering Task Force. *Clarifications for the Use of REFER with RFC 6665*. R. Sparks, A. Roach. RFC 7647, October 14, 2015.

[48] Internet Engineering Task Force. *Session Recording Protocol (SIPREC)*. L. Portman, et. al. RFC 7866, May 2016.

[49] Internet Engineering Task Force. *Basic Network Media Services with SIP*. E. Burger, Ed., et. al. RFC 4240, December 2005.

[50] Internet Engineering Task Force. *RPID: Rich Presence Extensions to the Presence Information Data Format (PIDF)*. H. Schulzrinne, et. al. RFC 4480, July 2006.

DRAFT

[51] Internet Engineering Task Force. *CIPID: Contact Information for the Presence Information Data Format*. H. Schulzrinne. RFC 4482, July 2006.

[52] Internet Engineering Task Force. *Session Initiation Protocol Service Example -- Music on Hold*. D. Worley. RFC 7088, February 2014.

[53] Internet Engineering Task Force. *Negotiating Human Language in Real-Time Communications*. R. Gellens. RFC 8373, May 2018.

[54] Internet Engineering Task Force. *Presence Information Data Format (PIDF)*. H. Sugano, et. al. RFC 3863, August 2004.

[55] Internet Engineering Task Force. *A Session Initiation Protocol (SIP) Event Notification Extension for Resource Lists*. A. B. Roach, et. al. RFC 4662, August 2006.

[56] Internet Engineering Task Force. *An Extensible Markup Language (XML)-Based Format for Event Notification Filtering*. H. Khartabil, et. al. RFC 4661, September 2006.

[57] Internet Engineering Task Force. *A Presence-based GEOPRIV Location Object Format*. J Peterson. RFC 4119, December 2005.

[58] Internet Engineering Task Force. *Revised Civic Location Format for Presence Information Data Format Location Object (PIDF-LO)*. M. Thomson, J. Winterbottom. RFC 5139, February 2008.

[59] Internet Engineering Task Force. *GEOPRIV Presence Information Data Format Location Object (PIDF-LO) Usage Clarification, Considerations, and Recommendations*. J. Winterbottom, M. Thomson. RFC 5491, March 2009.

[60] Internet Engineering Task Force. *Representation of Uncertainty and Confidence in the Presence Information Data Format Location Object (PIDF-LO)*. M. Thomson, J. Winterbottom. RFC 7459, February 2015.

[61] Internet Engineering Task Force. *Session Initiation Protocol (SIP) Extension for Event State Publication*. A. Niemi, Ed. RFC 3903, October 2004.

[62] World Wide Web Consortium (W3C). *Voice Browser Call Control, CCXML Version 1*. R.J. Auburn, Paolo Baggia, Mark Scott. https://www.w3.org/TR/ccxml/, July 5, 2011.

[63] World Wide Web Consortium (W3C). *Voice Extensible Markup Language (VoiceXML) Version 2.0*. S. McGlashan, et. al. https://www.w3.org/TR/voicexml20/, March 16, 2004.

[64] Internet Engineering Task Force. *SIP Interface to VoiceXML Media Services*. D. Burke, M. Scott. RFC 5552, May 2009.

[65] Internet Engineering Task Force. *A Data Model for Presence*. J. Rosenberg. RFC 4479, July 2006.

[66] Internet Engineering Task Force. *Session Initiation Protocol (SIP)-Specific Event Notification*. A. B. Roach. RFC 3265, June 2002.

[67] Internet Engineering Task Force. *Real-Time Streaming Protocol (RTSP) Version 2.0*. H. Schulzrinne, et. al. RFC 7826, December 2016.

[68] Internet Engineering Task Force. *Real Time Streaming Protocol (RTSP) RTSP 1.0*. H. Schulzrinne, et. al. RFC 2326, April 1998.

[69] National Emergency Number Association. *NG9-1-1 Call Processing Metrics Standard*. NENA-STA-019.2-2022. Alexandria, VA: NENA, approved September 21, 2022.

[70] 3rd Generation Partnership Project (3GPP). *Technical Specification Group Services and System Aspects; Mission Critical Push To Talk (MCPTT); Stage 1 (Release 20)*. 3GPP Rel-20, March 28, 2025.

[71] Internet Engineering Task Force. *Session Initiation Protocol (SIP) Recording Metadata*. R. Ravindranath, et. al. RFC 7865, May 2016.

[72] National Emergency Number Association. *NENA Registry System (NRS) Standard*. NENA STA 008.2-2014. Alexandria, VA: NENA, approved October 6, 2014.

[73] Internet Engineering Task Force. *Guidelines for Writing an IANA Considerations Section in RFCs*. M. Cotton, et. al. RFC 8126, June 2017.

[74] Internet Engineering Task Force. *Simple Mail Transfer Protocol*. J. Klensin. RFC 5321, October 2008.

Commented [MS173]: TODO: This *was* referenced in a footnote that referred to “Media”; the footnote was deleted. RFC 4479 defines <device>, which we don’t currently use, but maybe we should - it allows you to specify a unique deviceID. Section 7.1 in 4479 gives an example. Should we be using <device> in our PIDF? Our <FE> element defines the device *type*, not the user’s specific device. Discuss.

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[75] 3rd Generation Partnership Project (3GPP). *Codec for Enhanced Voice Services (EVS)*. https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=1463, October 22, 2025.

[76] Internet Engineering Task Force. *RTP Payload for Text Conversation*. G. Hellstrom, P. Jones. RFC 4103, June 2006.

[77] Internet Engineering Task Force. *SIP: Session Initiation Protocol*. J. Rosenberg, et. al. RFC 3261, June 2002.

[78] Internet Engineering Task Force. *Uniform Resource Identifier (URI): Generic Syntax*. T. Berners-Lee, et. al. RFC 3986, January 2005.

[79] Internet Engineering Task Force. *URN Syntax*. R. Moats. RFC 2141, May 1997.

[80] Internet Assigned Numbers Authority (IANA). *Session Initiation Protocol (SIP) Event Types Namespace*. https://www.iana.org/assignments/sip-events/sip-events.xhtml, May 2, 2014.

[81] Internet Assigned Numbers Authority (IANA). *Media Types*. Accessed December 11, 2025. https://www.iana.org/assignments/media-types/media-types.xhtml, March 24, 2026.

[82] Internet Engineering Task Force. *Early Media and Ringing Tone Generation in the Session Initiation Protocol (SIP)*. G. Camarillo, H. Schulzrinne. RFC 3960, December 2004.

[83] National Emergency Number Association. *NENA Managing & Monitoring NG9-1-1 Information Document*. NENA-INF-040.2-2022. Alexandria, VA: NENA, approved 07/27/2022.

[84] NG9-1-1 Interoperability Oversight Commission (NIOC) website.

[85] OASIS Open. *eXtensible Access Control Markup Language (XACML) Version 2.0*. OASIS eXtensible Access Control Markup Language (XACML) Technical Committee. XACML-V2.0. February 1, 2005.

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- *NENA i3 Standard for Next Generation 9-1-1* [5]
- NENA-STA-010 [5]

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11 APPENDICES

11.1 Appendix A—DRM Policy Examples

11.1.1 IRR Policy Example

Commented [MS174]: TODO: This example will have to be edited to agree with whatever i3 adopts for XACML policies. Brian has proposed standardizing the attribute types for interfaces, subject IDs, and data items. He also proposed that 3.1 specifies XACML 3.0 as OPTIONAL and states that 2.0 will be deprecated in a future revision. Should the PSAP specify XACML 3.0, period? PSAPs and NGCS operators won't be using each others policies. NG PSAP and i3v4 will be published at different times.

This example XACML 2.0 policy document is named "IrrPolicy". It defines access control rules for 9-1-1 call takers to replay or display media from recent emergency calls using the Instant Recall Recorder (IRR) function, based on specific requirements including roles, agent IDs, and time constraints.

The resource being protected is "Irr", which only allows "Read" access.

The rules in this example provide the following permissions:

- A Supervisor can replay or display media from any call
- An Agent with the "Call Taking" role can replay or display media only from their own calls, for up to an hour after the call has ended.
- Other attempts to replay or display media are denied.

The name of the policy is "IrrPolicy".

Commented [MS175]: policyType?

"AgentAgentId" is the ID of the requesting Agent

"CallAgentId" is the ID of the Agent that handled the call in question

"Role" in the rule represents the role string from the Agent's PCA certificate Subject Alt Name field.

"CallEndTime" represents the date and time the call ended (from the CallEndLogEvent timestamp), if the call has ended. The current date and time could be used if the call was still in progress.

The XACML policy document is shown below:

Formatted: Body Text

```
<?xml version="1.0" encoding="UTF-8"?>
<Policy xmlns="urn:oasis:names:tc:xacml:2.0:policy:schema:os"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    PolicyId="IrrPolicy"
    RuleCombiningAlgId="urn:oasis:names:tc:xacml:1.0:rule-combining-algorithm:first-
applicable">

  <Description>
    This policy controls whether a 9-1-1 call taker is permitted to replay media from recent
emergency calls.
  </Description>

  <Target>
    <!-- Policy applies to Read actions on Irr resources -->
    <Resources>
      <Resource>
        <ResourceMatch MatchId="urn:oasis:names:tc:xacml:1.0:function:string-equal">
          <AttributeValue
DataType="http://www.w3.org/2001/XMLSchema#string">Irr</AttributeValue>
          <ResourceAttributeDesignator
```

Commented [MS176]: See xacml 1.0 7.8. Hierarchical resources - it specifies how a PEP should interpret data items with children. I think we need to add urn:oasis:names:tc:xacml:1.0:resource:scope attribute to what Brian specified in his contribution email. See: oasis-xacml-1.0.pdf

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```
            AttributeId="resource-type"
            DataType="http://www.w3.org/2001/XMLSchema#string"/>
        </ResourceMatch>
      </Resource>
    </Resources>
    <Actions>
      <Action>
        <ActionMatch MatchId="urn:oasis:names:tc:xacml:1.0:function:string-equal">
          <AttributeValue
DataType="http://www.w3.org/2001/XMLSchema#string">Read</AttributeValue>
          <ActionAttributeDesignator
            AttributeId="action-id"
            DataType="http://www.w3.org/2001/XMLSchema#string"/>
        </ActionMatch>
      </Action>
    </Actions>
  </Target>

  <!-- Rule 1: Supervisors can replay any call -->
  <Rule RuleId="supervisor-access" Effect="Permit">
    <Description>Supervisors can replay any call media</Description>
    <Condition>
      <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:string-is-in">
        <AttributeValue
DataType="http://www.w3.org/2001/XMLSchema#string">Supervisor</AttributeValue>
        <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:string-one-and-only">
          <SubjectAttributeDesignator
            AttributeId="Role"
            DataType="http://www.w3.org/2001/XMLSchema#string"/>
        </Apply>
      </Apply>
    </Condition>
  </Rule>

  <!-- Rule 2: Call takers can replay their own calls within one hour -->
  <Rule RuleId="calltaker-own-recent-call" Effect="Permit">
    <Description>Call takers can replay their own calls within one hour of call end
time</Description>
    <Condition>
      <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:and">
        <!-- Check if Role contains "Call Taking" -->
        <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:string-is-in">
          <AttributeValue DataType="http://www.w3.org/2001/XMLSchema#string">Call
Taking</AttributeValue>
          <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:string-one-and-only">
            <SubjectAttributeDesignator
              AttributeId="Role"
              DataType="http://www.w3.org/2001/XMLSchema#string"/>
          </Apply>
        </Apply>

        <!-- Check if AgentAgentId equals CallAgentId -->
        <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:string-equal">
```

DRAFT

```
     <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:string-one-and-only">
      <SubjectAttributeDesignator
       AttributeId="AgentAgentId"
       DataType="http://www.w3.org/2001/XMLSchema#string"/>
     </Apply>
     <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:string-one-and-only">
      <ResourceAttributeDesignator
       AttributeId="CallAgentId"
       DataType="http://www.w3.org/2001/XMLSchema#string"/>
     </Apply>
    </Apply>

    <!-- Check if CurrentTime <= CallEndTime + 1 hour -->
    <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:dateTime-less-than-or-
equal">
     <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:dateTime-one-and-only">
      <EnvironmentAttributeDesignator
       AttributeId="current-dateTime"
       DataType="http://www.w3.org/2001/XMLSchema#dateTime"/>
     </Apply>
     <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:dateTime-add-
dayTimeDuration">
      <Apply FunctionId="urn:oasis:names:tc:xacml:1.0:function:dateTime-one-and-only">
       <ResourceAttributeDesignator
        AttributeId="CallEndTime"
        DataType="http://www.w3.org/2001/XMLSchema#dateTime"/>
      </Apply>
      <AttributeValue
DataType="http://www.w3.org/2001/XMLSchema#dayTimeDuration">PT1H</AttributeValue>
     </Apply>
    </Apply>
   </Apply>
  </Condition>
 </Rule>

 <!-- Rule 3: Default deny rule -->
 <Rule RuleId="default-deny" Effect="Deny">
  <Description>Default rule - deny access to call media replay</Description>
 </Rule>

</Policy>
```

11.2 Appendix B—PIDF Examples with Activity State Extensions

Commented [SD177]: Please review the coding below. It looks like it may have shifted and I'm unsure if this is a concern.

Commented [MS177R2]: TODO: change this yaml back to two spaces per tab.

This section contains examples of PIDF documents that contain the activityState, agentRoles, and reasonCode extensions to PIDF that support functionality of the Presence Server FE. These extensions are defined in the Presentity Activity States for the Presence Server section of this document.

The following PIDF example shows what might be sent to the Presence Server by a CallHandling FE at the beginning of an emergency voice+video call. The "sessions" array shows that the Agent is in a single call (a "session") that has incoming voice and video media, and outgoing voice media without video (as would be the case if no camera was present on the Agent's workstation).

- The "supportedIncomingMedia" member indicates the FE supports incoming voice, video, and instant messaging, but does NOT support incoming real time text media – note that rttMedia is omitted in the example.
- The "supportedOutgoingMedia" member indicates outgoing voice and instant messaging are supported, but outgoing video and real time text are NOT supported – note that videoMedia and rttMedia are omitted.
- Note that the CallHandling FE MAY provide the Agent's availability for a particular medium from its point of view. Our example shows imMedia as NOT available. The Presence Server will specify Agent availability according to policy and report that value to subscribers.
- The Presentity is "John.Smith@myagency.mycounty.gov".
- John's contact URI is "sip:jsmith@agency911.gov".
- John agentRoles are "CallTaking,Supervisor".

See RFC 3863 [54] for PIDF specifications.

```
<?xml version="1.0" encoding="UTF-8"?>
<impp:presence
        xmlns:xs="http://www.w3.org/2001/XMLSchema"
        xmlns:impp="urn:ietf:params:xml:ns:pidf"
        xmlns:as="urn:emergency:xml:ns:pidfExt:ActivityState"
  entity="John.Smith@myagency.mycounty.gov" >
        <impp:tuple id="sg89ae">
                <impp:status>
                        <impp:basic>open</impp:basic>
                        <!-- Overall Agent availability -->
                        <as:activityState>
                                <as:fe impp:mustUnderstand="true">CallHandling</as:fe>
                                <as:supportedIncomingMedia>
                                        <as:voiceMedia>
                                                <as:isAvailable>false</as:isAvailable>
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                        </as:voiceMedia>
                                        <as:videoMedia>
                                                <as:isAvailable>false</as:isAvailable>
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                        </as:videoMedia>
                                        <as:imMedia>
                                                <as:isAvailable>true</as:isAvailable>
                                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                                        </as:imMedia>
                                        <as:rttMedia>
                                        </as:rttMedia>
                                </as:supportedIncomingMedia>
                                <as:supportedOutgoingMedia>
                                        <as:voiceMedia>
                                                <as:isAvailable>false</as:isAvailable>
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                        </as:voiceMedia>
```

Commented [MS178]: TODO: How do we refer to CallHandling activityState as opposed to PttSystemServices activityState? I think we want to be able to do that by specifying something like this: PresenceServer~SIMPLEpresence.presence.tuple.status.activityState.CallHandling
See the example in the DRM section.

```
                        <as:imMedia>
                                <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                        </as:imMedia>
                </as:supportedOutgoingMedia>
                <as:sessions>
                        <!--array of 0 or more session elements-->
                        <as:session>
                                <as:incomingMedia>
                                        <as:mediaType>Voice</as:mediaType>
                                        <as:mediaType>Video</as:mediaType>
                                         </as:incomingMedia>
                                <as:outgoingMedia>
                                        <as:mediaType>Voice</as:mediaType>
                                         </as:outgoingMedia>
                                <as:isEmergency>true</as:isEmergency>
                        </as:session>
                </as:sessions>
            </as:activityState>
        </impp:status>
        <impp:contact priority="0.8">sip:jsmith@agency911.gov</impp:contact>
        <as:agentRoles>CallTaking,Supervisor</as:agentRoles>
        <impp:timestamp>2025-01-06T16:49:29Z</impp:timestamp>
    </impp:tuple>
</impp:presence>
```

If Agency policy doesn't allow the Agent to receive instant messages while on a voice+video call, the Presence Server could change <isAvailable> and <isAvailableForEmergency> to "false" for <imMedia> before sending to subscribers.

The standard PIDF element <basic> denotes whether the Agent is available or unavailable in general. <basic> MUST be included in any PIDF sent to or from the Presence Server. The value of <basic> is either "closed" (unavailable) or "open" (available). The "closed" value is used when the Agent can't be readily contacted, like when on a break or out for a meal.

The next example shows how an FE can set <basic> to "closed" when the agent is on break, wrapping up after a call, at lunch, etc. The Presence Server could also set <basic> to "closed" if dictated by policy when notifying subscribers of a change in Agent status. If either the Presence Server or another FE sets the value of <basic> to "closed", denoting that the Agent is not available for any call, it MAY supply an optional reasonCode from the list in section ?.? giving the reason the Agent is not available.

The following PIDF example denotes that the Agent in question is at lunch: <basic> is set to "closed" and the <reasonCode> "Lunch" indicates the reason the Agent is not available. The <sessions> array is empty since there are no current sessions.

Commented [MS179]: "Lunch" is customarily used to denote any meal break in existing agent state implementations.

```
<?xml version="1.0" encoding="UTF-8"?>
<impp:presence
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:impp="urn:ietf:params:xml:ns:pidf"
    xmlns:as="urn:emergency:xml:ns:pidfExt:ActivityState"
  entity="John.Smith@myagency.mycounty.gov" >
    <impp:tuple id="sg89ae">
        <impp:status>
```

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```
				<impp:basic>closed</impp:basic>
				<!-- Overall Agent availability -->
				<as:reasonCode>Lunch</as:reasonCode>
				<as:activityState>
					<as:fe impp:mustUnderstand="1">CallHandling</as:fe>
					<as:supportedIncomingMedia>
						<as:voiceMedia>
							<as:isAvailable>false</as:isAvailable>
				<as:isAvailableForEmergency>false</as:isAvailableForEmergency>
						</as:voiceMedia>
						<as:videoMedia>
							<as:isAvailable>false</as:isAvailable>
				<as:isAvailableForEmergency>false</as:isAvailableForEmergency>
						</as:videoMedia>
						<as:imMedia>
							<as:isAvailable>false</as:isAvailable>
				<as:isAvailableForEmergency>false</as:isAvailableForEmergency>
						</as:imMedia>
					</as:supportedIncomingMedia>
					<as:supportedOutgoingMedia>
						<as:voiceMedia>
							<as:isAvailable>false</as:isAvailable>
				<as:isAvailableForEmergency>false</as:isAvailableForEmergency>
						</as:voiceMedia>
						<as:imMedia>
							<as:isAvailable>false</as:isAvailable>
				<as:isAvailableForEmergency>false</as:isAvailableForEmergency>
						</as:imMedia>
					</as:supportedOutgoingMedia>
					<as:sessions>
						<!--array of 0 or more session elements-->
					</as:sessions>
				</as:activityState>
			</impp:status>
			<impp:contact priority="0.8">sip:jsmith@agency911.gov</impp:contact>
			<as:agentRoles>CallTaking,Supervisor</as:agentRoles>
			<impp:timestamp>2025-01-06T16:49:29Z</impp:timestamp>
		</impp:tuple>
</impp:presence>
```

In the following example, the Agent is in an emergency Instant Media (IM) session but is available for Emergency voiceMedia, videoMedia, imMedia, or rttMedia. The FE uses PUBLISH to send the following PIDF to the Presence Server.

```
<?xml version="1.0" encoding="UTF-8"?>
<impp:presence
	xmlns:xs="http://www.w3.org/2001/XMLSchema"
	xmlns:impp="urn:ietf:params:xml:ns:pidf"
	xmlns:as="urn:emergency:xml:ns:pidfExt:ActivityState"
  entity="John.Smith@myagency.mycounty.gov" >
	<impp:tuple id="sg89ae">
		<impp:status>
			<impp:basic>open</impp:basic>
			<!-- Overall Agent availability -->
```

```xml
            <as:activityState>
                <as:fe impp:mustUnderstand="true">CallHandling</as:fe>
                <as:supportedIncomingMedia>
                    <as:voiceMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:voiceMedia>
                    <as:videoMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:videoMedia>
                    <as:imMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:imMedia>
                    <as:rttMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:rttMedia>
                </as:supportedIncomingMedia>
                <as:supportedOutgoingMedia>
                    <as:voiceMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:voiceMedia>
                    <as:imMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:imMedia>
                    <as:rttMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:rttMedia>
                </as:supportedOutgoingMedia>
                <as:sessions>
                    <!--array of 0 or more session elements-->
                    <as:session>
                        <as:incomingMedia>
                            <as:mediaType>IM</as:mediaType>
                             </as:incomingMedia>
                        <as:outgoingMedia>
                            <as:mediaType>IM</as:mediaType>
                             </as:outgoingMedia>
                        <as:isEmergency>true</as:isEmergency>
                    </as:session>
                </as:sessions>
            </as:activityState>
        </impp:status>
        <impp:contact priority="0.8">sip:jsmith@agency911.gov</impp:contact>
        <as:agentRoles>CallTaking,Supervisor</as:agentRoles>
        <impp:timestamp>2025-01-06T16:49:29Z</impp:timestamp>
    </impp:tuple>
</impp:presence>
```

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When the Presence Server receives the PIDF above it applies the Agency's policy, which specifies that an Agent in an emergency IM session cannot take Voice, Video, or RTT calls, but can take another emergency or non-emergency IM call. The Presence Server sets availability for Voice, Video, and RTT media to false, and IM media to true. Subscribers to the Presence Server will then receive the following policy-adjusted PIDF. Note that the Presence Server has changed the availability values according to the Agency's policy:

```xml
<?xml version="1.0" encoding="UTF-8"?>
<impp:presence
        xmlns:xs="http://www.w3.org/2001/XMLSchema"
        xmlns:impp="urn:ietf:params:xml:ns:pidf"
        xmlns:as="urn:emergency:xml:ns:pidfExt:ActivityState"
  entity="John.Smith@myagency.mycounty.gov" >
        <impp:tuple id="sg89ae">
                <impp:status>
                        <impp:basic>open</impp:basic>
                        <!-- Overall Agent availability -->
                        <as:activityState>
                                <as:fe impp:mustUnderstand="true">CallHandling</as:fe>
                                <as:supportedIncomingMedia>
                                        <as:voiceMedia>
                                                <as:isAvailable>false</as:isAvailable>
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                        </as:voiceMedia>
                                        <as:videoMedia>
                                                <as:isAvailable>false</as:isAvailable>
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                        </as:videoMedia>
                                        <as:imMedia>
                                                <as:isAvailable>true</as:isAvailable>
                                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                                        </as:imMedia>
                                        <as:rttMedia>
                                                <as:isAvailable>false</as:isAvailable>
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                        </as:rttMedia>
                                </as:supportedIncomingMedia>
                                <as:supportedOutgoingMedia>
                                        <as:voiceMedia>
                                                <as:isAvailable>false</as:isAvailable>
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                        </as:voiceMedia>
                                        <as:imMedia>
                                                <as:isAvailable>true</as:isAvailable>
                                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                                        </as:imMedia>
                                        <as:rttMedia>
                                                <as:isAvailable>false</as:isAvailable>
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                        </as:rttMedia>
                                </as:supportedOutgoingMedia>
                                <as:sessions>
                                        <!--array of 0 or more session elements-->
```

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```
                        <as:session>
                            <as:incomingMedia>
                                <as:mediaType>IM</as:mediaType>
                            </as:incomingMedia>
                            <as:outgoingMedia>
                                <as:mediaType>IM</as:mediaType>
                            </as:outgoingMedia>
                            <as:isEmergency>true</as:isEmergency>
                        </as:session>
                    </as:sessions>
                </as:activityState>
            </impp:status>
            <impp:contact priority="0.8">sip:jsmith@agency911.gov</impp:contact>
            <as:agentRoles>CallTaking,Supervisor</as:agentRoles>
            <impp:timestamp>2025-01-06T16:49:29Z</impp:timestamp>
        </impp:tuple>
    </impp:presence>
```

In the following example, the application publishing the PIDF document implements two FEs – CallHandling and Collaboration. The activityState for each FE is in its own separate tuple. The Agent is on an emergency call with voice+video, and simultaneously in a non-emergency IM (i.e., MSRP) Collaboration session. Note the difference in contact URIs for the two FEs due to the difference in protocols.

```
<?xml version="1.0" encoding="UTF-8"?>
<impp:presence
        xmlns:xs="http://www.w3.org/2001/XMLSchema"
        xmlns:impp="urn:ietf:params:xml:ns:pidf"
        xmlns:as="urn:emergency:xml:ns:pidfExt:ActivityState"
  entity="John.Smith@myagency.mycounty.gov" >
        <impp:tuple id="sg89ae">
            <impp:status>
                <impp:basic>open</impp:basic>
                <!-- Overall Agent availability -->
                <as:activityState>
                    <as:fe impp:mustUnderstand="true">CallHandling</as:fe>
                    <as:supportedIncomingMedia>
                        <as:voiceMedia>
                            <as:isAvailable>false</as:isAvailable>
                <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                        </as:voiceMedia>
                        <as:videoMedia>
                            <as:isAvailable>false</as:isAvailable>
                <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                        </as:videoMedia>
                        <as:imMedia>
                            <as:isAvailable>true</as:isAvailable>
                    <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                        </as:imMedia>
                    </as:supportedIncomingMedia>
                    <as:supportedOutgoingMedia>
                        <as:voiceMedia>
                            <as:isAvailable>false</as:isAvailable>
```

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```
                        <as:isAvailableForEmergency>false</as:isAvailableForEmergency>
                                </as:voiceMedia>
                                <as:imMedia>
                                        <as:isAvailable>true</as:isAvailable>
                            <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                                </as:imMedia>
                        </as:supportedOutgoingMedia>
                        <as:sessions>
                                <!--array of 0 or more session elements-->
                                <as:session>
                                        <as:incomingMedia>
                                                <as:mediaType>Voice</as:mediaType>
                                                <as:mediaType>Video</as:mediaType>
                                                 </as:incomingMedia>
                                        <as:outgoingMedia>
                                                <as:mediaType>Voice</as:mediaType>
                                                 </as:outgoingMedia>
                                        <as:isEmergency>true</as:isEmergency>
                                </as:session>
                        </as:sessions>
                </as:activityState>
        </impp:status>
        <impp:contact priority="0.8">sip:jsmith@agency911.gov</impp:contact>
        <as:agentRoles>CallTaking,Supervisor</as:agentRoles>
        <impp:timestamp>2025-01-06T16:49:29Z</impp:timestamp>
    </impp:tuple>
    <impp:tuple id="sg90bf">
        <impp:status>
                <impp:basic>open</impp:basic>
                <as:activityState>
                        <as:fe impp:mustUnderstand="true">Collaboration</as:fe>
                        <as:supportedIncomingMedia>
                                <as:voiceCollaborationMedia>
                                        <as:isAvailable>true</as:isAvailable>
                        <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                                </as:voiceCollaborationMedia>
                                <as:videoCollaborationMedia>
                                        <as:isAvailable>true</as:isAvailable>
                        <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                                </as:videoCollaborationMedia>
                                <as:imCollaborationMedia>
                                        <as:isAvailable>true</as:isAvailable>
                        <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                                </as:imCollaborationMedia>
                                <as:rttCollaborationMedia>
                                        <as:isAvailable>true</as:isAvailable>
                        <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                                </as:rttCollaborationMedia>
                        </as:supportedIncomingMedia>
                        <as:supportedOutgoingMedia>
                                <as:voiceCollaborationMedia>
                                        <as:isAvailable>true</as:isAvailable>
                        <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
```


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```
                    </as:voiceCollaborationMedia>
                    <as:videoCollaborationMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:videoCollaborationMedia>
                    <as:imCollaborationMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:imCollaborationMedia>
                    <as:rttCollaborationMedia>
                        <as:isAvailable>true</as:isAvailable>
                <as:isAvailableForEmergency>true</as:isAvailableForEmergency>
                    </as:rttCollaborationMedia>
                </as:supportedOutgoingMedia>
                <as:sessions>
                    <!--array of 0 or more session elements-->
                    <as:session>
                        <as:incomingMedia>
                            <as:mediaType>IM</as:mediaType>
                             </as:incomingMedia
                        <as:outgoingMedia>
                            <as:mediaType>IM</as:mediaType>
                             </as:outgoingMedia>
                        <as:isEmergency>false</as:isEmergency>
                    </as:session>
                </as:sessions>
            </as:activityState>
    </imp:status>
    <impp:contact priority="0.5">msrp://jsmith@agency911.gov;tcp</impp:contact>
        <as:agentRoles>CallTaking,Supervisor</as:agentRoles>
        <impp:timestamp>2025-01-06T16:49:29Z</impp:timestamp>
    </impp:tuple>
</impp:presence>
```


11.3 Appendix C—Media Proxy Media Types

The following Media Types from the IANA Media Types registry [81] MUST be supported by the Media Proxy FE:

Table 11-1 Media Proxy Media Type

Commented [SD180]: Table name okay?

Media Type
application/gzip
application/json
application/mp4
application/ogg
application/pgp-encrypted
application/pdf
application/pidf-dff+xml
application/pidf+xml

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application/postscript
application/rtf
application/vnd.google-earth.kml+xml
application/vnd.google-earth.kmz
application/vnd.mcd
application/vnd.medcalcdata
application/vnd.ms-excel
application/vnd.ms-htmlhelp
application/vnd.ms-powerpoint
application/vnd.oasis.opendocument.chart
application/vnd.oasis.opendocument.database
application/vnd.oasis.opendocument.image
application/vnd.oasis.opendocument.presentation
application/vnd.oasis.opendocument.spreadsheet
application/vnd.oasis.opendocument.text
application/vnd.openxmlformats-officedocument.presentationml.presentation
application/vnd.openxmlformats-officedocument.spreadsheetml.sheet
application/vnd.openxmlformats-officedocument.wordprocessingml.document
application/x-abiword
audio/3gpp
audio/3gpp2
audio/aac
audio/aiff
audio/basic
audio/ogg
audio/midi
audio/mp4
audio/mp4a-latm
audio/mpeg
audio/mpeg4-generic
audio/mp3
audio/vnd.wav
audio/wav
audio/x-wav
audio/vnd.wave
audio/wave
audio/x-pn-wav
audio/x-pn-realaudio
audio/vorbis
image/bmp
image/gif
image/jpeg

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image/pjpeg
image/png
image/pwg-raster
image/svg+xml
image/tiff
image/vnd.adobe.photoshop
image/webp
image/x-canon-cr2
image/x-canon-crw
image/x-cmu-raster
image/x-icon
image/x-kodak-k25
image/x-kodak-kdc
image/x-minolta-mrw
image/x-nikon-nef
image/x-olympus-orf
image/x-panasonic-raw
image/x-pentax-pef
image/x-pict
image/x-portable-anymap
image/x-portable-bitmap
image/x-portable-graymap
image/x-portable-pixmap
image/x-rgb
image/x-sony-sr2
message/http
message/rfc822
message/sip
message/sipfrag
multipart/encrypted
text/csv
text/html
text/markdown
text/plain
text/sgml
text/tab-separated-values
video/3gpp
video/3gpp2
video/h261
video/h263
video/h264
video/mp4

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video/mpeg
video/mpeg4-generic
video/ogg
video/quicktime
video/raw
video/vnd.motorola.video
video/vnd.youtube.yt
video/x-flv
video/x-m4v
video/x-ms-wmv
video/x-msvideo

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The NG9-1-1 PSAP Systems Working Group is part of the NENA Development Group that is led by:

- Wendi Rooney, ENP, and Lisa Dodson, ENP, Development Steering Council Co-Chairs
- Brandon Abley, ENP, Chief Technology Officer
- April Heinze, ENP, VP, Chief of 9-1-1 Operations

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GENERAL DOCUMENT EDITING HELP

Commented [SD181]: Delete prior to WG Approval Ballot.

NOTE to document editor: NENA documents follow the Chicago Manual of Style (CMOS). Document editors are to use the following guidelines when editing NENA documents. If there are any questions regarding this section, contact the CRM team.

Delete this entire section from the document before submitting the document for external review.

Navigating the Styles Gallery and Styles Pane

NOTE: Document editors MUST use the styles listed in the **Styles Gallery** in this template. These predefined Word styles are set to follow NENA's required font and formatting options.

- The most frequently used styles are shown in the **Styles Gallery** on the **Home** tab.
- Open the **Styles Panel** to see additional styles available in the document by clicking on the small arrow in the bottom right corner of the **Styles Gallery**.
- To apply a style, select the text to be formatted, and then click on the desired style in the **Style Gallery** or **Styles Panel.**
- Use the **Font** options on the **Home** tab to apply formatting (e.g., bold, italics).

Watermark

- All pages must include a **DRAFT** watermark.

Font

- **Verdana 10 pt**
- When copying in text from a document, email, chat box, or other source, use the **Keep Text Only** paste option so text is added with the correct font and formatting.

Paragraph Formatting

- Use the Word Style **Body Text** for text in the document's main body. This adds visual space between paragraphs, eliminating the need for a blank line.

Messages/Code in Text

- **Font:** Lucida Console 10pt
- **Left indent:** .5
- **XML elements:** shown in brackets (<...>)
- **JSON objects:** shown in double quotes with brackets ({"..."})
- **Values:** shown in double quotes ("...")
- **Examples:**

```
<note>
      <to>Dern Handy</to>
      <from>Brandon</from>
      <heading>Reminder</heading>
      <body>Don't forget to book!</body>
</note>

{
    "iFoo":12,
    "sBar":"blurdybloop"
}

“urn:service:sos”
```

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Footnotes

- Cited documents[20], websites, and other references MUST be cited in the References section, not in footnotes. If something is important enough to include as a footnote, it is strongly recommended that the information be included in the document text.
- **Numbers:** Superscript Verdana 7pt
- **Text:** Verdana 8pt

Punctuation

- A comma is placed before a conjunction (e.g., and, or) in a series of three or more.
- Periods are followed by a **single** space.
- Periods and commas are placed **within** closing quotation marks.
    - **Exception:** If file names, words, or other strings to be typed are within quotation marks, any punctuation that belongs to the surrounding sentence should appear outside the quotation marks.
- Colons, semicolons, question marks, and exclamation points are placed **outside** closing quotation marks (unless they belong within the quoted matter).
- CMOS prefers square brackets [ ] within parentheses instead of parentheses within parentheses.
- The abbreviations **i.e.** ("that is") and **e.g.** ("for example") are placed within parentheses, where they are followed by a comma.
- CMOS prefers to limit the abbreviation **etc.** ("and other things") to parentheses. If used in a sentence, **etc.** is preceded by a comma. Do NOT use **etc.** at the end of a list that begins with *for example*, *such as*, *e.g.*, and the like.

Non-breaking Hyphens

- The CRM Team will insert non-breaking hyphens for 9-1-1 and all hyphenated words at time of publication. Editors should use regular hyphens during document development to avoid copy/paste of non-breaking hyphens being changed to spaces (no hyphen).
- **alt+shift+hyphen** inserts a non-breaking hyphen which is treated as a letter, keeping the word on a single line.

Vertical Lists—bulleted vs. numbered

- Numbered lists are used to indicate the order in which tasks should be done, to suggest chronology or relative importance among the items, or to facilitate text references.
- A vertical list is best introduced by a grammatically complete sentence, followed by a colon.
- All items should be styled with the same syntax, either sentences or sentence fragments.
- All items should use the same verb tense.
- For bulleted lists, unless the items themselves form complete sentences, use lowercase for the first letter of each item (unless it is a proper noun) and omit periods.
- For bulleted lists whose items require more prominence, capitalization may be preferred; choose one approach and follow it consistently.
- Capitalize items in a numbered list even if the items do not consist of complete sentences. Closing punctuation is used only if items consist of complete sentences.
- If the items in a vertical list complete a sentence that began in the introductory text, commas or semicolons may be used between the items (use semicolons if the items include internal punctuation), a period should follow the final item, and no colon is used in the introductory sentence. A conjunction (e.g., and, or) before the final item is optional.

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[20] Example text to show footnotes with 7pt font for the footnote number and 8pt font for the footnote text.

- Items in a vertical list should be short and to the point.

Vertical List Symbology and Indentation

- **Hanging Indents:** All use .25”
- **Paragraph Indents and Spacing:**
    - Don’t add space between paragraphs of the same style (check box)
- **Paragraph Line and Page Breaks:**
    - Widow/Orphan Control (keep box checked) *Note: This only affects a bulleted paragraph consisting of 2-3 lines. Bullets with a single line are not affected.
    - Keep with next (uncheck box)

Bulleted and numbered lists can be formatted by using the **Format Painter** tool to apply the formatting below to the text in the core document. They can also be created by using the styles listed in the **Styles** gallery on the **Home** tab or by using the **Bullets** and **Numbering** formatting options within the **Paragraph** group on the **Home** tab.

<u>Bulleted List</u>

- Left Indent .25
- Left Indent .25
    - Left Indent .75
    - Left Indent .75
        - Left Indent 1.25
        - Left Indent 1.25
            - Left Indent 1.75
            - Left Indent 1.75

<u>Numbered List</u>

1. Left Indent .25
2. Left Indent .25
    a. Left Indent .75
    b. Left Indent .75
        i. Left Indent 1.25
        ii. Left Indent 1.25
            1. Left Indent 1.75
            2. Left Indent 1.75

<u>Alpha List</u>

A. Left Indent .25
B. Left Indent .25 (AA-ZZ: Hanging indent .38”)
    a. Left Indent .75
    b. Left Indent .75
        i. Left Indent 1.25
        ii. Left Indent 1.25
            1. Left Indent 1.75
            2. Left Indent 1.75

Tables

The example tables below MUST be copied and pasted into the document and adjusted as needed.

- **Table Alignment:** Centered on page
- **Table Width:** 6.75” (maximum)
- **Table Caption:** Centered above table, Verdana 10pt Bold, Keep with Next pagination

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- **Cell Margins:** R/L: 0.08”; Top: 0”; Bottom: 0.04”
- **Table Text Wrapping:** Around→Positioning→ Bottom 0.2”
- **Header Row:** Bold, repeats every page

To update the Table Number to reflect the correct section it was copied into, highlight the table number, right-click on the number, and select **Update Field**.

Table 11-2 General Document Editing Example Table with 3 Headings (3 columns)

Table 11-3 General Document Editing Example Table with 4 Headings (4 columns)

Figures

Figures are typically inserted between paragraphs or are placed at the end of a numbered section. Figure numbers should update automatically as they are inserted into the document, but if needed, they can be manually updated. To do so, highlight the figure number, right-click on the number, and select **Update Field**.

Insert figures on their own line using the following **Layout** settings:

- **Text Wrapping:** Top and bottom
- **Size:** Use **Absolute** settings with 6.75” maximum width
- **Position**
  - **Horizontal:** Alignment Centered relative to column
  - **Vertical:** Absolute position 0” below Line
  - **Options:** Check **Move object with text** and **Lock anchor**

Add captions once the figures are properly placed and sized. If figure size or position is adjusted later, captions may need to be redone for proper spacing between the caption and figure. Use the following **Caption** settings:

- **Label:** Figure
- **Position:** Below selected item
- **Style:** Caption (Verdana 10pt, bold)

Alt Text: Add a brief sentence to describe the diagram for someone who is blind or low vision. Example Alt text has been added to the figures below.

Copyrighted figure from an external source: Permission must be granted by the owner and may require a “used with permission” statement.

Store source files on NENA Workspace: Upload source files for all diagrams, images, and illustrations (e.g., Visio, Illustrator, PowerPoint) into NENA Workspace for future editing.

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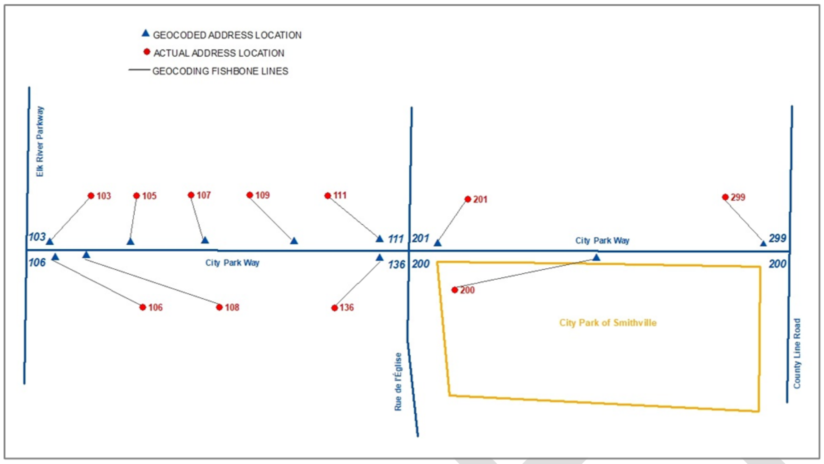

Figure 11-1 General Document Editing Effect of Geocoding Addresses against Actual Address Ranges

9-1-1 [Call]	Assumed IP Call Transport; may include Geolocation Header/PIDF-LO
Signaling+Media (Data)	User 9-1-1 call -> OSP -> NGCS -> i3 PSAP. Media may include voice, messaging, and video.
OSP	Originating Service Provider (can include an Application Service Provider, ASP)
NGCS	Next Generation Core Services (in the context of an ESInet-Emergency Services IP Network)
PSAP (i3)	NG9-1-1 Capable Public Safety Answering Point

Figure 11-2 General Document Editing NENA i3 NG9-1-1 Reference Architecture

Cross-References—How to Jump Between the Citations

- Use <Ctrl+click> on a cross-reference to jump to the cited reference in the References section or to the targeted section, table, or figure in the document.
- To return to the clicked cross-reference, use <Alt+Left Arrow key>.

Cross-References—How to Add

To add a Cross-reference **to a cited document, web page, or other cited item listed in the References section**, use these steps to insert the Reference number:

- Insert the curser after the cited reference in the text and on the **References** tab, select **Cross-reference**.
- To add the cited reference number, in the **Cross-reference** pop-up box, set **Reference type:** to **Numbered item**, and set **Insert reference to:** to **Paragraph number**.
- Scroll through the **For which numbered item:** list and select the appropriate Reference that is being cited.
- Select the **Insert** button.
- Change the font color of the inserted reference, including the brackets, to Blue.

To add a Cross-reference **to a cited section** in the document, use these steps to insert the number and title of the section:

- Insert the curser where you want to cite the section in the text, and on the **References** tab, select **Cross-reference**.
- In the **Cross-reference** pop-up box, set **Reference type:** to **Heading**, and set **Insert reference to:** to **Heading Number**.
- Scroll through the **For which heading:** list and select the appropriate section heading that is being cited.
- Select the **Insert** button.
- Add a space to separate the section heading number from the section heading title.
- In the **Cross-reference** pop-up box, set **Insert reference to:** to **Heading text**.
- In the **For which heading:** list, select the appropriate section heading that is being cited.
- Select the **Insert** button.
- Change the font color of the inserted number and title of the section to Blue.

To add a Cross-reference **to a cited table** in the document, use these steps to insert the number and title of the table:

- Insert the curser where you want to cite the table in the text, and on the **References** tab, select **Cross-reference**.
- In the **Cross-reference** pop-up box, set **Reference type:** to **Table**, and set **Insert reference to:** to **Entire caption**.
- Scroll through the **For which caption:** list and select the appropriate table that is being cited.
- Select the **Insert** button.
- Change the font color of the inserted table caption to Blue.

To add a Cross-reference **to a cited figure** in the document, use these steps to insert the number and title of the figure:

- Insert the curser where you want to cite the figure in the text, and on the **References** tab, select **Cross-reference**.
- In the **Cross-reference** pop-up box, set **Reference type:** to **Figure**, and set **Insert reference to:** to **Entire caption**.
- Scroll through the **For which caption:** list and select the appropriate figure that is being cited.
- Select the **Insert** button.
- Change the font color of the inserted figure caption to Blue.

Cross-References—How to Update

When cited references are added or removed from the References section, all cross-references in the document need to be updated. Similarly, when sections are added or removed, section numbers will change and all cross-references to sections, tables, and figures will need to be updated. These can be updated individually or all at once.

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- To update all cross-references at once, select the entire document <Ctrl+A>, then press **F9**. It is important to confirm that all cross-references were updated and go to the correct cited reference, section, table, or figure by selecting each updated cross-reference <Ctrl+click>.

To update an individual cross-reference, right-click on the specific cross-reference and select **Update Field**. Confirm the cross-reference was updated and goes to the correct cited reference, section, table, or figure by selecting on the updated cross-reference <Ctrl+click>.

Navigation Shortcuts

Use the **Navigation Pane** to quickly jump to a section or specific text in the document rather than scrolling through the pages.

On the **View** tab, check the **Navigation Pane** box to display it.

- **Search for a section:**
  - **Headings:** Click on a section title to jump to that section.
  - **Pages:** Scroll through page thumbnail images to quickly find a table, image, or other content when a section title is unknown and a visual clue is needed. Click on the thumbnail image of interest to jump to that page.
- **Search for a word or phrase:** Type the text in the **Search document** at the top of the Navigation Pane.
  - **Headings:** Section titles containing the typed text are highlighted in yellow.
  - **Pages:** Thumbnail images are displayed for only those pages containing the typed text.
  - **Results:** Each individual search result is displayed with some of its surrounding text.

Check Accessibility

Use Microsoft's **Accessibility Assistant** to identify and fix common accessibility issues that make it difficult for blind or low vision people. The **Status Bar** at the bottom of the window will display **Accessibility: Investigate** if there is an accessibility issue.

On the **Review** tab, select **Check Accessibility** to open the Accessibility pane on the right. Issues and suggested resolutions are listed in the Accessibility pane.

- **Hard-to-read text contrast:** If the text color is hard to see. Use the styles listed in the **Styles** gallery as they do not cause accessibility issues.
- **Missing Alt Text:** A short 1-2 sentence description is needed to describe what the image is for someone who is blind or has low vision.
- **Use of merged or split cells:** Cells should not be merged or split as it makes it difficult for screen readers to convey the information.

Inclusive Language

- Review the document to confirm that terms used in the document meet NENA's [Use of Inclusive Language Policy], found in the NENA Workspace Administrative Procedure, Templates & Training Material.

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Page 44: [1] Commented [MLS69R2]	Michael Smith	17/10/2024 16:58:00

Agreed to propose text in the Transferring Calls section that allows you to include last known caller location in an EIDO, when transferring a callback call to another agency.

Page 44: [2] Commented [MLS69]	Michael Smith	09/10/2024 14:35:00

Discuss this:
When transferring a callback call to another agency, the locationReference may no longer return the latest location. Per 2024 JCM discussion, we could allow CHFE to provide location by value in the EIDO within the call to the other agency - an exception to our rule of not providing data by value that we got by reference.

Page 61: [3] Commented [100]	Michael Smith	07/10/2021 14:16:00

If the CHFE or a logger provides silent monitoring without involving a Bridge or a separate call leg, we still want to log it, so we probably need a separate LogEvent for that purpose.

Page 61: [4] Commented [99]	Michael Smith	07/10/2021 14:11:00

We have partyAdd, partyRemove, and callBargeIn callStateChanges. How about addind partyMute and partyUnMute? The Bridge or CHFE could log these.

Page 61: [5] Commented [98]	Michael Smith	07/10/2021 13:47:00

STA-010 needs to detail how media/events get logged by the bridge – different users can have different streams. What if the Bridge is not the SRC? A BCF would probably just be logging the caller's leg.

Page 61: [6] Commented [101]	Michael Smith	21/10/2021 13:52:00

The Logger must also log this if it provides monitoring. Brian also proposed standardizing a method for initiating silent monitoring from, for example, the Management Console.

Page 61: [7] Commented [108]	Michael Smith	21/10/2021 13:52:00

The Logger must also log this if it provides monitoring. Brian also proposed standardizing a method for initiating silent monitoring from, for example, the Management Console.

Page 61: [8] Commented [MS110R2]	Michael Smith	30/04/2026 12:52:00

TODO: study this and propose a solution.
Suresh: leave it up to implementation.

Page 61: [9] Commented [111]	Dan	30/09/2021 01:27:00

This needs to be specified in STA-010 Bridging FE section. For ad hoc it is simple, CHFE simply creates a new conference and move all three party to the conference. But for an Answer All Calls at a Bridge, how is the Conference session URI obtained?

Page 61: [10] Deleted	Daniel Mongrain	22/06/2026 14:46:00